

SUPERPOSITION

8 Superposition

8.1 Stationary waves

Candidates should be able to:

- 1 explain and use the principle of superposition
- 2 show an understanding of experiments that demonstrate stationary waves using microwaves, stretched strings and air columns (it will be assumed that end corrections are negligible; knowledge of the concept of end corrections is not required)
- 3 explain the formation of a stationary wave using a graphical method, and identify nodes and antinodes
- 4 understand how wavelength may be determined from the positions of nodes or antinodes of a stationary wave

8.2 Diffraction

Candidates should be able to:

- 1 explain the meaning of the term diffraction
- 2 show an understanding of experiments that demonstrate diffraction including the qualitative effect of the gap width relative to the wavelength of the wave; for example diffraction of water waves in a ripple tank

8.3 Interference

Candidates should be able to:

- 1 understand the terms interference and coherence
- 2 show an understanding of experiments that demonstrate two-source interference using water waves in a ripple tank, sound, light and microwaves
- 3 understand the conditions required if two-source interference fringes are to be observed
- 4 recall and use $\lambda = ax/D$ for double-slit interference using light

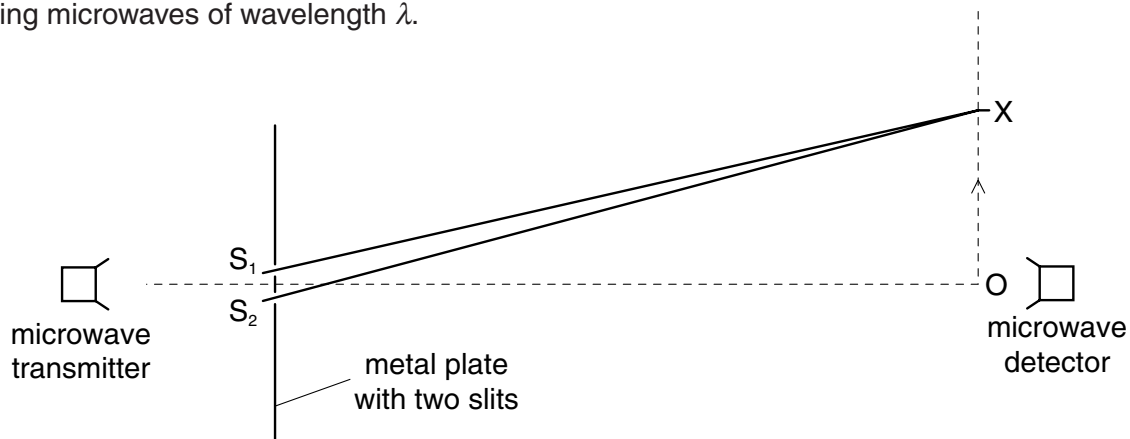
8.4 The diffraction grating

Candidates should be able to:

- 1 recall and use $d \sin \theta = n\lambda$
- 2 describe the use of a diffraction grating to determine the wavelength of light (the structure and use of the spectrometer are not included)

1 9702/1/M/J/02/Q28

The diagram shows an experiment which has been set up to demonstrate two-source interference, using microwaves of wavelength λ .



The detector is moved from O in the direction of the arrow. The signal detected decreases until the detector reaches the point X, and then starts to increase again as the detector moves beyond X.

Which equation correctly determines the position of X?

- A** $OX = \lambda/2$ **B** $OX = \lambda$ **C** $S_2X - S_1X = \lambda/2$ **D** $S_2X - S_1X = \lambda$

2 9702/1/M/J/02/Q29

Two progressive waves of frequency 300 Hz are superimposed to produce a stationary wave in which adjacent nodes are 1.5 m apart.

What is the speed of the progressive waves?

- A** 100 m s^{-1} **B** 200 m s^{-1} **C** 450 m s^{-1} **D** 900 m s^{-1}

3 9702/1/O/N/02/Q28

Coherent monochromatic light illuminates two narrow parallel slits and the interference pattern that results is observed on a screen some distance beyond the slits.

Which change increases the separation between the dark lines of the interference pattern?

- A** using monochromatic light of higher frequency
B using monochromatic light of a longer wavelength
C decreasing the distance between the screen and the slits
D increasing the distance between the slits

4 9702/1/O/N/02/Q29

Monochromatic light of wavelength 590 nm is incident normally on a diffraction grating. The angle between the two second-order diffracted beams is 43° .

What is the spacing of the lines on the grating?

- A** $0.87 \mu\text{m}$ **B** $1.6 \mu\text{m}$ **C** $1.7 \mu\text{m}$ **D** $3.2 \mu\text{m}$

5 9702/01/O/N/03/Q28

Microwaves of wavelength 3.00 cm are incident normally on a row of parallel metal rods. The separation of the rods is 8.00 cm. The first order diffraction maximum is observed at an angle of 22.0° to the direction of the incident waves.

What is the angle between the first and second order diffraction maxima?

- A** 22.0° **B** 26.6° **C** 44.0° **D** 48.6°

6 9702/01/M/J/03/Q27

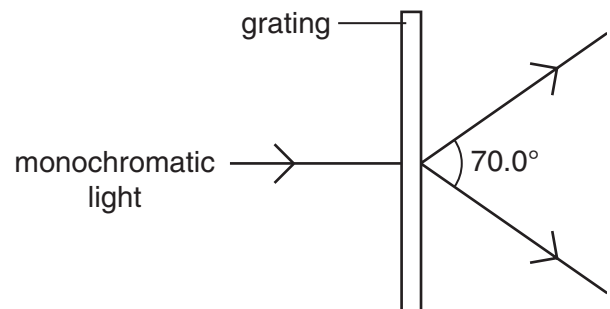
When the light from two lamps falls on a screen, no interference pattern can be obtained.

Why is this?

- A** The lamps are not point sources.
- B** The lamps emit light of different amplitudes.
- C** The light from the lamps is not coherent.
- D** The light from the lamps is white.

7 9702/01/M/J/03/Q28**

A diffraction grating is used to measure the wavelength of monochromatic light, as shown in the diagram.



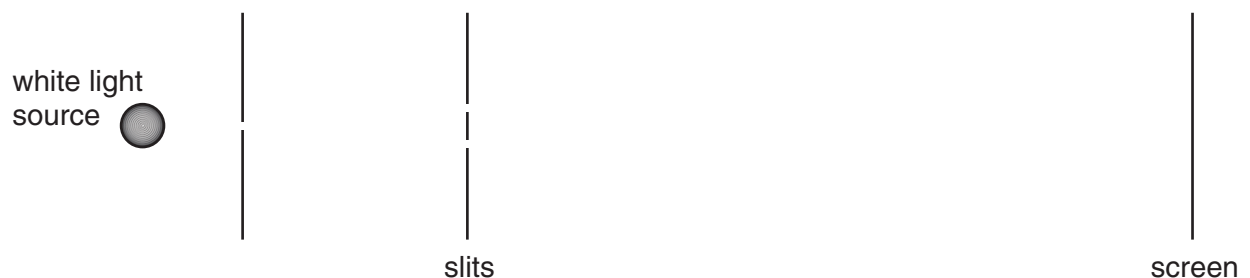
The spacing of the slits in the grating is $1.00 \times 10^{-6} \text{ m}$. The angle between the first order diffraction maxima is 70.0° .

What is the wavelength of the light?

- A** 287 nm **B** 470 nm **C** 574 nm **D** 940 nm

8 9702/01/O/N/03/Q27

In an interference experiment, two slits are illuminated with white light.



What is seen on the screen?

- A** The central fringe is black with black and white fringes on each side.
- B** The central fringe is black with coloured fringes on each side.
- C** The central fringe is white with black and white fringes on each side.
- D** The central fringe is white with coloured fringes on each side.

9 9702/01/O/N/04/Q28

Fringes of separation y are observed on a screen 1.00 m from a Young's slit arrangement that is illuminated by yellow light of wavelength 600 nm.

At which distance from the slits would fringes of the same separation y be observed when using blue light of wavelength 400 nm?

- A** 0.33 m **B** 0.67 m **C** 0.75 m **D** 1.50 m

10 9702/13/M/J/14/Q26

What is the approximate range of frequencies of infra-red radiation?

- A** $1 \times 10^3 \text{ Hz}$ to $1 \times 10^9 \text{ Hz}$ **C** $1 \times 10^{11} \text{ Hz}$ to $1 \times 10^{14} \text{ Hz}$
B $1 \times 10^9 \text{ Hz}$ to $1 \times 10^{11} \text{ Hz}$ **D** $1 \times 10^{14} \text{ Hz}$ to $1 \times 10^{17} \text{ Hz}$

11 9702/01/M/J/04/Q28

The lines of a diffraction grating have a spacing of $1.6 \times 10^{-6} \text{ m}$. A beam of light is incident normally on the grating. The first order maximum makes an angle of 20° with the undeviated beam.

What is the wavelength of the incident light?

- A** 210 nm **B** 270 nm **C** 420 nm **D** 550 nm

12 9702/01/M/J/05/Q27*

T is a microwave transmitter placed at a fixed distance from a flat reflecting surface S.



A small microwave receiver is moved steadily from T towards S and receives signals of alternate maxima and minima of intensity.

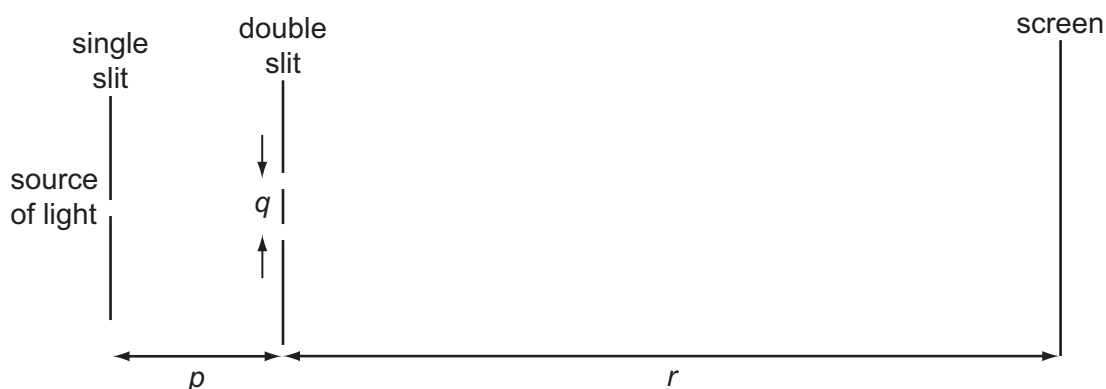
The distance between successive maxima is 15 mm.

What is the frequency of the microwaves?

- A** $1.0 \times 10^7 \text{ Hz}$ **B** $2.0 \times 10^7 \text{ Hz}$ **C** $1.0 \times 10^{10} \text{ Hz}$ **D** $2.0 \times 10^{10} \text{ Hz}$

13 9702/01/M/J/05/Q28*

A teacher sets up the apparatus shown to demonstrate a two-slit interference pattern on the screen.



Which change to the apparatus will increase the fringe spacing?

- A** decreasing the distance p **C** decreasing the distance r
B decreasing the distance q **D** decreasing the wavelength of the light

14 9702/01/M/J/05/Q29

A parallel beam of white light is incident normally on a diffraction grating. It is noted that the second-order and third-order spectra partially overlap.

Which wavelength in the third-order spectrum appears at the same angle as the wavelength of 600 nm in the second-order spectrum?

- A** 300 nm **B** 400 nm **C** 600 nm **D** 900 nm

15 9702/01/O/N/05/Q27

In which situation does diffraction occur?

- A** A wave bounces back from a surface. **B** A wave passes from one medium into another.
C A wave passes through an aperture. **D** Waves from two identical sources are superposed.

16 9702/01/O/N/05/Q28

Light of wavelength 700 nm is incident on a pair of slits, forming fringes 3.0 mm apart on a screen.

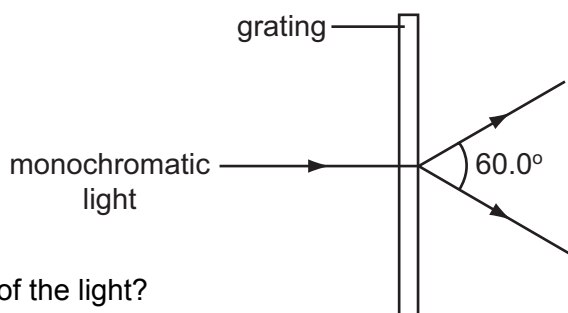
What is the fringe spacing when light of wavelength 350 nm is used and the slit separation is doubled?

- A** 0.75 mm **B** 1.5 mm **C** 3.0 mm **D** 6.0 mm

17 9702/11/O/N/16/Q29

A diffraction grating is used to measure the wavelength of monochromatic light.

The spacing of the slits in the grating is 1.15×10^{-6} m. The angle between the first order diffraction maxima is 60.0° , as shown in the diagram.



What is the wavelength of the light?

- A** 288 nm **B** 498 nm **C** 575 nm **D** 996 nm

18 9702/01/M/J/06/Q27

Monochromatic light is incident on a diffraction grating and a diffraction pattern is observed.

Which row shows possible effects of replacing the grating with one that has twice as many lines per millimetre?

	number of orders of diffraction visible	angle between first and second orders of diffraction
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

19 9702/01/O/N/06/Q27

The interference patterns from a diffraction grating and a double slit are compared.

Using the diffraction grating, yellow light of the first order is seen at 30° to the normal to the grating.

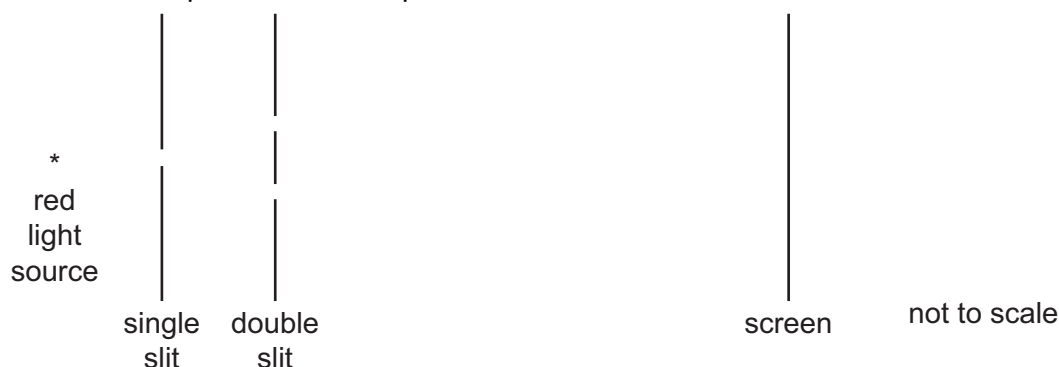
The same light produces interference fringes on a screen 1.0 m from the double slit. The slit separation is 500 times greater than the line spacing of the grating.

What is the fringe separation on the screen?

- A** 2.5×10^{-7} m **B** 1.0×10^{-5} m **C** 1.0×10^{-3} m **D** 1.0×10^{-1} m

20 9702/01/M/J/06/Q28

A double-slit interference experiment is set up as shown.



Fringes are formed on the screen. The distance between successive bright fringes is found to be 4 mm.

Two changes are then made to the experimental arrangement. The double slit is replaced by another double slit which has half the spacing. The screen is moved so that its distance from the double slit is twice as great.

What is now the distance between successive bright fringes?

- A** 1mm **B** 4mm **C** 8mm **D** 16mm

21 9702/01/M/J/07/Q26

A two-slit arrangement is set up to produce interference fringes on a screen. The fringes are too close together for convenient observation when a monochromatic source of violet light is used.

In which way would it be possible to increase the separation of the fringes?

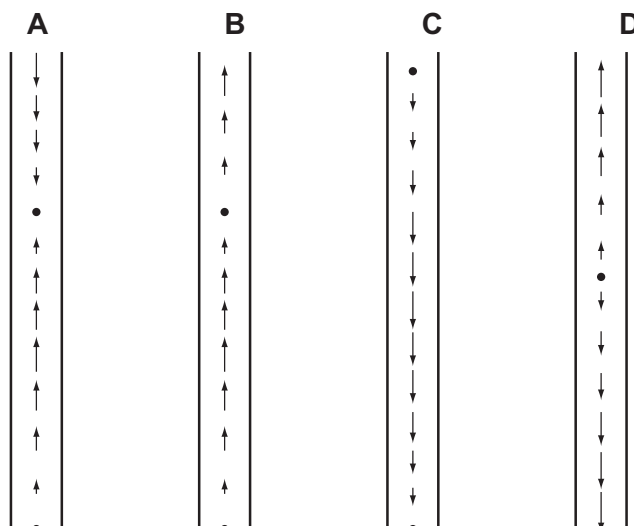
- A** Decrease the distance between the screen and the slits.
B Increase the distance between the two slits.
C Increase the width of each slit.
D Use a monochromatic source of red light.

22 9702/01/M/J/07/Q27

A stationary longitudinal wave is set up in a pipe.

In the diagrams below, the length of each arrow represents the amplitude of the motion of the air molecules, and the arrow head shows the direction of motion at a particular instant.

Which diagram shows a stationary wave in which there are two nodes and two antinodes?



23 9702/01/O/N/07/Q24

Monochromatic light illuminates two narrow parallel slits. The interference pattern which results is observed on a screen some distance beyond the slits.

Which change increases the separation between the dark lines of the interference pattern?

- A** decreasing the distance between the screen and the slits
- B** increasing the distance between the slits
- C** using monochromatic light of higher frequency
- D** using monochromatic light of longer wavelength

24 9702/01/O/N/07/Q25

A narrow beam of monochromatic light is incident normally on a diffraction grating. Third-order diffracted beams are formed at angles of 45° to the original direction.

What is the highest order of diffracted beam produced by this grating?

- A** 3rd
- B** 4th
- C** 5th
- D** 6th

25 9702/01/M/J/08/Q28

A diffraction grating has N lines per unit length and is placed at 90° to monochromatic light of wavelength λ .

What is the expression for θ , the angle to the normal to the grating at which the third order diffraction peak is observed?

- A** $\sin \theta = \frac{1}{3N\lambda}$
- B** $\sin \theta = 3N\lambda$
- C** $\sin \theta = \frac{N\lambda}{3}$
- D** $\sin \theta = \frac{3\lambda}{N}$

26 9702/01/M/J/08/Q29

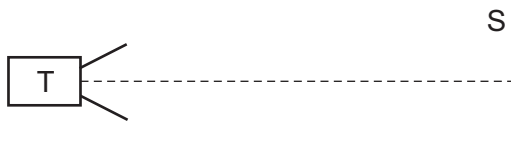
Light of wavelength 700 nm is incident on a pair of slits, forming fringes 3.0 mm apart on a screen.

What is the fringe spacing when light of wavelength 350 nm is used and the slit separation is doubled?

- A** 0.75 mm
- B** 1.5 mm
- C** 3.0 mm
- D** 6.0 mm

27 9702/01/O/N/08/Q27

T is a microwave transmitter placed at a fixed distance from a flat reflecting surface S.



A small microwave receiver is moved from T towards S and receives signals of alternate maxima and minima of intensity.

The distance between one maximum and the next is 15 mm.

What is the frequency of the microwaves?

- A** 1.0×10^7 Hz
- B** 2.0×10^7 Hz
- C** 1.0×10^{10} Hz
- D** 2.0×10^{10} Hz

28 9702/12/O/N/09/Q25

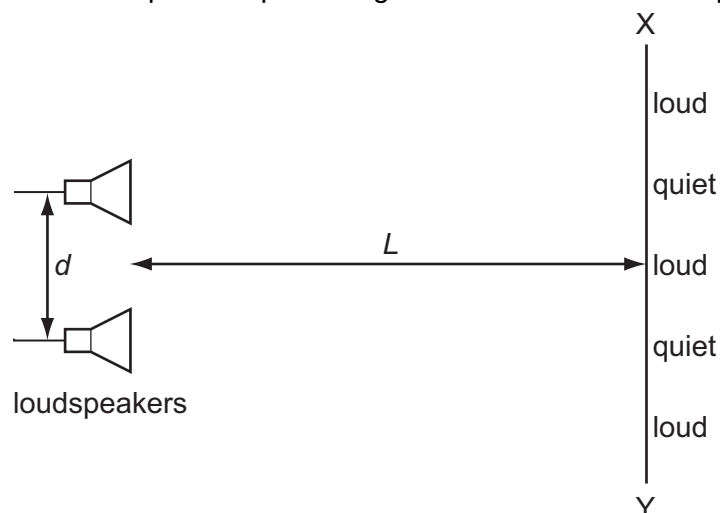
A parallel beam of light of wavelength 450 nm falls normally on a diffraction grating which has 300 lines/mm.

What is the total number of transmitted maxima?

- A** 7
- B** 8
- C** 14
- D** 15

29 9702/01/O/N/08/Q28

The diagram shows two loudspeakers producing sound waves that are in phase.



As a student moves from X to Y, the intensity of the note she hears is alternately loud and quiet.

The distance between adjacent loud and quiet regions may be reduced by

- | | |
|------------------------------------|------------------------------------|
| A decreasing distance d . | C decreasing the amplitude. |
| B increasing distance L . | D increasing the frequency. |

30 9702/13/O/N/13/Q28*

Diffraction is the name given to the

- A** addition of two coherent waves to produce a stationary wave pattern.
- B** bending of waves round an obstacle.
- C** change of direction when waves cross the boundary between one medium and another.
- D** splitting of white light into colours.

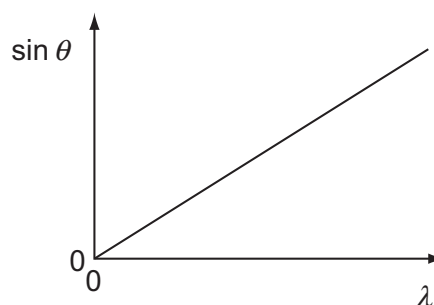
31 9702/13/M/J/16/Q27

A diffraction grating with N lines per metre is used to deflect light of various wavelengths λ .

The diagram shows a relation between the deflection angles θ for different values of λ in the n^{th} order interference pattern.

What is the gradient of the graph?

- | | |
|------------------------|-------------------------|
| A Nn | B $\frac{N}{n}$ |
| C $\frac{n}{N}$ | D $\frac{1}{Nn}$ |



32 9702/12/M/J/10/Q25

Using monochromatic light, interference fringes are produced on a screen placed a distance D from a pair of slits of separation a . The separation of the fringes is x .

Both a and D are now doubled.

What is the new fringe separation?

- | | | | |
|------------------------|--------------|---------------|---------------|
| A $\frac{x}{2}$ | B x | C $2x$ | D $4x$ |
|------------------------|--------------|---------------|---------------|

33 9702/11/O/N/10/Q27

Which electromagnetic wave would cause the most significant diffraction effect for an atomic lattice of spacing around 10^{-10} m?

- A infra-red B microwave C ultraviolet D X-ray

34 9702/12/O/N/10/26

A microwave transmitter emits waves towards a metal plate. The waves strike the plate and are reflected back along their original path.



A microwave detector is moved along the line PT.

Points P, Q, R, S and T are the positions where minima of intensity are observed. These points are found to be 15 mm apart.

What is the frequency of the microwaves?

- A 5.0 GHz B 6.7 GHz C 10 GHz D 20 GHz

35 9702/12/O/N/10/Q27

A double slit experiment, using light of wavelength 600 nm, results in fringes being produced on a screen. The fringe separation is found to be 1.0 mm.

When the distance between the double slits and the viewing screen is increased **by** 2.0 m, the fringe separation increases **to** 3.0 mm.

What is the separation of the double slits producing the fringes?

- A 0.4 mm B 0.6 mm C 0.9 mm D 1.2 mm

36 9702/11/M/J/14/Q27

The basic principle of note production in a horn is to set up a stationary wave in an air column.



For the lowest note produced by a horn, a node is formed at the mouthpiece and the antinode is formed at the bell. The frequency of this note is 75 Hz.

What are the frequencies of the next two higher notes for this air column?

	first higher note / Hz	second higher note / Hz
A	113	150
B	150	225
C	150	300
D	225	375

37 9702/11/M/J/11/Q26

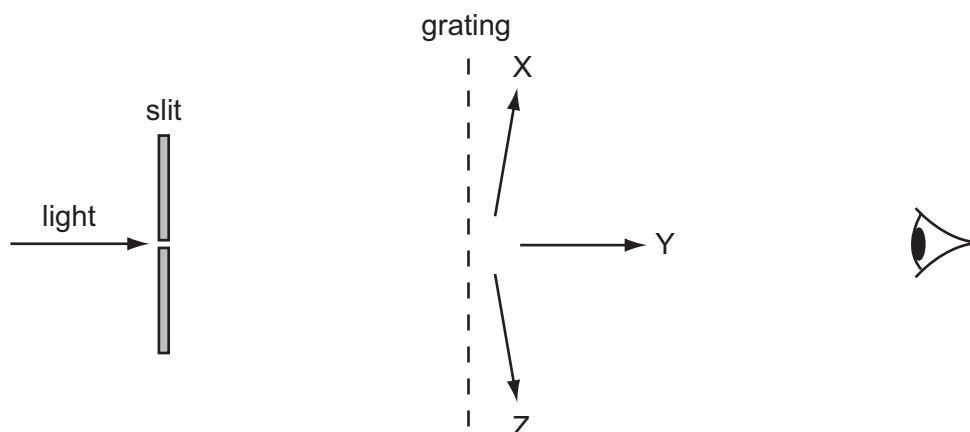
Which electromagnetic wave phenomenon is needed to explain the spectrum produced when white light falls on a diffraction grating?

- A coherence B interference C polarisation D refraction

38 9702/11/M/J/11/Q27

A diffraction grating with 500 lines per mm is used to observe diffraction of monochromatic light of wavelength 600 nm.

The light is passed through a narrow slit and the grating is placed so that its lines are parallel to the slit. Light passes through the slit and then the grating.



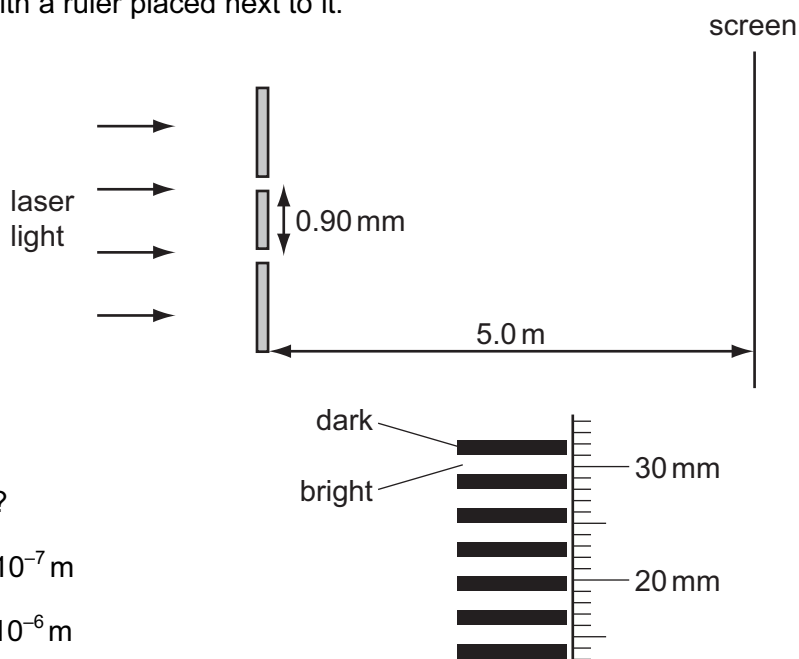
An observer views the slit through the grating at different angles, moving his head from X parallel to the grating, through Y, opposite the slit, to Z parallel to the grating on the opposite side.

How many images of the slit does he see?

- A 3 B 4 C 6 D 7

39 9702/12/M/J/11/Q29

The diagrams show the arrangement of apparatus for a Young's slits experiment and also part of the pattern formed on the screen with a ruler placed next to it.



What is the wavelength of the light?

- A $4.8 \times 10^{-7} \text{ m}$ B $5.4 \times 10^{-7} \text{ m}$
C $3.2 \times 10^{-6} \text{ m}$ D $3.4 \times 10^{-6} \text{ m}$

40 9702/12/M/J/11/Q25

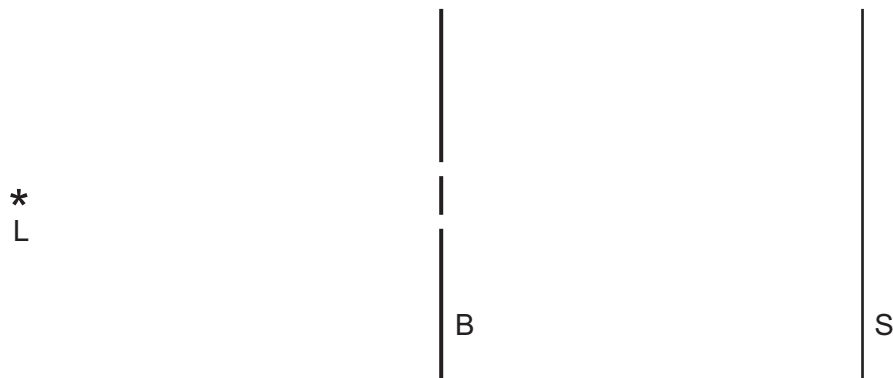
In which order of magnitude are the frequencies of electromagnetic waves in the visible spectrum?

- A 10^{12} Hz B 10^{13} Hz C 10^{14} Hz D 10^{15} Hz

41 9702/11/O/N/11/Q28

The diagram shows a view from above of a double slit interference demonstration.

L is a monochromatic light source with a vertical filament. B is a barrier with two narrow vertical slits and S is a screen upon which interference fringes form.



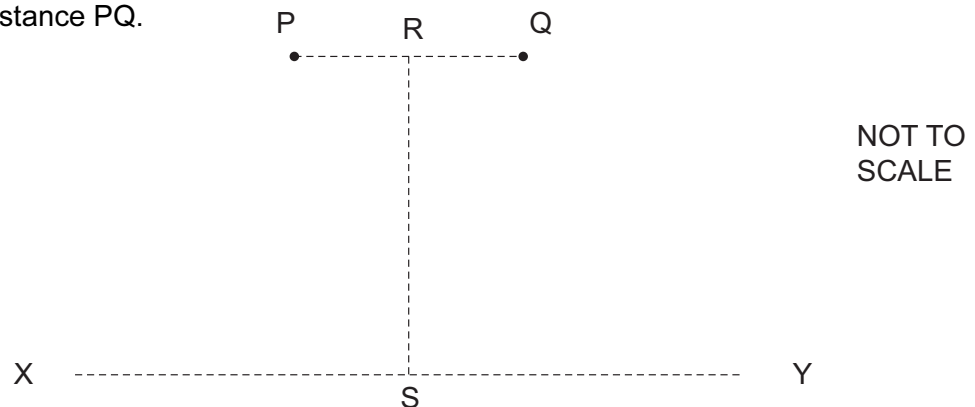
The intensity is I at a point on the screen where the centre of the fringe pattern forms.

What is the intensity, at the same point, when one of the slits is covered up?

- A $\frac{I}{\sqrt{2}}$ B $\frac{I}{2}$ C $\frac{I}{2\sqrt{2}}$ D $\frac{I}{4}$

42 9702/11/O/N/11/Q30

Coherent waves are produced at P and at Q and travel outwards in all directions. The line RS is halfway between P and Q and perpendicular to the line joining P and Q. The distance RS is much greater than the distance PQ.



Along which line, or lines, is an interference pattern observed?

- A both RS and XY B RS only C XY only D neither RS nor XY

43 9702/12/O/N/11/Q28

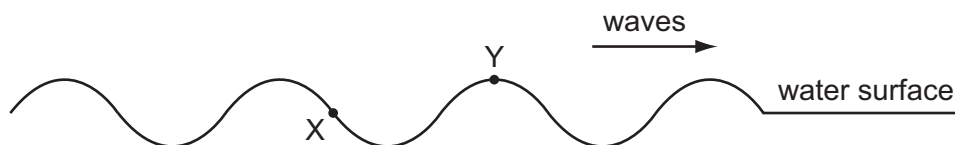
Two light sources produce visible interference fringes only in certain circumstances.

Which condition enables visible interference fringes to be formed?

- A using a white light source
B using incoherent sources
C using one light source which is polarised at right angles to light from the other source
D using sources from which the light does not overlap

44 9702/12/M/J/12/Q28

X and Y are two points on the surface of water in a ripple tank. A source of waves of constant frequency begins to generate waves which then travel past X and Y, causing them to oscillate.



What is the phase difference between X and Y?

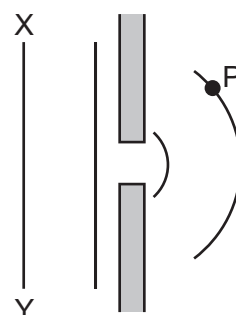
- A 45° B 135° C 180° D 270°

45 9702/12/M/J/12/Q29

A monochromatic plane wave of speed c and wavelength λ is diffracted at a small aperture. The diagram illustrates successive wavefronts.

After what time will some portion of the wavefront XY reach point P?

- A $\frac{3\lambda}{2c}$ B $\frac{2\lambda}{c}$
C $\frac{3\lambda}{c}$ D $\frac{4\lambda}{c}$



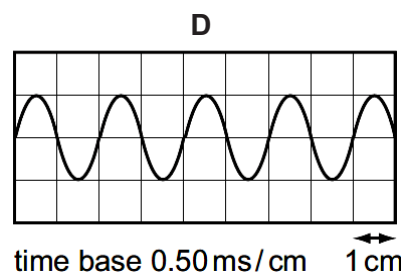
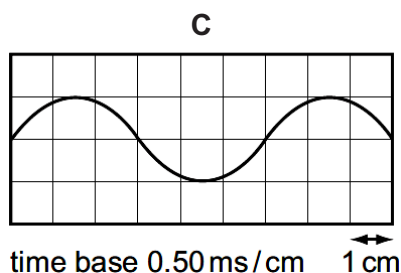
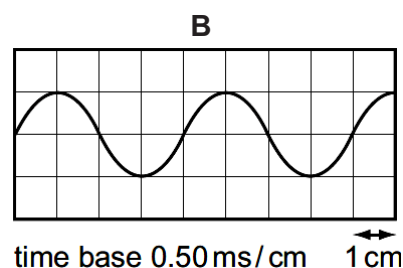
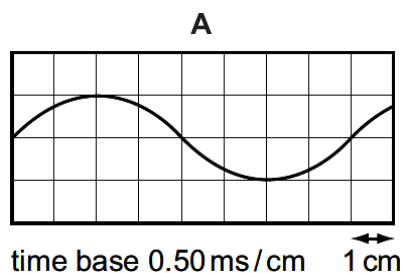
46 9702/12/M/J/12/Q30

A standing sound wave is set up between a loudspeaker and a wall.

A microphone is connected to a cathode-ray oscilloscope (c.r.o.) and is moved along a line directly between the loudspeaker and the wall. The amplitude of the trace on the c.r.o. rises to a maximum at a position X, falls to a minimum and then rises once again to a maximum at a position Y.

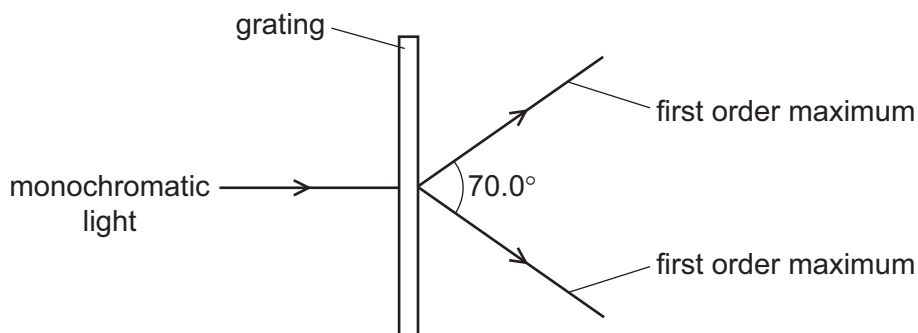
The distance between X and Y is 33 cm. The speed of sound in air is 330 ms^{-1} .

Which diagram represents the c.r.o. trace of the sound received at X?



47 9702/13/M/J/12/Q28

A diffraction grating is used to measure the wavelength of monochromatic light, as shown in the diagram.



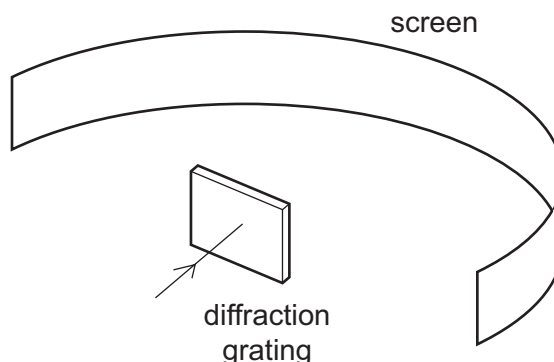
The spacing of the slits in the grating is 1.00×10^{-6} m. The angle between the first order diffraction maxima is 70.0° .

What is the wavelength of the light?

- A 287 nm B 470 nm C 574 nm D 940 nm

48 9702/12/O/N/12/Q30

Monochromatic light of wavelength 690 nm passes through a diffraction grating with 300 lines per mm, producing a series of maxima on a screen.



What is the greatest number of maxima that can be observed?

- A 4 B 5 C 8 D 9

49 9702/13/O/N/12/Q28

Diffraction can be observed when a wave passes an obstruction. The diffraction effect is greatest when the wavelength and the obstruction are similar in size.

For waves travelling through air, what is the combination of wave and obstruction that could best demonstrate diffraction?

- A microwaves passing a steel post C sound waves passing a human hair
B radio waves passing a copper wire D visible light waves passing a gate post

50 9702/12/M/J/13/Q29

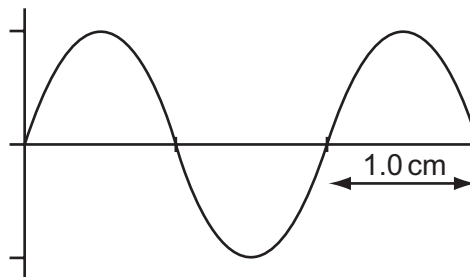
Monochromatic light of wavelength 5.30×10^{-7} m is incident normally on a diffraction grating. The first order maximum is observed at an angle of 15.4° to the direction of the incident light.

What is the angle between the first and second order diffraction maxima?

- A 7.6° B 15.4° C 16.7° D 32.0°

51 9702/12/O/N/12/Q31

The diagram shows a cathode-ray oscilloscope display of an electromagnetic wave.



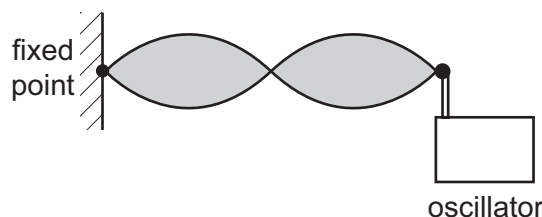
The time base setting is $0.20 \mu\text{s cm}^{-1}$.

Which statement is correct?

- A The frequency of the wave is 2.5 MHz and it lies in the radio wave region of the electromagnetic spectrum.
- B The frequency of the wave is 2.5 MHz and it lies in the microwave region of the electromagnetic spectrum.
- C The frequency of the wave is 5.0 MHz and it lies in the radio wave region of the electromagnetic spectrum.
- D The frequency of the wave is 5.0 MHz and it lies in the microwave region of the electromagnetic spectrum.

52 9702/11/M/J/13/Q28

The speed of a transverse wave on a stretched string can be changed by adjusting the tension of the string. A stationary wave pattern is set up on a stretched string using an oscillator set at a frequency of 650 Hz.



How must the wave be changed to maintain the same stationary wave pattern if the applied frequency is increased to 750 Hz?

- A Decrease the speed of the wave on the string.
- B Decrease the wavelength of the wave on the string.
- C Increase the speed of the wave on the string.
- D Increase the wavelength of the wave on the string.

53 9702/11/M/J/13/Q29

Noise reduction headphones actively produce their own sound waves in order to cancel out external sound waves.

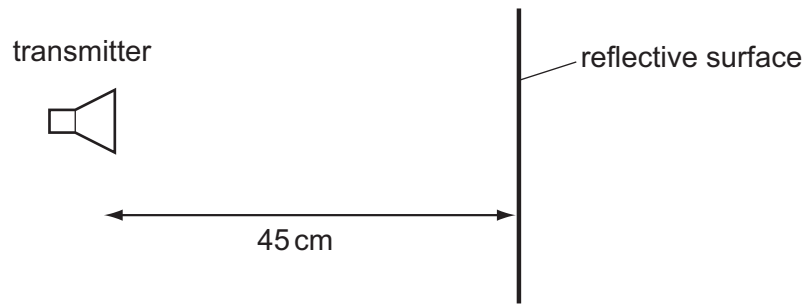
A microphone in the headphones receives waves of one frequency. A loudspeaker in the headphones then produces a wave of that frequency but of a different phase.

What is the phase difference between the external sound wave and the wave produced by the loudspeaker in the headphones?

- A 90°
- B 180°
- C 270°
- D 360°

54 9702/12/M/J/13/Q27

A transmitter of electromagnetic waves is placed 45 cm from a reflective surface.



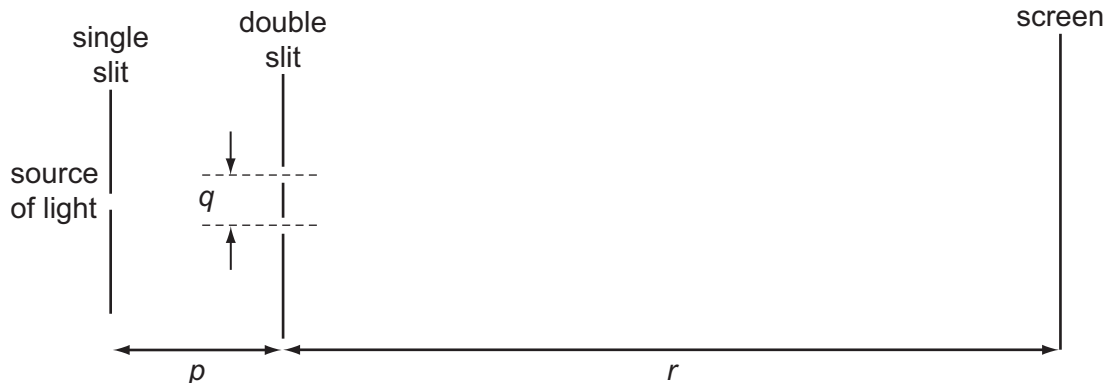
The emitted waves have a frequency of 1.00 GHz. A stationary wave is produced with a node at the transmitter and a node at the surface.

How many antinodes are in the space between the transmitter and the surface?

- A 1 B 2 C 3 D 4

55 9702/01/M/J/05/Q28

A teacher sets up the apparatus shown to demonstrate a two-slit interference pattern on the screen.



Which change to the apparatus will increase the fringe spacing?

- A decreasing the distance p C decreasing the distance r
 B decreasing the distance q D decreasing the wavelength of the light

56 9702/13/M/J/13/Q27

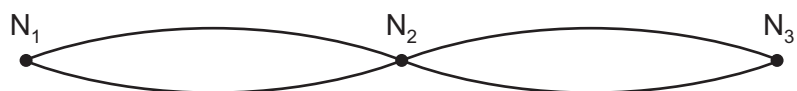
A parallel beam of red light of wavelength 700 nm is incident normally on a diffraction grating that has 400 lines per millimetre.

What is the total number of transmitted maxima?

- A 3 B 4 C 6 D 7

57 9702/13/M/J/13/Q28

The diagram shows a standing wave on a string. The standing wave has three nodes N_1 , N_2 and N_3 .

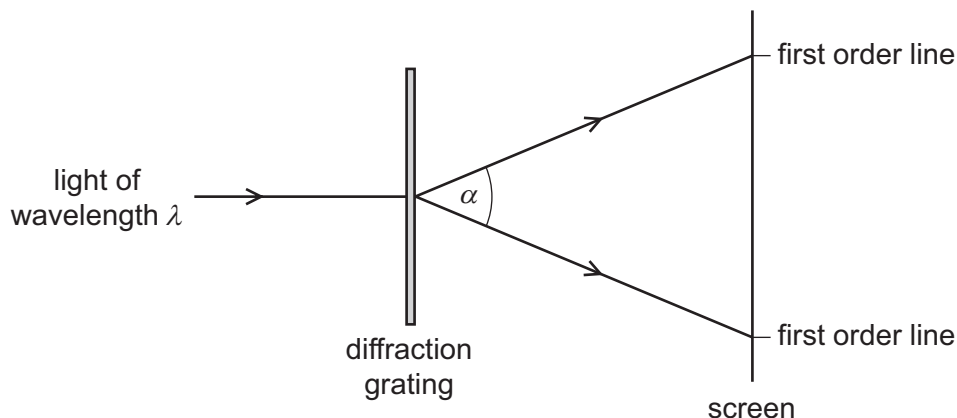


Which statement is correct?

- A All points on the string vibrate in phase.
 B All points on the string vibrate with the same amplitude.
 C Points equidistant from N_2 vibrate with the same frequency and in phase.
 D Points equidistant from N_2 vibrate with the same frequency and the same amplitude.

58 9702/12/O/N/13/Q28

Light of wavelength λ passes through a diffraction grating with slit spacing d . A series of lines is observed on a screen.

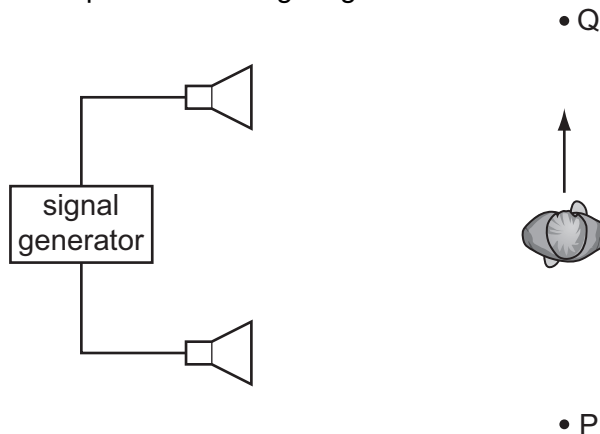


What is the angle α between the two first order lines?

- A $\sin^{-1}\left(\frac{\lambda}{2d}\right)$ B $\sin^{-1}\left(\frac{\lambda}{d}\right)$ C $2 \sin^{-1}\left(\frac{\lambda}{2d}\right)$ D $2 \sin^{-1}\left(\frac{\lambda}{d}\right)$

59 9702/12/O/N/13/Q29

A student connects two loudspeakers to a signal generator.



As the student walks from P to Q, he notices that the loudness of the sound rises and falls repeatedly.

What causes the loudness of the sound to vary?

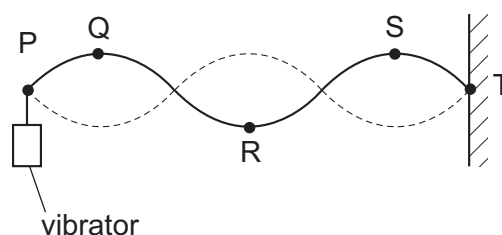
- A diffraction of the sound waves C polarisation of the sound waves
B interference of the sound waves D reflection of the sound waves

60 9702/11/M/J/14/Q25

A stationary wave on a stretched string is set up between two points P and T.

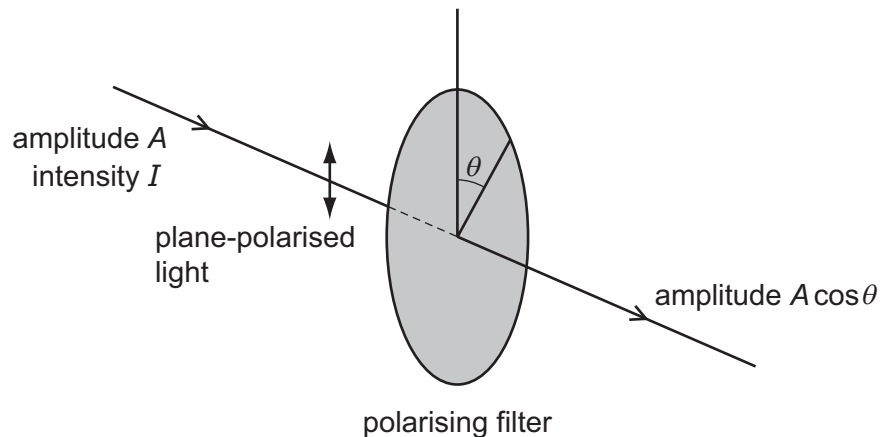
Which statement about the wave is correct?

- A Point R is at a node.
B Points Q and S vibrate in phase.
C The distance between P and T is three wavelengths.
D The wave shown has the lowest possible frequency.



61 9702/12/O/N/13/Q30

When plane-polarised light of amplitude A is passed through a polarising filter as shown, the amplitude of the light emerging is $A \cos \theta$.



The intensity of the initial beam is I .

What is the intensity of the emerging light when θ is 60.0° ?

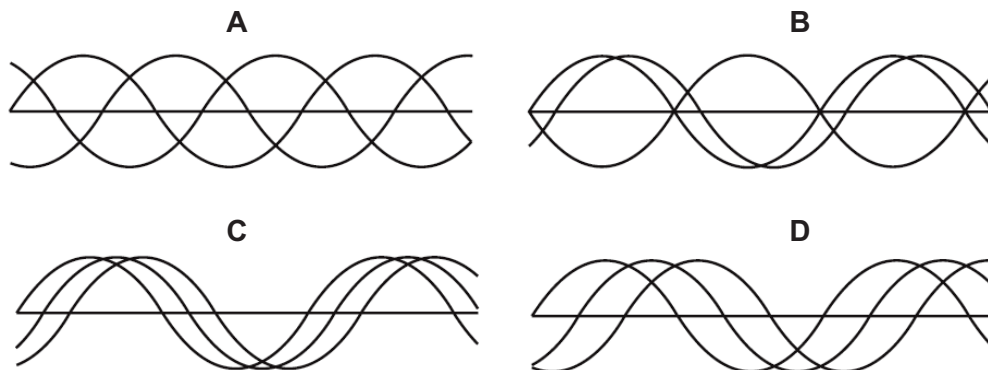
- A** $0.250 I$ **B** $0.500 I$ **C** $0.750 I$ **D** $0.866 I$

62 9702/13/O/N/13/Q26

The three waves shown in each diagram have the same amplitude and frequency but differ in phase.

They are added together to give a resultant wave.

In which case is the resultant wave zero?



63 9702/13/O/N/13/Q27

A stationary sound wave has a series of nodes. The distance between the first and the sixth node is 30.0 cm.

What is the wavelength of the sound wave?

- A** 5.0 cm **B** 6.0 cm **C** 10.0 cm **D** 12.0 cm

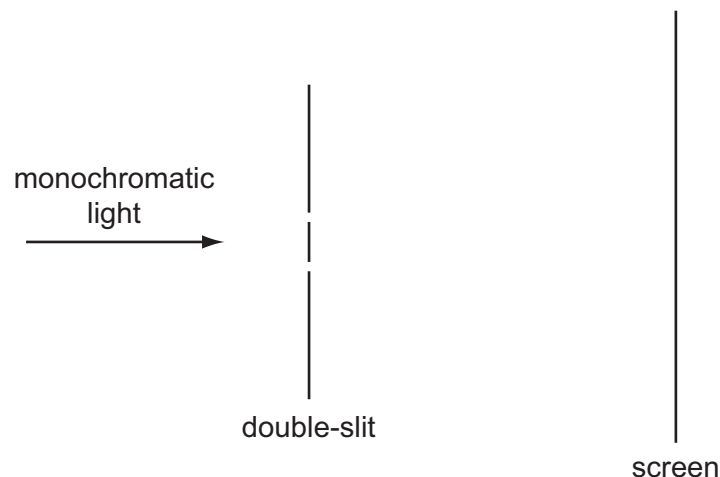
64 9702/13/O/N/13/Q28

What is meant by diffraction?

- A** Addition of two coherent waves to produce a stationary wave pattern.
B Bending of waves round an obstacle.
C Change of direction when waves cross the boundary between one medium and another.
D Splitting of white light into colours.

65 9702/13/O/N/13/Q29

A student sets up apparatus to observe the double-slit interference of monochromatic light, as shown.



Interference fringes are formed on the screen.

Which change would increase the distance between adjacent fringes?

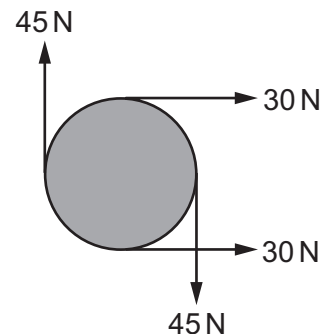
- A Decrease the distance between the two slits.
- B Decrease the width of each slit.
- C Move the screen closer to the double-slit.
- D Use light of a higher frequency.

66 9702/12/M/J/14/Q13

The diagram shows four forces applied to a circular object.

Which of the following describes the resultant force and resultant torque on the object?

	resultant force	resultant torque
A	non-zero	non-zero
B	non-zero	zero
C	zero	non-zero
D	zero	zero



67 9702/12/M/J/14/Q25

The principle of superposition states that a certain quantity is added when two or more waves meet at a point.

What is this quantity?

- A amplitude
- B displacement
- C intensity
- D wavelength

68 9702/13/M/J/14/Q28

A student attempts to show the interference of light using two identical green LEDs.

Which statement explains why the experiment will **not** succeed?

- A The light waves from the sources are not coherent.
- B The light waves from the sources do not have the same amplitude.
- C The light waves from the sources have a range of wavelengths.
- D The light waves from the sources are not monochromatic.

69 9702/12/M/J/14/Q26

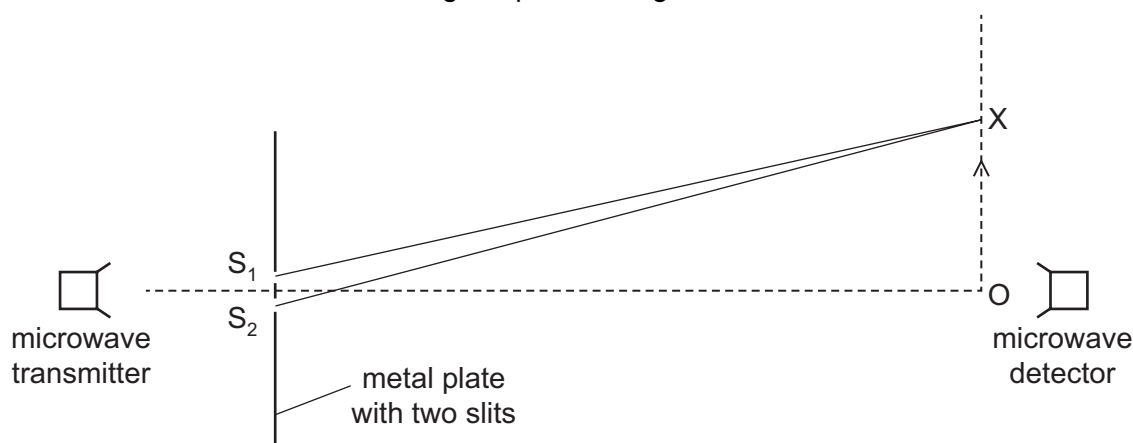
Light passes through a diffraction grating ruled at 1000 lines per cm and the same wavelength of light also passes through two narrow slits 0.5 mm apart. Both situations produce intensity maxima and minima on a screen.

Which statement about the separation of the maxima on the screen and the sharpness of the maxima is correct?

- A The diffraction grating maxima are less widely spaced and are less sharp than the two-slit maxima.
- B The diffraction grating maxima are less widely spaced and are sharper than the two-slit maxima.
- C The diffraction grating maxima are more widely spaced and are less sharp than the two-slit maxima.
- D The diffraction grating maxima are more widely spaced and are sharper than the two-slit maxima.

70 9702/12/M/J/14/Q27

The diagram shows an experiment which has been set up to demonstrate two-source interference. Microwaves of wavelength λ pass through two slits S_1 and S_2 .



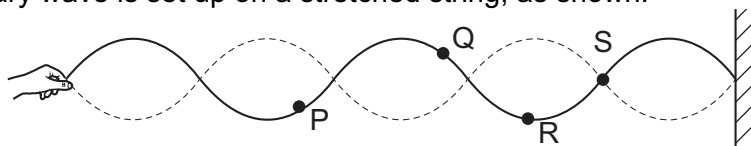
The detector is moved from point O in the direction of the arrow. The signal detected decreases until the detector reaches point X, and then starts to increase again as the detector moves beyond X.

Which equation correctly determines the position of X?

- A $OX = \lambda$
- B $OX = \lambda/2$
- C $S_2X - S_1X = \lambda$
- D $S_2X - S_1X = \lambda/2$

71 9702/13/M/J/14/Q29

A stationary wave is set up on a stretched string, as shown.



Which statement about the points on the string is correct?

- A Point Q vibrates with the largest amplitude.
- B Points P and R vibrate in phase.
- C Point S is an antinode.
- D The horizontal distance between R and S is half the wavelength.

72 9702/13/M/J/14/Q30

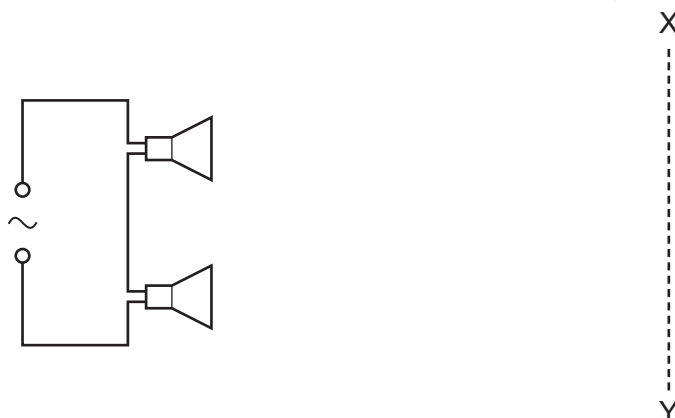
Monochromatic light is incident on a diffraction grating and a diffraction pattern is observed.

Which line of the table gives the effect of replacing the grating with one that has more lines per metre?

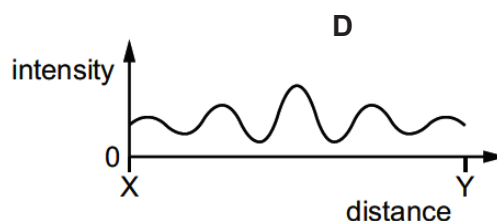
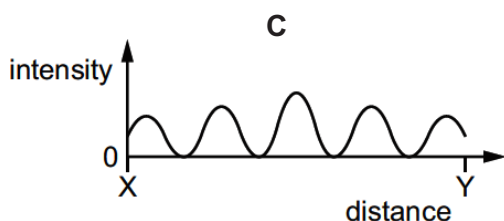
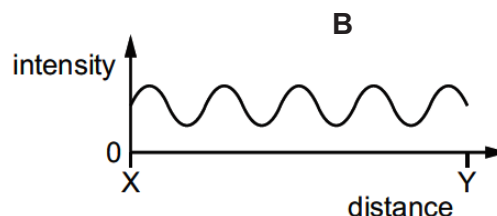
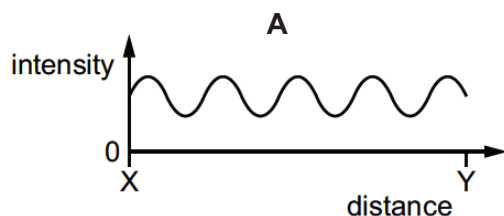
	number of orders of diffraction visible	angle between first and second orders of diffraction
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

73 9702/11/O/N/14/Q26

Two identical loudspeakers are connected in series to an a.c. supply, as shown.



Which graph best shows the variation of the intensity of the sound with distance along the line XY?



74 9702/13/O/N/14/Q30

Interference fringes are produced on a screen by double-slit interference using light of wavelength 600 nm. The fringe separation is 4.0 mm and the separation of the slits is 0.60 mm.

What is the distance between the double slit and the screen?

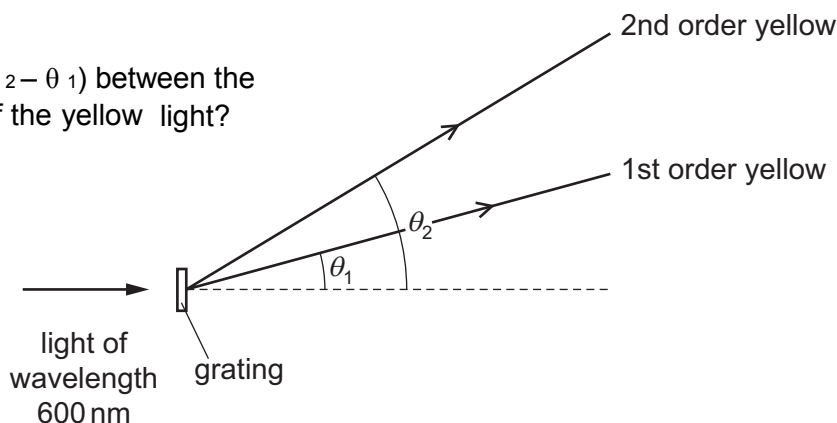
- A** 0.25 m **B** 0.40 m **C** 2.5 m **D** 4.0 m

75 9702/11/O/N/14/Q27

A diffraction grating experiment is set up using yellow light of wavelength 600 nm. The grating has a slit separation of $2.00 \mu\text{m}$.

What is the angular separation ($\theta_2 - \theta_1$) between the first and second order maxima of the yellow light?

- A 17.5° B 19.4°
C 36.9° D 54.3°



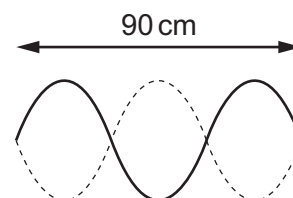
76 9702/11/M/J/15/Q28

The diagram shows a stationary wave on a string at two instants of maximum vertical displacement.

The frequency of the wave is 12 Hz.

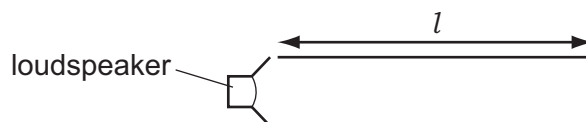
What is the speed of the wave?

- A 3.6 ms^{-1} B 7.2 ms^{-1} C 360 ms^{-1} D 720 ms^{-1}



77 9702/11/M/J/15/Q29

A loudspeaker emitting sound of frequency f is placed at the open end of a pipe of length l which is closed at the other end. A standing wave is set up in the pipe.



A series of pipes are then set up with either one or two loudspeakers of frequency f . The pairs of loudspeakers vibrate in phase with each other.

Which pipe contains a standing wave?

- A
- B
- C
- D

78 9702/11/M/J/15/Q30

In a double-slit experiment the distance between the fringes, on a screen, was too small to measure.

What would increase the distance between the fringes?

- A increasing the distance between the light source and the slits
- B increasing the distance between the slits and the screen
- C increasing the distance between the slits
- D increasing the frequency of the light source

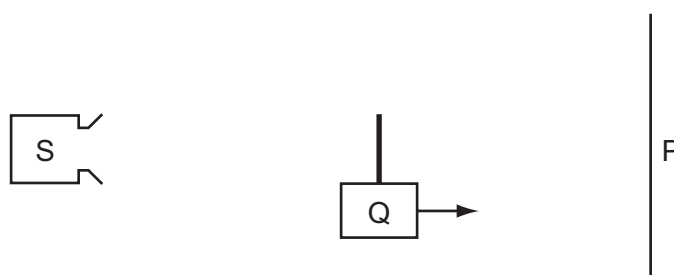
79 9702/12/M/J/15/Q26

What is **not** an **essential** condition for an observable interference pattern to occur between the waves from two sources?

- A The frequencies of the two sources must be equal.
- B The sources must be coherent.
- C The sources must emit waves of equal amplitude.
- D The waves from the two sources must overlap.

80 9702/12/M/J/15/Q27

Source S emits microwaves with a constant amplitude. The microwaves hit a metal screen P and are reflected. A stationary wave is formed between S and P. The wavelength of the microwaves is much smaller than the distance between S and P.



A detector Q is moved at a slow, constant speed from S to P.

What happens to the amplitude of the signal detected by Q?

- A decreases steadily
- B increases and decreases regularly
- C increases steadily
- D remains constant

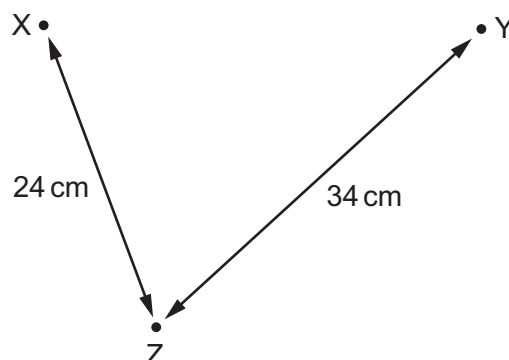
81 9702/13/M/J/15/Q29

Wave generators at points X and Y produce water waves of the same wavelength. At point Z, the waves from X have the same amplitude as the waves from Y. Distances XZ and YZ are as shown.

When the wave generators operate in phase, the amplitude of oscillation at Z is zero.

What could be the wavelength of the waves?

- A 2 cm
- B 3 cm
- C 4 cm
- D 6 cm

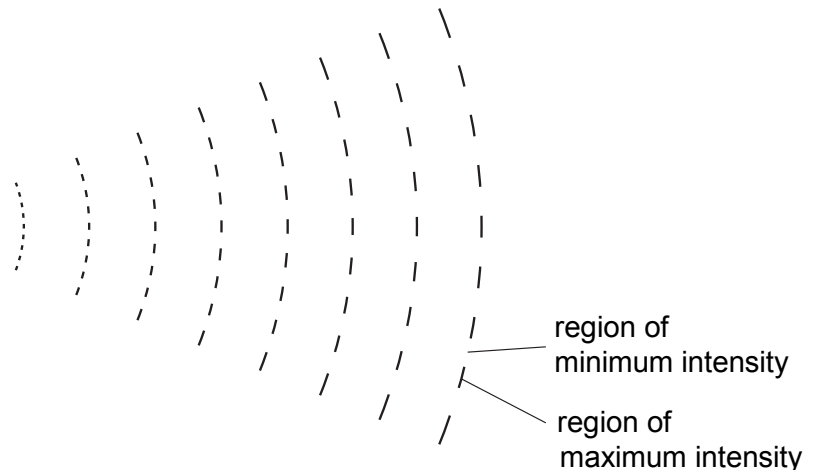


82 9702/12/M/J/15/Q28

A pattern of waves was observed without being able to view the source of the waves. The pattern is represented in the diagram.

What can cause this pattern?

- A coherence only
- B diffraction and interference
- C diffraction only
- D interference only



83 9702/11/M/J/17/Q28

A parallel beam of light of wavelength 600 nm is incident normally on a diffraction grating. The grating has 300 lines per millimetre.

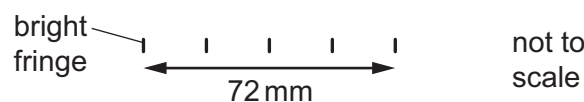
What is the total number of intensity maxima from the grating?

- A 1
- B 3
- C 11
- D 13

84 9702/11/M/J/17/Q29

A pattern of interference fringes is produced using a red laser, a double slit and a screen. The screen is 3.5 m from the double slit. The light from the laser has a wavelength of 640 nm.

The pattern of fringes is shown.



What is the separation of the slits?

- A $1.2 \times 10^{-4} \text{ m}$
- B $1.6 \times 10^{-4} \text{ m}$
- C $3.1 \times 10^{-5} \text{ m}$
- D $3.3 \times 10^{-9} \text{ m}$

85 9702/12/M/J/17/Q28

A parallel beam of red light of wavelength 700 nm is incident normally on a diffraction grating that has 400 lines per millimetre.

What is the total number of intensity maxima from the grating?

- A 6
- B 7
- C 8
- D 9

86 9702/12/M/J/17/Q29

Two wave sources are oscillating in phase. Each source produces a wave of wavelength λ . The two waves from the sources meet at point X with a phase difference of 90° .

What is a possible difference in the distances from the two wave sources to point X?

- A $\frac{\lambda}{8}$
- B $\frac{\lambda}{4}$
- C $\frac{\lambda}{2}$
- D λ

87 9702/11/O/N/17/Q24

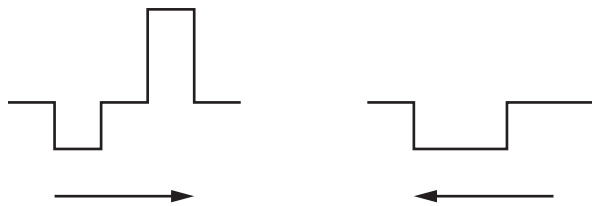
Diffraction can be observed when a wave passes an obstruction. The diffraction effect is greatest when the wavelength and the obstruction are similar in size.

For waves travelling through air, what is the combination of wave and obstruction that could best demonstrate diffraction?

- A** microwaves passing a steel post **C** sound waves passing a human hair
B radio waves passing a copper wire **D** visible light waves passing a gate post

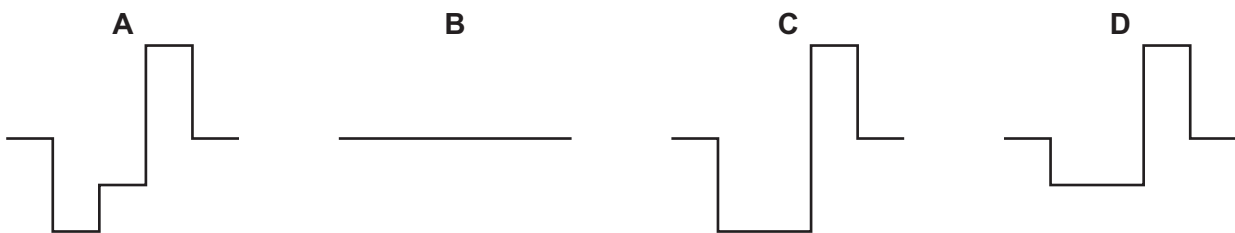
88 9702/11/O/N/17/Q27

Two signals approach each other, as shown.



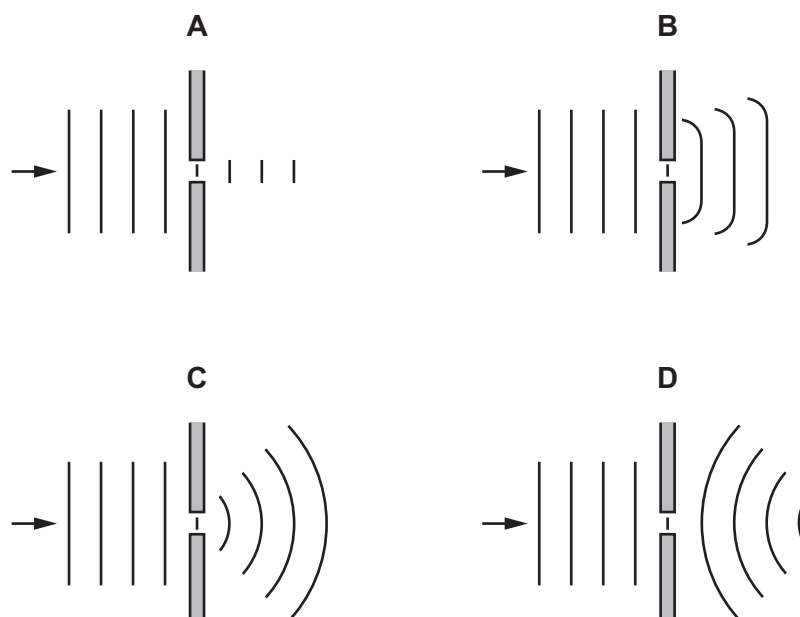
At one instant, the signals completely overlap.

According to the principle of superposition, what is the shape of the resulting signal at this instant?



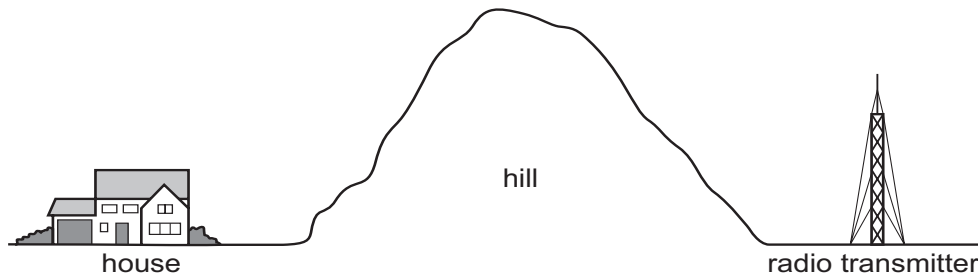
89 9702/12/O/N/17/Q28

Which diagram shows the diffraction of water waves in a ripple tank?



90 9702/12/O/N/17/Q29

A hill stands between a radio transmitter and a house, as shown.



The radio transmitter cannot be seen from the house, but radio waves from the transmitter are received at the house.

Why is this?

- A The wavelength of light is longer than the wavelength of radio waves so there is more diffraction of light over the hill.
- B The wavelength of light is shorter than the wavelength of radio waves so there is more diffraction of light over the hill.
- C The wavelength of radio waves is longer than the wavelength of light so there is more diffraction of radio waves over the hill.
- D The wavelength of radio waves is shorter than the wavelength of light so there is more diffraction of radio waves over the hill.

91 9702/12/O/N/17/Q30

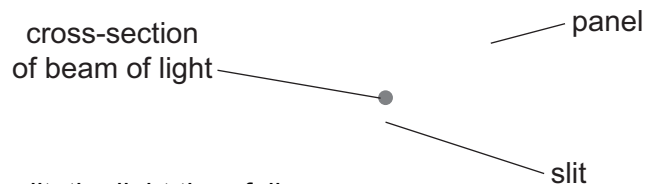
In an experiment to demonstrate double-slit interference using light, the distance from the slits to the screen is doubled and the slit separation is halved. The wavelength of the light is kept constant.

By which factor does the separation of adjacent bright fringes change?

- A $\frac{1}{4}$
- B $\frac{1}{2}$
- C 2
- D 4

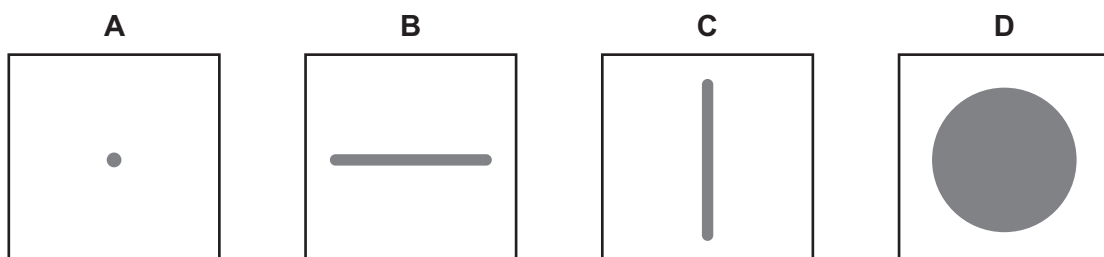
92 9702/13/O/N/17/Q29

A beam of laser light is directed towards a narrow slit.

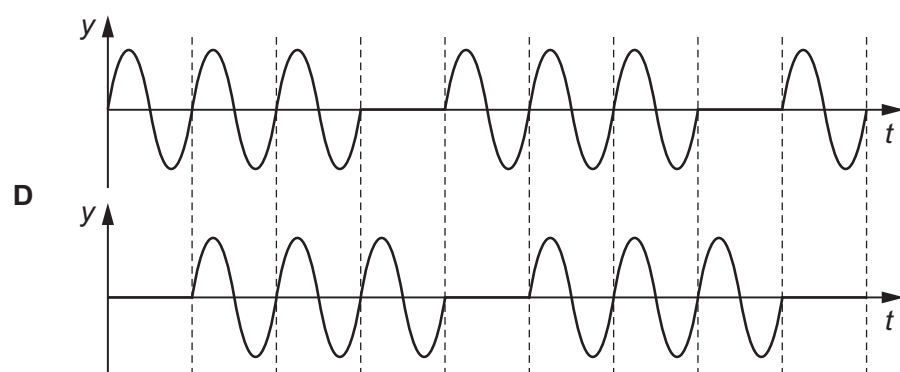
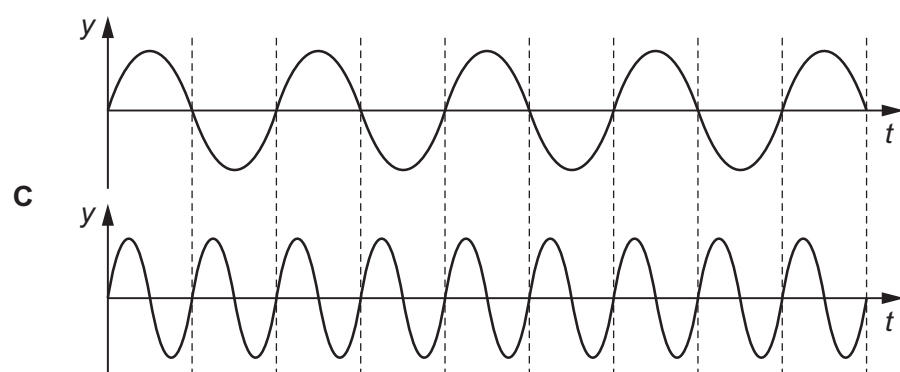
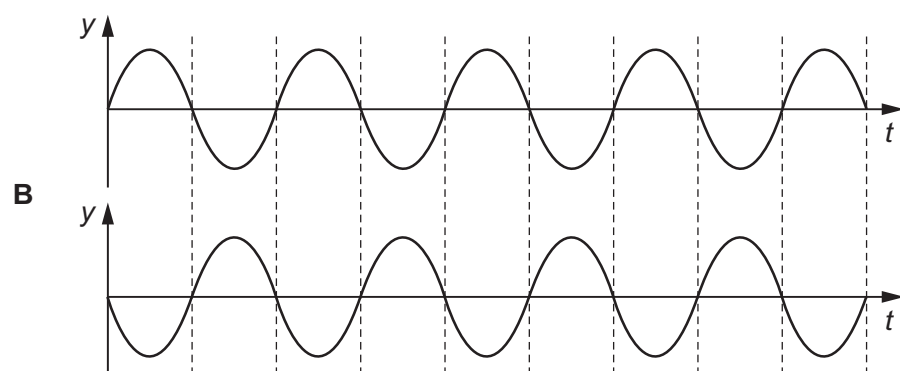
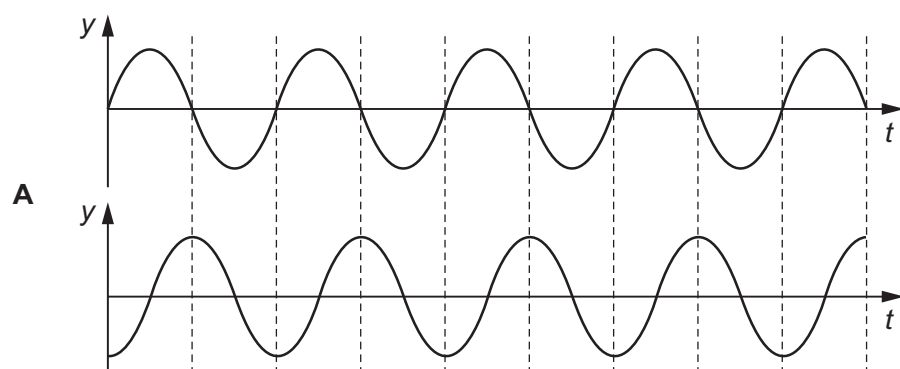


After emerging from the other side of the slit, the light then falls on a screen.

What is the pattern of light seen on the screen?

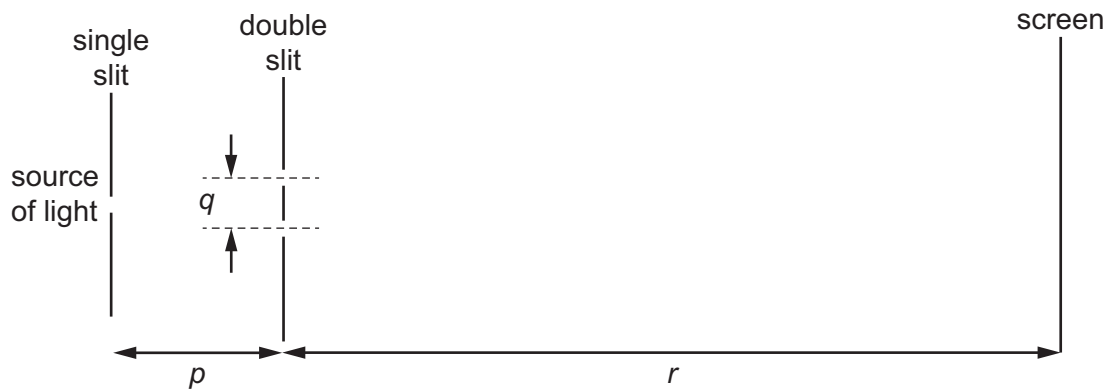


The diagrams show four pairs of waves. In each case the displacement y measured at a fixed point is plotted against time t . Which pair of waves is **not** coherent?



94 9702/11/M/J/18/Q25

A teacher sets up the apparatus shown to demonstrate a double-slit interference pattern on the screen.

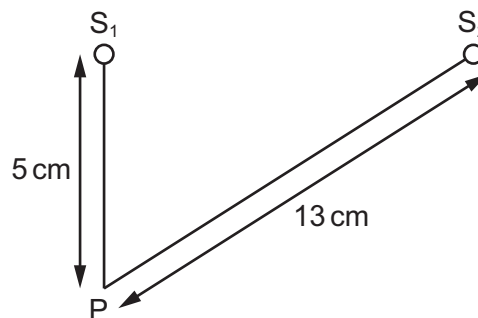


Which change to the apparatus will increase the fringe spacing?

- A** decreasing the distance p **C** decreasing the distance r
B decreasing the distance q **D** decreasing the wavelength of the light

95 9702/11/M/J/18/Q26

The diagram shows two sources of waves S_1 and S_2 . The sources oscillate with a phase difference of 180° .



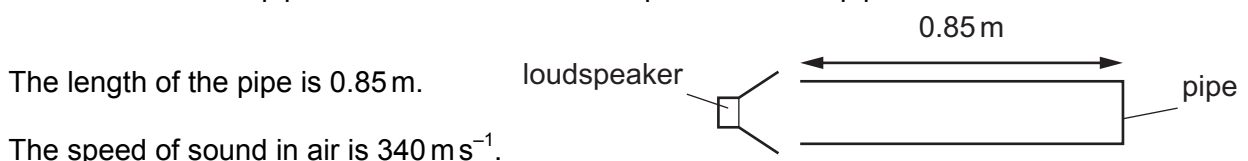
The sources each generate a wave of wavelength 2.0 cm. Each source produces a wave that has amplitude x_0 when it reaches point P.

What is the amplitude of the oscillation at P?

- A** 0 **B** $\frac{x_0}{2}$ **C** x_0 **D** $2x_0$

96 9702/11/M/J/18/Q27

A pipe, closed at one end, has a loudspeaker at the open end. A stationary sound wave is formed in the air within the pipe with an antinode at the open end of the pipe.



Which frequency of sound from the loudspeaker would **not** produce a stationary wave?

- A** 100 Hz **B** 200 Hz **C** 300 Hz **D** 500 Hz

97 9702/12/M/J/18/Q26

The diagram shows apparatus for the measurement of the frequency of a sound wave.



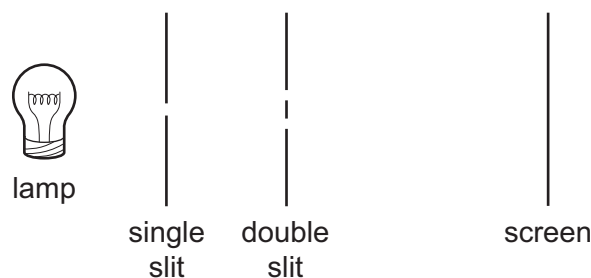
Sound of the unknown frequency is reflected back from a metal plate. A microphone placed at a distance D from the metal plate detects the sound intensity. A minimum intensity is detected with $D = 12.0$ cm. The plate is moved further away from the microphone until the next minimum is detected with $D = 15.0$ cm.

The speed of sound in air is 336 m s^{-1} . What is the frequency of the sound?

- A** 56 Hz **B** 112 Hz **C** 5600 Hz **D** 11 200 Hz

98 9702/12/M/J/18/Q29

A two-source interference experiment uses the apparatus shown.



What is the main purpose of the single slit?

- A** to make a narrow beam of light
B to make the same amplitude of light incident on each slit
C to provide coherent light
D to provide monochromatic light

99 9702/13/M/J/18/Q25

A water wave in a ripple tank is diffracted as it passes through a gap in a barrier.

Which two factors affect the angle of diffraction of the wave?

- A** the amplitude and frequency of the incident wave
B the amplitude of the incident wave and the width of the gap
C the wavelength and amplitude of the incident wave
D the wavelength of the incident wave and the width of the gap

100 9702/13/M/J/18/Q26

A double-slit interference pattern using red light of wavelength $7.0 \times 10^{-7} \text{ m}$ has a fringe spacing of 3.5 mm.

Which fringe spacing would be observed for the same arrangement of apparatus but using blue light of wavelength $4.5 \times 10^{-7} \text{ m}$?

- A** 2.3 mm **B** 3.5 mm **C** 5.4 mm **D** 9.0 mm

101 9702/12/F/M/18/Q26

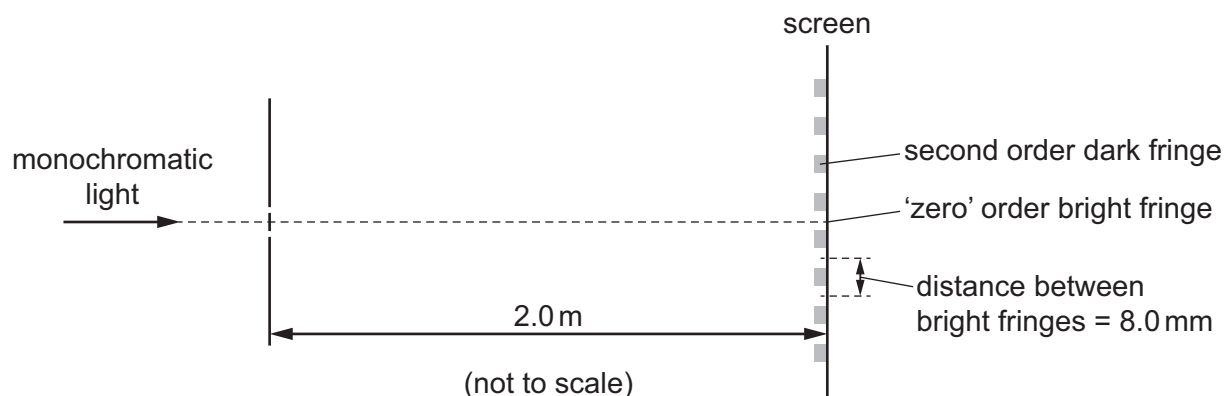
In a double-slit interference experiment, light of frequency $6.0 \times 10^{14} \text{ Hz}$ is incident on a pair of slits. Bright fringes that are 3.0 mm apart are observed on a screen some distance away.

What is the separation of the bright fringes when the frequency of the light is changed to $5.0 \times 10^{14} \text{ Hz}$?

- A** 1.8 mm **B** 2.5 mm **C** 3.0 mm **D** 3.6 mm

102 9702/12/F/M/18/Q27

Monochromatic light is incident on a pair of narrow slits a distance of 0.1 mm apart. A series of bright and dark fringes are observed on a screen a distance of 2.0 m away. The distance between adjacent bright fringes is 8.0 mm.



What is the path difference between the light waves from the two slits that meet at the second order dark fringe?

- A** $2.0 \times 10^{-7} \text{ m}$ **B** $4.0 \times 10^{-7} \text{ m}$ **C** $6.0 \times 10^{-7} \text{ m}$ **D** $8.0 \times 10^{-7} \text{ m}$

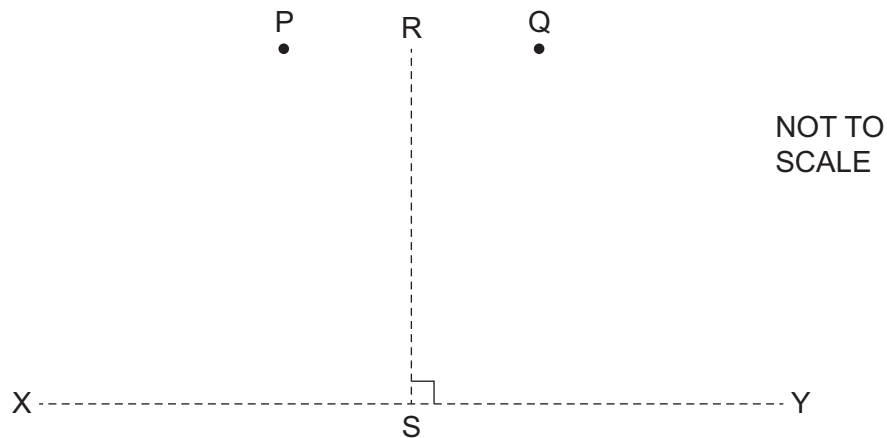
103 9702/12/F/M/18/Q29

Which statement gives a condition that enables diffraction to occur?

- A** A source of waves moves towards a stationary observer.
B A wave is partially blocked by an obstacle.
C Two coherent waves are superposed.
D Two waves of equal speed and frequency are travelling through the same part of a medium in opposite directions.

104 9702/11/O/N/17/Q28

Coherent waves are produced at P and at Q and travel outwards in all directions. The line RS is half-way between P and Q and perpendicular to the line joining P and Q. The distance RS is much greater than the distance PQ.



Along which of the lines shown is an interference pattern observed?

- A** both RS and XY **C** XY only
B RS only **D** neither RS nor XY

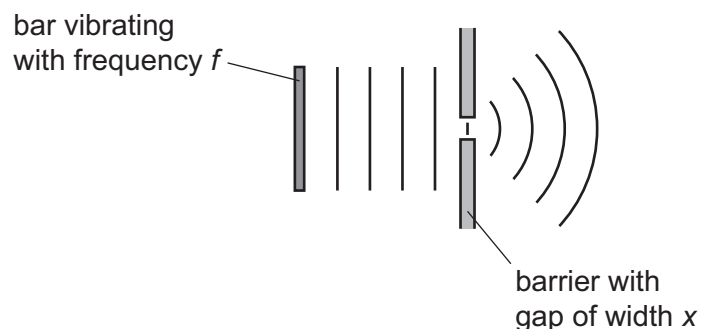
105 9702/11/O/N/17/Q29

A bar vibrates with frequency f to produce water waves in a ripple tank.

The waves pass through a gap of width x in a barrier so that diffraction occurs.

Which combination of vibration frequency and gap width will produce the smallest angle of diffraction?

	vibration frequency	gap width
A	$\frac{f}{2}$	$\frac{x}{2}$
B	$\frac{f}{2}$	$2x$
C	$2f$	$\frac{x}{2}$
D	$2f$	$2x$

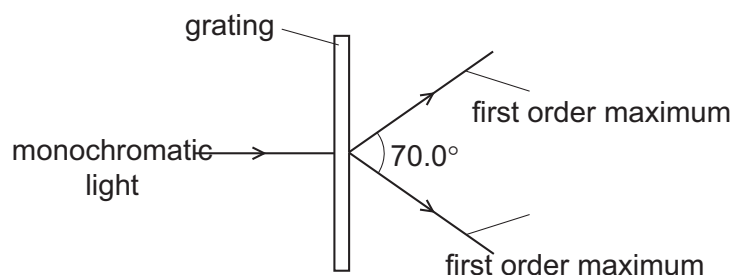


106 9702/11/O/N/17/Q30

A diffraction grating is used to measure the wavelength of monochromatic light, as shown in the diagram.

The spacing of the slits in the grating is 1.00×10^{-6} m. The angle between the first order diffraction maxima is 70.0° .

What is the wavelength of the light?



- A** 287 nm **C** 574 nm **B** 470 nm **D** 940 nm

107 9702/11/O/N/18/Q26

What may be used to produce stationary waves?

- A** blowing air over the top of an empty bottle
- B** making a loud sound near a mountain
- C** passing monochromatic light through a double slit
- D** passing water waves through a narrow slit

108 9702/11/O/N/18/Q27

What is an example of the diffraction of a wave?

- A** laser light travelling along an optic fibre
- B** light waves forming images on a cinema screen
- C** microwaves passing the edge of a metal plate
- D** sound waves diverging as they pass through air

109 9702/11/O/N/18/Q28

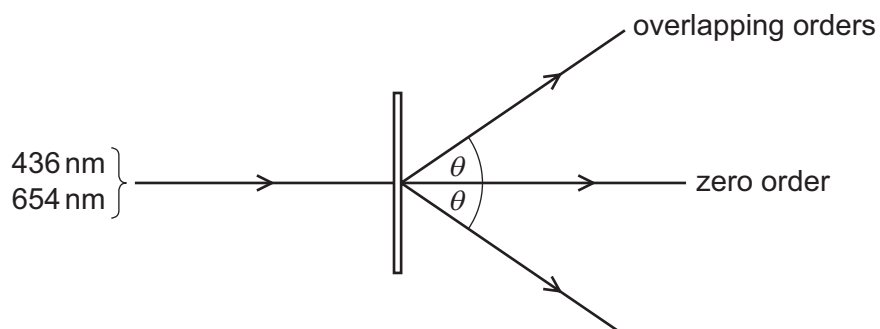
When the light from two lamps falls on a screen, no interference pattern can be obtained.

Why is this?

- A** The lamps are not point sources.
- B** The lamps emit light of different amplitudes.
- C** The light from the lamps is not coherent.
- D** The light from the lamps is white.

110 9702/11/O/N/18/Q29

A beam of light consists of two wavelengths of 436 nm and 654 nm. A diffraction grating of $5.00 \times 10^5 \text{ lines m}^{-1}$ produces a diffraction pattern in which the second order of one of these wavelengths occurs at the same angle θ as the third order of the other wavelength.



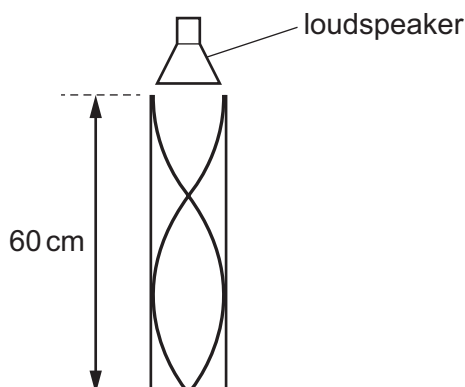
What is the angle θ ?

- A** 19.1°
- B** 25.8°
- C** 40.8°
- D** 78.8°

111 9702/12/O/N/18/Q24

The sound from a loudspeaker placed above a tube causes resonance of the air in the tube.

A stationary wave is formed with two nodes and two antinodes as shown.



The speed of sound in the air is 340 m s^{-1} .

What is the frequency of the sound?

- A** 430 Hz **B** 570 Hz **C** 850 Hz **D** 1700 Hz

112 9702/12/O/N/18/Q26

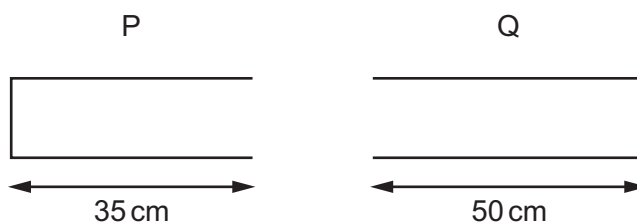
A surveyor's device emits a pulse of light. The light is reflected from a wall 150 m away.

What is the time taken for the pulse to travel from the device to the wall and then back to the device?

- A** 0.05 ns **B** 0.10 ns **C** 0.50 μs **D** 1.0 μs

113 9702/12/O/N/18/Q27

Progressive sound waves of wavelength 20 cm enter the air columns in a closed pipe P and an open pipe Q. The lengths of the pipes are shown.



In which pipe or pipes are stationary waves formed?

- A** P and Q
B P only
C Q only
D neither P nor Q

114 9702/12/O/N/18/Q28

What happens when waves pass through a gap equal to their wavelength?

- A** There is diffraction and the wavelength decreases.
- B** There is diffraction and the wavelength stays the same.
- C** There is no diffraction and the wavelength decreases.
- D** There is no diffraction and the wavelength stays the same.

115 9702/12/O/N/18/Q29

Two sources of microwaves P and Q produce coherent waves with a phase difference of 180° . The waves have the same wavelength λ .



At the point S there is a minimum in the interference pattern produced by waves from the two sources. The distance (QS – PS) is called the path difference.

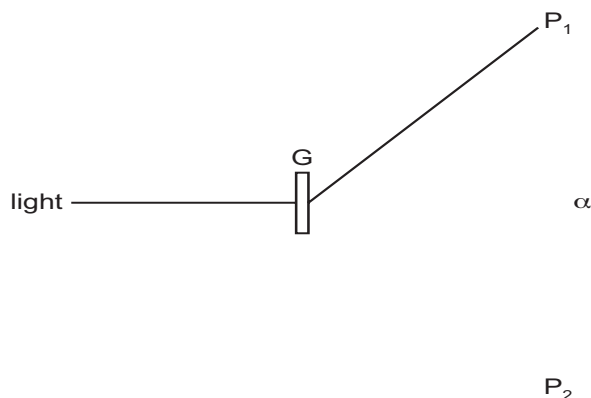
In the expressions shown, n is an integer.

Which expression represents the path difference?

- A** $n\lambda$
- B** $\frac{1}{2}n\lambda$
- C** $(n + \frac{1}{2})\lambda$
- D** $(2n + \frac{1}{2})\lambda$

116 9702/12/O/N/18/Q30

A parallel beam of monochromatic light of wavelength λ is incident normally on a diffraction grating G. The angle between the directions of the two second-order diffracted beams at P_1 and at P_2 is α , as shown.

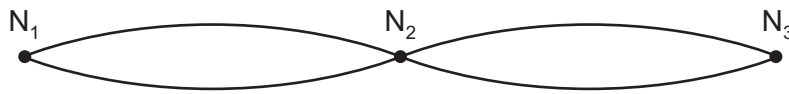


What is the spacing of the lines on the grating?

- A** $\frac{2\lambda}{\sin \alpha}$
- B** $\frac{\lambda}{\sin \alpha}$
- C** $\frac{2\lambda}{\sin(\alpha/2)}$
- D** $\frac{\lambda}{\sin(\alpha/2)}$

117 9702/13/O/N/18/Q26

The diagram shows a stationary wave on a string. The stationary wave has three nodes N_1 , N_2 and N_3 .



Which statement is correct?

- A All points on the string vibrate in phase.
- B All points on the string vibrate with the same amplitude.
- C Points equidistant from N_2 vibrate with the same frequency and in phase.
- D Points equidistant from N_2 vibrate with the same frequency and the same amplitude.

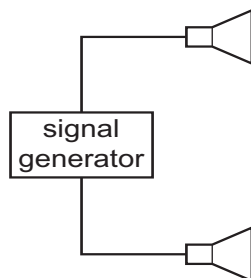
118 9702/13/O/N/18/Q27

In which situation does diffraction occur?

- A A wave bounces back from a surface.
- B A wave passes from one medium into another.
- C A wave passes through an aperture.
- D Waves from two identical sources are superposed.

119 9702/13/O/N/18/Q28

A student connects two loudspeakers to a signal generator.



As the student walks from P to Q, he notices that the loudness of the sound rises and falls repeatedly.

What causes the loudness of the sound to vary?

- A diffraction of the sound waves
- B Doppler shift of the sound waves
- C interference of the sound waves
- D reflection of the sound waves

120 9702/13/O/N/18/Q29

A parallel beam of white light is incident normally on a diffraction grating. The second-order and third-order spectra partially overlap.

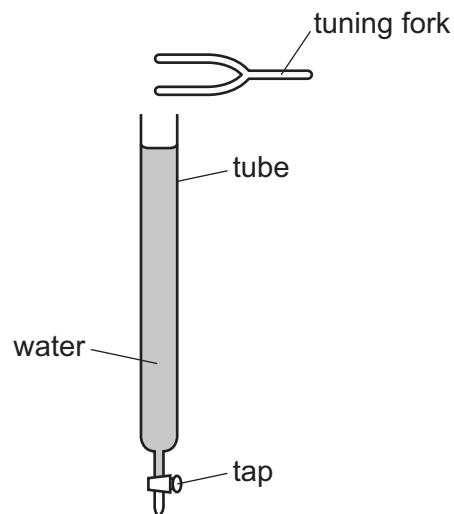
Which wavelength in the third-order spectrum appears at the same angle as the wavelength of 600 nm in the second-order spectrum?

- A** 300 nm **B** 400 nm **C** 600 nm **D** 900 nm

121 9702/11/M/J/19/Q24

A long tube, filled with water, has a tap fitted at its base, as shown.

A tuning fork is sounded above the tube and the water is allowed to run gradually out of the tube.



A louder sound is heard at intervals as the water runs out of the tube. The change in water level between louder sounds is 32 cm.

What is the wavelength of the sound in the tube?

- A** 16 cm **B** 32 cm **C** 64 cm **D** 128 cm

122 9702/11/M/J/19/Q27

Two progressive waves of frequency 300 Hz superpose to produce a stationary wave in which adjacent nodes are 1.5 m apart.

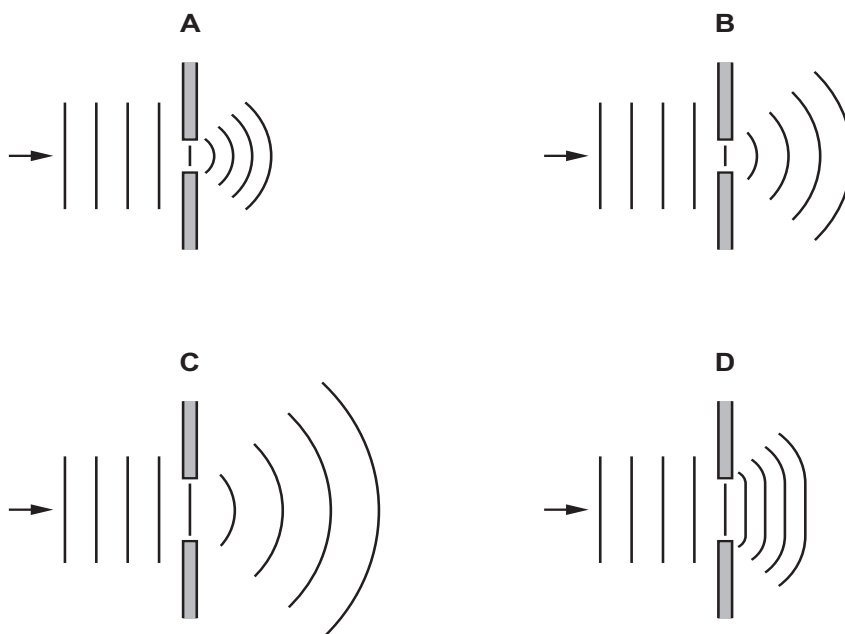
What is the speed of the progressive waves?

- A** 100 m s^{-1} **B** 200 m s^{-1} **C** 450 m s^{-1} **D** 900 m s^{-1}

123 9702/11/M/J/19/Q28

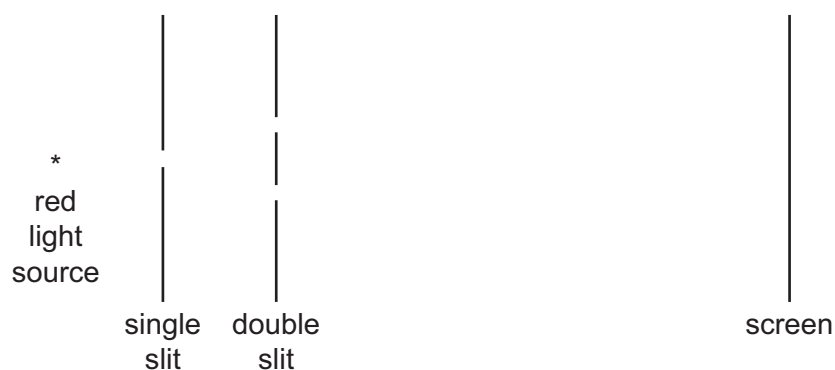
The diagrams show the diffraction of water waves in a ripple tank as they pass through a gap between two barriers.

Which diagram is correct?



124 9702/11/M/J/19/Q29

A double-slit interference experiment is set up as shown.



not to scale

Fringes are formed on the screen. The distance between successive bright fringes is found to be 4 mm.

Two changes are then made to the experimental arrangement. The double slit is replaced by another double slit which has half the spacing. The screen is moved so that its distance from the double slit is twice as great.

What is now the distance between successive bright fringes?

- A** 1 mm **B** 4 mm **C** 8 mm **D** 16 mm

125 9702/11/M/J/19/Q30

The interference patterns from a diffraction grating and a double slit are compared.

Using the diffraction grating, yellow light of the first order is seen at 30° to the normal to the grating.

The same light produces interference fringes on a screen 1.0 m from the double slit. The slit separation is 500 times greater than the line spacing of the grating.

What is the fringe separation on the screen?

- A** $2.5 \times 10^{-7} \text{ m}$
- B** $1.0 \times 10^{-5} \text{ m}$
- C** $1.0 \times 10^{-3} \text{ m}$
- D** $1.0 \times 10^{-1} \text{ m}$

126 9702/12/M/J/19/Q25

In an experiment to determine the wavelength of sound in air, a stationary wave is set up in an air column.

The distance between a node and an adjacent antinode is L .

What is the wavelength of the sound?

- A** $\frac{1}{2}L$ **B** L **C** $2L$ **D** $4L$

127 9702/12/M/J/19/Q26

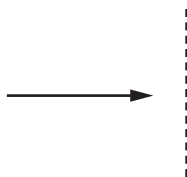
In one of the first experiments to demonstrate the Doppler effect, a train was filled with trumpeters all playing a note of frequency 440 Hz. The difference in observed frequency of the note as the train directly approached a stationary observer was 22 Hz. The speed of sound was 340 m s^{-1} .

At which speed was the train moving?

- A** 15.4 m s^{-1} **B** 16.2 m s^{-1} **C** 17.0 m s^{-1} **D** 17.9 m s^{-1}

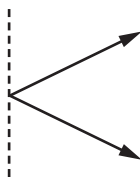
128 9702/12/M/J/19/Q29

Monochromatic light is directed at a diffraction grating, as shown.

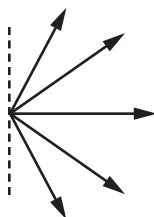


Which diagram could show all the possible directions of the light, after passing through the grating, that give maximum intensity?

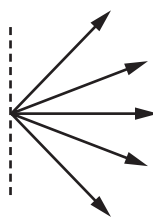
A



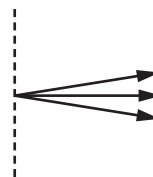
B



C



D



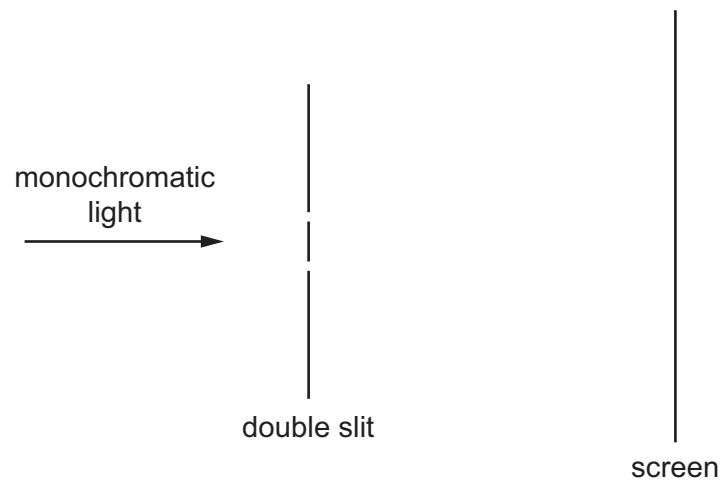
129 9702/12/M/J/19/Q30

Why can an observable interference pattern **never** be obtained between two monochromatic beams of light from different lamps?

- A The frequency of the light from the two lamps can never be the same.
- B The light from the two lamps can never be coherent.
- C The temperature of the filaments of the two lamps used can never be the same.
- D The wavelength of the light from the two lamps must always be different.

130 9702/12/M/J/19/Q31

A student sets up apparatus to observe the double-slit interference of monochromatic light, as shown.



Interference fringes are formed on the screen.

Which change would increase the distance between adjacent fringes?

- A** Decrease the distance between the two slits.
- B** Decrease the width of each slit.
- C** Move the screen closer to the double slit.
- D** Use light of a higher frequency.

131 9702/13/M/J/19/Q26

An elastic string is attached to an oscillator at one end and clamped at the other end so that the string is horizontal and in tension.

The oscillator is made to oscillate vertically. The frequency of oscillation is gradually increased from zero until a stationary wave is set up in the string. The frequency is then increased further to frequency f , when a second stationary wave is set up in the string.

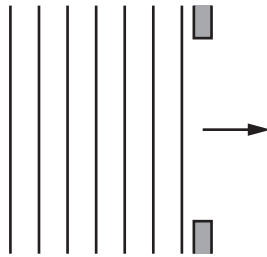
The frequency is then increased further.

At which frequency does a third stationary wave occur?

- A** $1.2f$ **B** $1.5f$ **C** $2.0f$ **D** $3.0f$

132 9702/13/M/J/19/Q27

In an experiment, water waves in a ripple tank are incident on a gap, as shown.



Some diffraction of the water waves is observed.

Which change to the experiment would provide a better demonstration of diffraction?

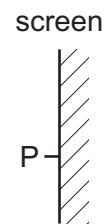
- A Increase the amplitude of the waves.
- B Increase the frequency of the waves.
- C Increase the wavelength of the waves.
- D Increase the width of the gap.

133 9702/13/M/J/19/Q28

Light of wavelength λ is emitted from two point sources R and S and falls onto a distant screen.

R •
S •

(not to scale)



At point P on the screen, the light intensity is zero.

What could explain the zero intensity at P?

- A Light from the two sources is emitted 180° out of phase and the path difference to P is $\frac{1}{2}\lambda$.
- B Light from the two sources is emitted in phase and the path difference to P is λ .
- C Light from the two sources is emitted 90° out of phase and the path difference to P is λ .
- D Light from the two sources is emitted in phase and the path difference to P is $\frac{1}{2}\lambda$.

134 9702/13/M/J/19/Q29

Apparatus is arranged to show double-slit interference using monochromatic light. The slit separation is 0.10 mm. The distance from the double slit to the screen where the interference pattern is observed is 2.4 m and the fringe width is 12 mm.

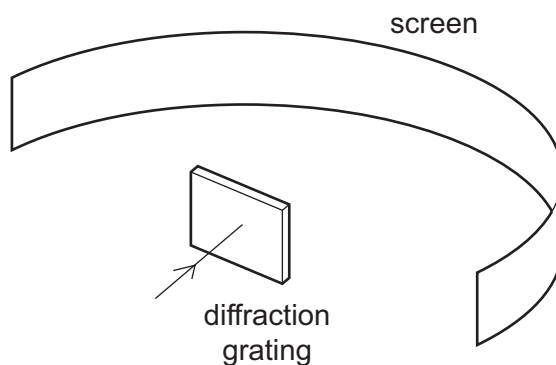
The distance to the screen is now changed to 1.8 m and the slit separation is doubled.

What is the new fringe width?

- A** 1.5 mm **B** 4.5 mm **C** 6.0 mm **D** 9.0 mm

135 9702/13/M/J/19/Q30

Monochromatic light of wavelength 690 nm passes through a diffraction grating with 300 lines per mm, producing a series of maxima (bright spots) on a screen.

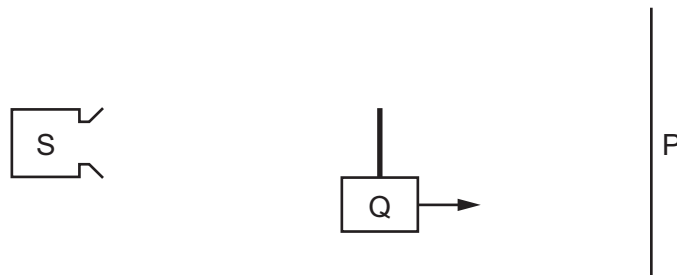


What is the greatest number of maxima that can be observed?

- A** 4 **B** 5 **C** 8 **D** 9

136 9702/11/O/N/19/Q23

Source S emits microwaves with a constant amplitude. The microwaves hit a metal screen P and are reflected. A stationary wave is formed between S and P. The wavelength of the microwaves is much smaller than the distance between S and P.



A detector Q is moved at a slow, constant speed from S to P.

What happens to the amplitude of the signal detected by Q?

- A** decreases steadily
- B** increases and decreases regularly
- C** increases steadily
- D** remains constant

137 9702/11/O/N/19/Q24

The siren of a moving police car emits a sound wave with a frequency of 440 Hz. A stationary observer hears sound of frequency 494 Hz. The speed of sound in the air is 340 m s^{-1} .

What could be the speed and the direction of movement of the car?

- A** 37 m s^{-1} directly towards the observer
- B** 37 m s^{-1} directly away from the observer
- C** 42 m s^{-1} directly towards the observer
- D** 42 m s^{-1} directly away from the observer

138 9702/11/O/N/19/Q25

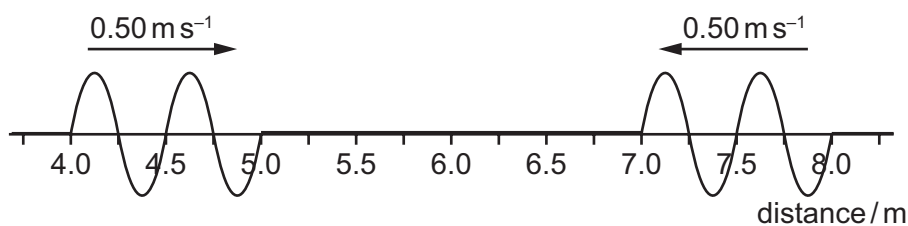
An electromagnetic wave has a wavelength of 138 pm in a vacuum.

To which region of the electromagnetic spectrum does this wave belong?

- A** radio wave
- B** microwave
- C** visible light
- D** X-ray

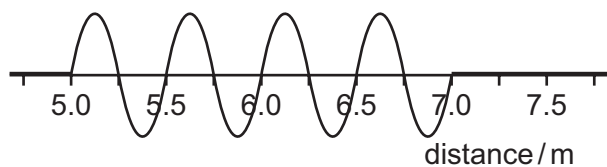
139 9702/11/O/N/19/Q26

Two wave pulses are travelling towards each other on a long rope. The pulses have the same amplitude and wavelength and are travelling at a speed of 0.50 m s^{-1} . The diagram shows the rope at time $t = 0$.

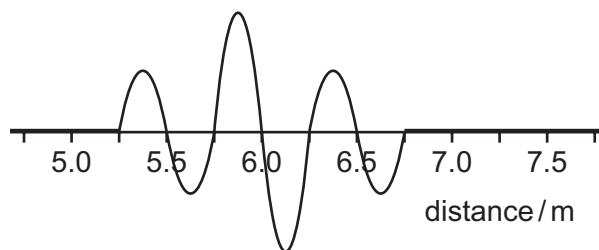


Which diagram shows the rope at time $t = 3.0 \text{ s}$?

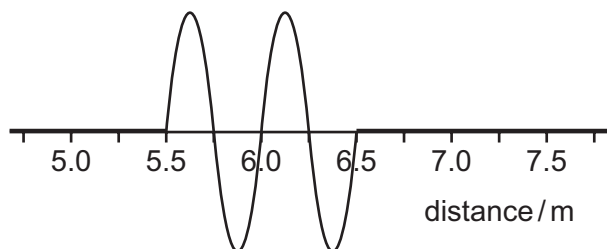
A



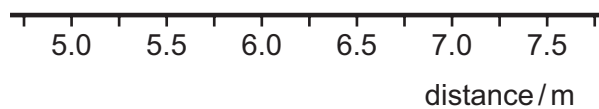
B



C

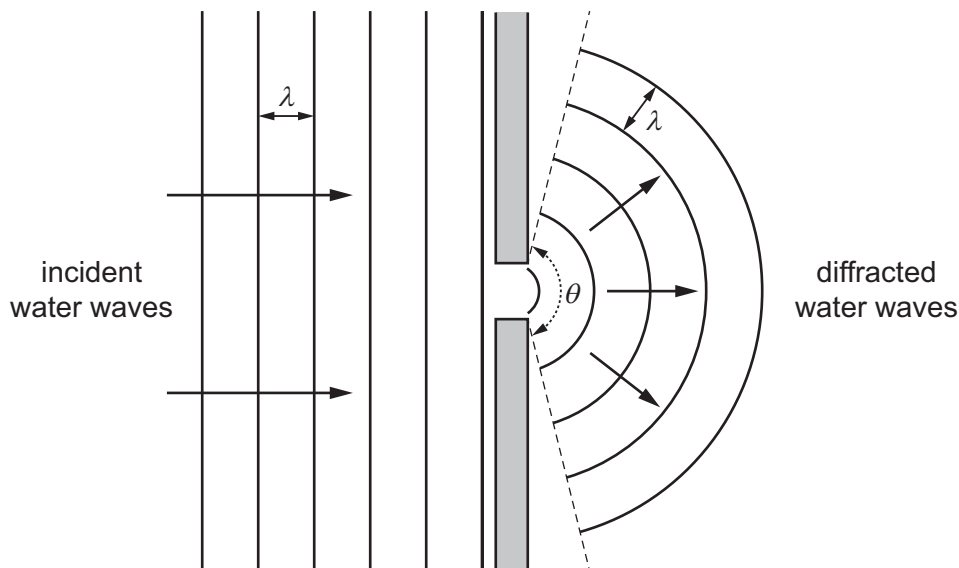


D



140 9702/11/O/N/19/Q27

Water waves of wavelength λ are incident normally on an obstacle with a narrow gap. The width of the gap is equal to λ . The waves from the gap emerge over an angle θ as shown.



The gap is slowly widened.

Which changes, if any, occur to θ and to the wavelength of the emerging waves?

	θ	wavelength
A	decreases	remains the same
B	increases	remains the same
C	remains the same	decreases
D	remains the same	increases

141 9702/11/O/N/19/Q28

Light of wavelength 720 nm from a laser X is incident normally on a diffraction grating and a diffraction pattern is observed. Light from a laser Y is then also incident normally on the same grating. The third-order maximum due to laser Y is seen at the same place as the second-order maximum due to laser X.

What is the wavelength of the light from laser Y?

- A** 480 nm **B** 540 nm **C** 720 nm **D** 1080 nm

142 9702/11/O/N/19/Q29

Monochromatic light of frequency f is incident on a diffraction grating of line spacing d . The speed of light is c .

Which expression can be used to determine the highest order of intensity maximum produced by the grating?

- A** $n = \frac{d}{cf}$ **B** $n = \frac{df}{c}$ **C** $n = \frac{dc}{f}$ **D** $n = \frac{c}{df}$

143 9702/12/O/N/19/Q23

A loudspeaker emitting a constant frequency of 2000 Hz is swung in a horizontal circle with a speed of 15.0 m s^{-1} .

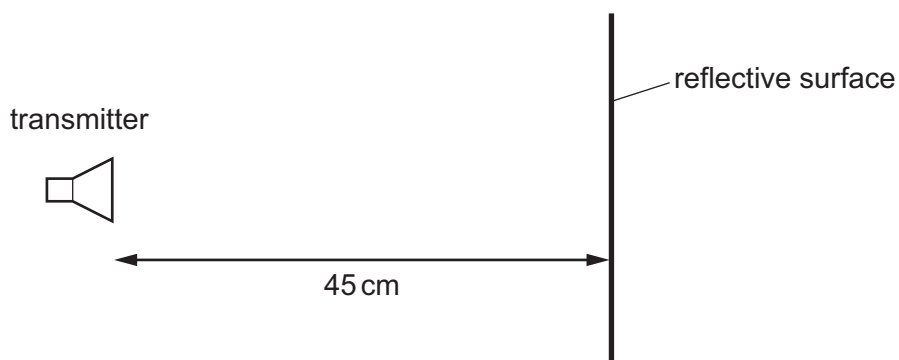
A stationary observer is level with the loudspeaker and situated a long distance from the loudspeaker. The observer hears a sound of varying frequency. The maximum frequency heard is 2097 Hz.

What is the speed of the sound in the air?

- A** 294 m s^{-1} **B** 309 m s^{-1} **C** 324 m s^{-1} **D** 330 m s^{-1}

144 9702/12/O/N/19/Q25

A transmitter of electromagnetic waves is placed 45 cm from a reflective surface.



The emitted waves have a frequency of 1.00 GHz. A stationary wave is produced with a node at the transmitter and a node at the surface.

How many antinodes are in the space between the transmitter and the surface?

- A** 1 **B** 2 **C** 3 **D** 4

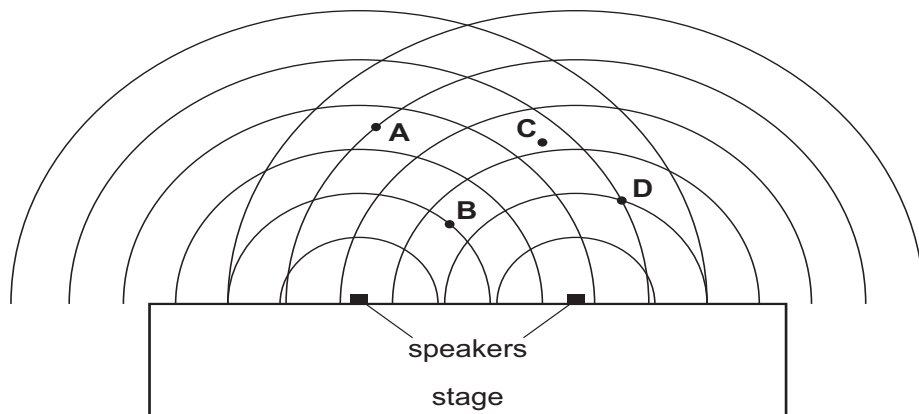
145 9702/12/O/N/19/Q27

An outdoor concert has two large speakers beside the stage for broadcasting music.

In order to test the speakers, they are made to emit sound of the same wavelength and the same amplitude.

The curved lines in the diagram represent wavefronts.

Where is the loudest sound heard?



146 9702/12/O/N/19/Q28

An electromagnetic wave is incident normally on a diffraction grating.

A second-order maximum is produced at an angle of 30° to a normal to the grating.

The grating has 5000 lines per cm.

What is the wavelength of the wave?

- A** $2.5 \times 10^{-7} \text{ m}$ **B** $5.0 \times 10^{-7} \text{ m}$ **C** $1.0 \times 10^{-6} \text{ m}$ **D** $5.0 \times 10^{-5} \text{ m}$

147 9702/13/O/N/19/Q24

A siren emits sound of frequency 1000 Hz. The siren moves at 20 m s^{-1} towards an observer who is standing still.

The speed of sound in the air is 330 m s^{-1} .

Which expression would correctly give the frequency heard by the observer?

- A** $\frac{1000 \times 330}{330 + 20}$
- B** $\frac{1000 \times 330}{330 - 20}$
- C** $\frac{1000 (330 + 20)}{330}$
- D** $\frac{1000 (330 - 20)}{330}$

148 9702/13/O/N/19/Q25

A source of sound of constant power P is situated in an open space. The intensity I of sound at distance r from this source is given by

$$I = \frac{P}{4\pi r^2}.$$

How does the amplitude a of the vibrating air molecules vary with the distance r from the source?

- A $a \propto \frac{1}{r}$ B $a \propto \frac{1}{r^2}$ C $a \propto r$ D $a \propto r^2$

149 9702/13/O/N/19/Q26

In an experiment to demonstrate a stationary wave, two microwave transmitters, emitting waves of wavelength 4 cm, are set facing each other, as shown.



A detector is moved along a straight line between the transmitters. It detects positions of maximum and minimum signal. The detector is a distance d from the left-hand transmitter.

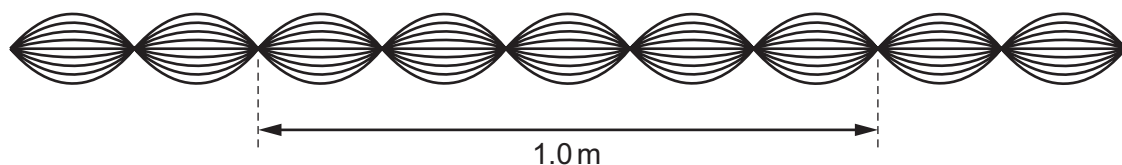
Assume that both transmitters are at antinodes of the stationary wave.

Which row gives a value of d for a maximum and for a minimum?

	value of d for a maximum / cm	value of d for a minimum / cm
A	46	48
B	47	48
C	48	47
D	49	47

150 9702/13/O/N/19/Q28

The diagram shows a sketch of a wave pattern over a short period of time.



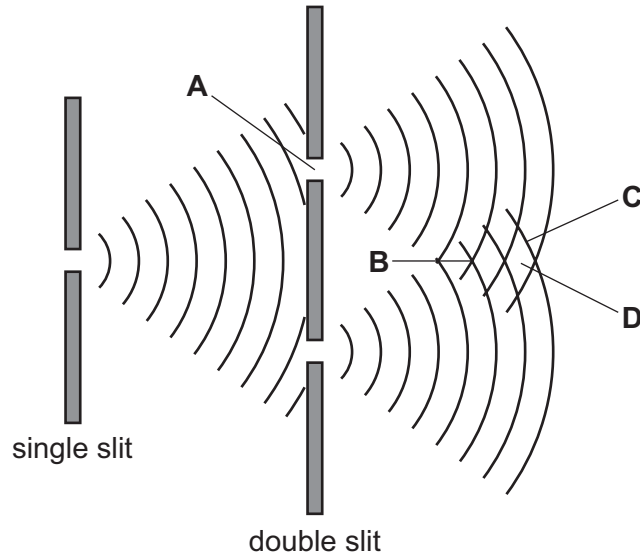
Which description of this wave is correct?

- A The wave is longitudinal, has a wavelength of 20 cm and is stationary.
 B The wave is transverse, has a wavelength of 20 cm and is stationary.
 C The wave is transverse, has a wavelength of 40 cm and is progressive.
 D The wave is transverse, has a wavelength of 40 cm and is stationary.

151 9702/13/O/N/19/Q29

The double-slit experiment demonstrates interference between two coherent sources of light waves. In the diagram, the curved lines represent wavefronts.

At which point does complete destructive interference (a minimum) occur?



152 9702/13/O/N/19/Q30

Light of wavelength 567 nm is incident normally on a diffraction grating. The grating has 400 lines per mm. A number of diffraction maxima are observed on the far side of the grating.

What is the angle between the second-order maximum and the third-order maximum?

- A** 13.1° **B** 13.9° **C** 15.9° **D** 27.0°

153. 9702/12/F/M/20/Q26

A musical instrument is made using a long tube with a mouthpiece at one end. The other end is open and flared, as shown.



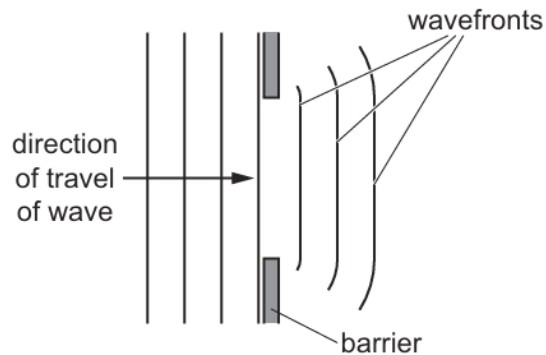
A musician maintains stationary sound waves with a node at the mouthpiece and an antinode at the other end. The lowest frequency of sound that the instrument can produce is 92 Hz.

Which different frequencies of sound can be produced by the instrument?

- A 92 Hz, 138 Hz, 184 Hz, 230 Hz
- B 92 Hz, 184 Hz, 276 Hz, 368 Hz
- C 92 Hz, 276 Hz, 460 Hz, 644 Hz
- D 92 Hz, 276 Hz, 828 Hz, 1288 Hz

154. 9702/12/F/M/20/Q27

A water wave passes through a gap between two barriers. The wavefronts spread out as shown.



What is the name of this phenomenon?

- A coherence
- B diffraction
- C interference
- D superposition

155. 9702/12/F/M/20/Q28

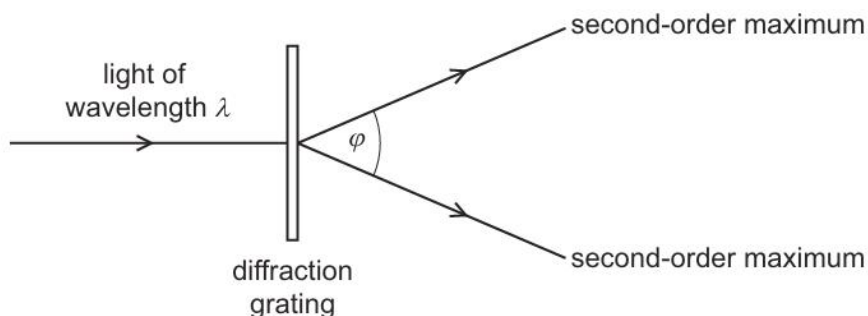
The table shows four possible combinations of values for the laser wavelength, slit separation and slit-screen distance in a two-slit interference experiment to show the interference of visible light on a white screen.

Which combination will result in visible fringes being observed?

	laser wavelength / nm	slit separation / mm	slit-screen distance / m
A	200	0.10	5.0
B	200	100	1.0
C	600	0.10	5.0
D	600	100	1.0

156. 9702/12/F/M/20/Q29

Light of wavelength λ is incident normally on a diffraction grating, as shown.



The angle between the two second-order maxima is ϕ .

Which expression gives the spacing of the lines on the diffraction grating?

- A** $\frac{\lambda}{\sin \phi}$ **B** $\frac{\lambda}{\sin (\phi/2)}$ **C** $\frac{2\lambda}{\sin \phi}$ **D** $\frac{2\lambda}{\sin (\phi/2)}$

157. 9702/11/M/J/20/Q30

A diffraction grating and a screen are used to determine the single wavelength λ of the light from a source.

What is an essential feature of this experiment?

- A** A curved screen must be used.
B The diffraction angle θ must be measured for at least two interference maxima.
C The light waves incident on the grating must be coherent.
D The third order intensity maximum must be produced.

158. 9702/11/M/J/20/Q29

Light of a single wavelength is incident normally on two slits that are 0.20 mm apart. Interference fringes are observed on a screen that is 5.4 m away from the slits. The distance between successive bright fringes is 12 mm.

What is the wavelength of the light?

- A** 440 nm **B** 540 nm **C** 650 nm **D** 900 nm

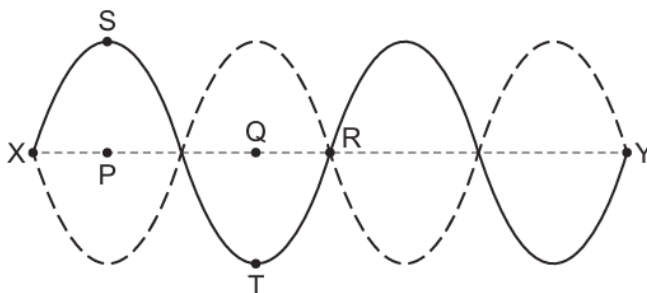
159. 9702/11/M/J/20/Q28

Which statement **must** be true for diffraction to occur when a wave passes through a gap?

- A** The wave is able to travel in a vacuum.
B The wave is progressive.
C The wave has a large amplitude.
D The wave has a long wavelength.

160. 9702/11/M/J/20/Q27

The diagram shows a string stretched between fixed points X and Y. There is a stationary wave on the string.



The solid curve shows the string at a position of maximum displacement. The dashed curve shows the other position of maximum displacement. The straight central dashed line shows the mean position of the string. Point S on the string is directly above point P. Point T on the string is directly below Q.

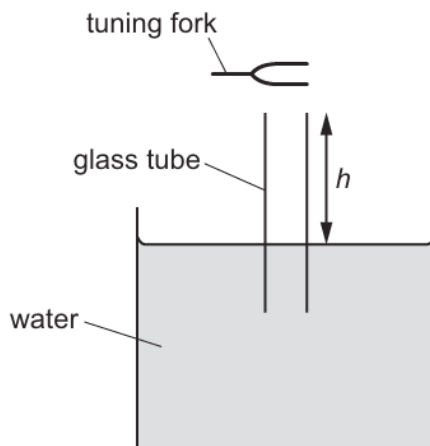
Which statement is correct?

- A** A short time later, point R on the string will be displaced.
B Points S and T on the string move in opposite directions.
C The distance between P and Q is one wavelength.
D Two points on the string that are equal distances from point R vibrate in phase.

161. 9702/11/M/J/20/Q24

A long glass tube is almost completely immersed in a large tank of water. A tuning fork is struck and held just above the open end of the tube as it is slowly raised.

A louder sound is first heard when the height h of the end of the tube above the water is 18.8 cm. A louder sound is next heard when h is 56.4 cm. The speed of sound in air is 330 m s^{-1} .

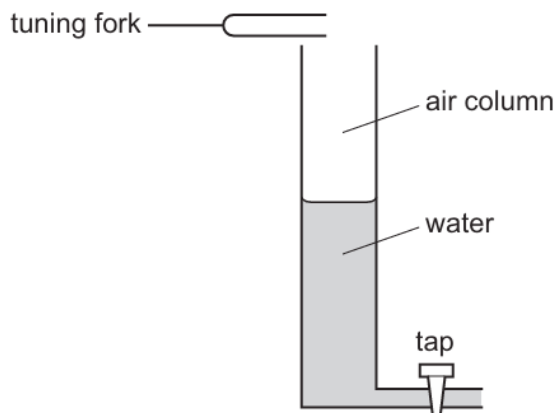


What is the frequency of the sound produced by the tuning fork?

- A** 220 Hz **B** 440 Hz **C** 660 Hz **D** 880 Hz

162. 9702/12/M/J/20/Q24

The diagram shows an experiment to produce a stationary wave in an air column. A tuning fork, placed above the column, vibrates and produces a sound wave. The length of the air column can be varied by altering the volume of the water in the tube.



The tube is filled and then water is allowed to run out of it. The first two stationary waves occur when the air column lengths are 0.14 m and 0.42 m.

What is the wavelength of the sound wave?

- A** 0.14 m **B** 0.28 m **C** 0.42 m **D** 0.56 m

163. 9702/12/M/J/20/Q27

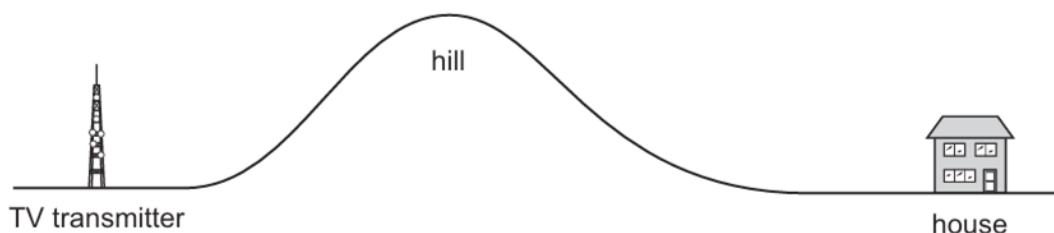
Two progressive waves meet at a point.

Which condition must be met for superposition of the waves to occur?

- A The waves must be coherent.
- B The waves must be of the same type.
- C The waves must be travelling in opposite directions.
- D The waves must meet in phase.

164. 9702/12/M/J/20/Q28

A hill separates a television (TV) transmitter from a house. The transmitter cannot be seen from the house. However, the house has good TV reception.

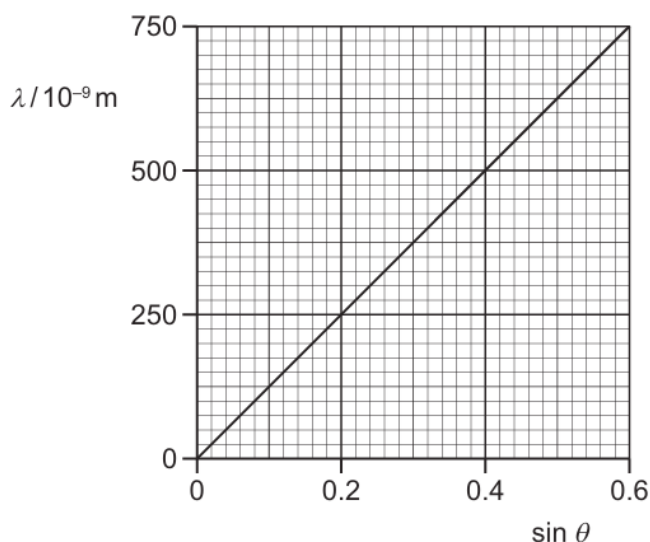


By which wave effect at the hill could the TV signal reach the house?

- A coherence
- B diffraction
- C interference

165. 9702/12/M/J/20/Q29

Light of wavelength λ is incident normally on a diffraction grating. The angle between the **second**-order maximum and the normal to the grating is θ . The variation with $\sin \theta$ of λ is shown on the graph.

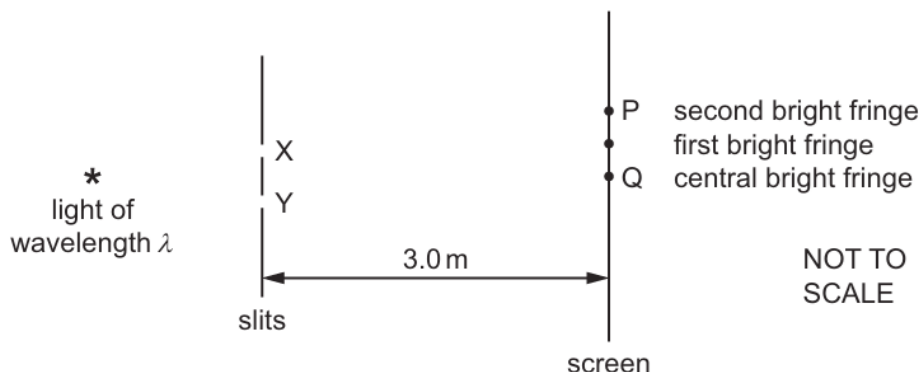


How many lines per millimetre are on the diffraction grating?

- A 400 mm^{-1}
- B 625 mm^{-1}
- C 800 mm^{-1}
- D 1250 mm^{-1}

166. 9702/12/M/J/20/Q30

The diagram shows an arrangement for demonstrating two-source interference using coherent light of a single wavelength λ .



An interference pattern is observed on a screen 3.0 m away from the slits X and Y, which have a separation of 1.0 mm.

The central bright fringe is at Q, and the **second** bright fringe from the centre is at P.

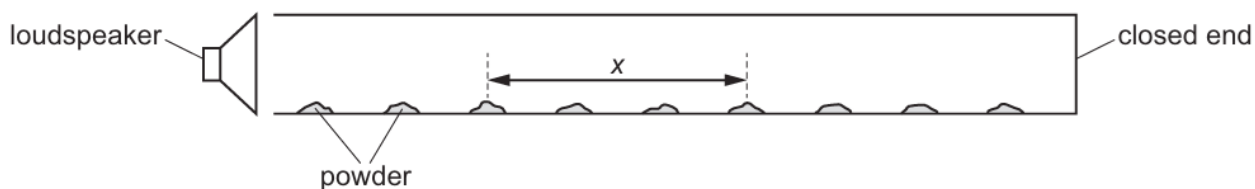
What is the distance between Q and P?

- A $6.0 \times 10^3 \lambda$
- B $3.0 \times 10^3 \lambda$
- C $6.7 \times 10^{-4} \lambda$
- D $3.3 \times 10^{-4} \lambda$

167. 9702/13/M/J/20/Q24

A loudspeaker is set up at the open end of a closed tube containing powder.

When the loudspeaker produces sound of frequency 1200 Hz, a stationary wave is produced in the tube. The powder gathers at the nodes of the stationary wave as shown.



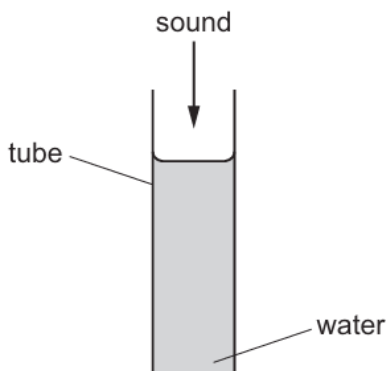
The speed of sound in the air is 336 m s^{-1} .

What is the value of distance x?

- A 28 cm
- B 42 cm
- C 84 cm
- D 112 cm

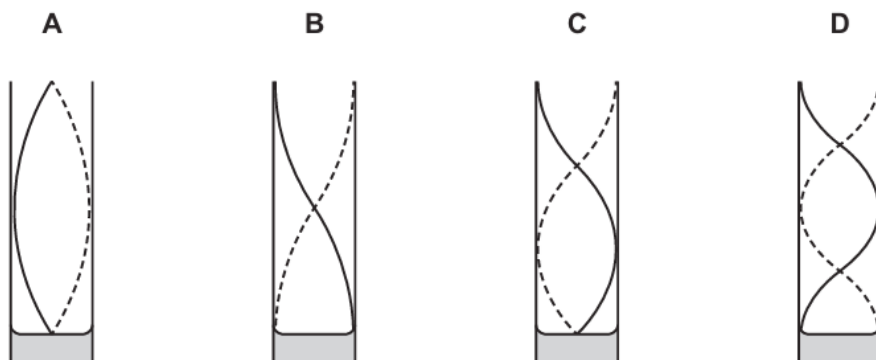
168. 9702/13/M/J/20/Q27

A vertical tube is partially filled with water. A sound wave moves down the tube and is reflected by the surface of the water. The frequency of the sound wave is gradually increased from zero until a much louder sound is heard.



Water is then removed from the tube until a second louder sound is heard.

Which diagram shows the new pattern of the stationary wave that is formed?



169. 9702/13/M/J/20/Q29

A double-slit interference experiment is set up using green light.

A pattern of interference fringes is formed on a screen.

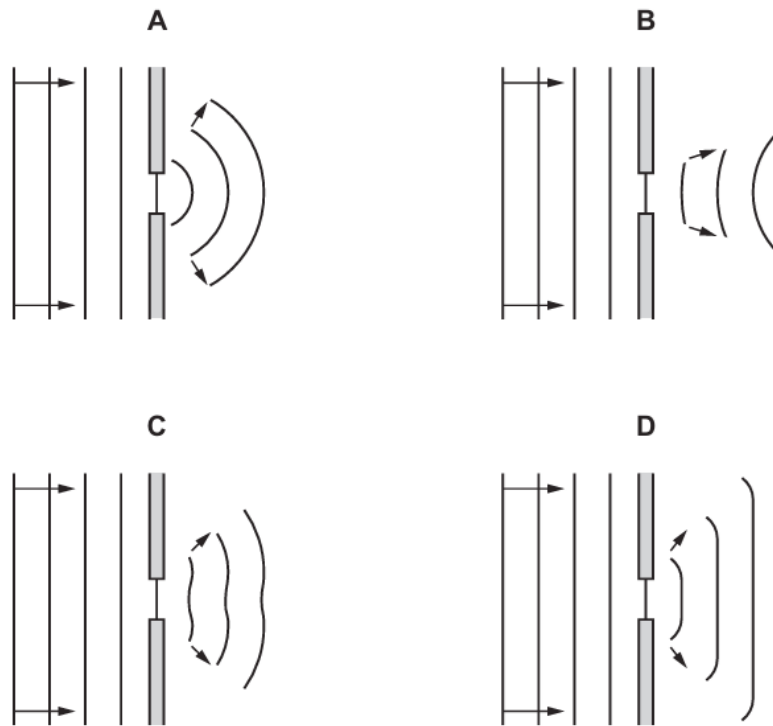
Which single change will increase the separation of the fringes?

- A** increase the width of each slit
- B** move the screen nearer to the double slit
- C** use slits that are further apart
- D** use red light instead of green light

170. 9702/13/M/J/20/Q28

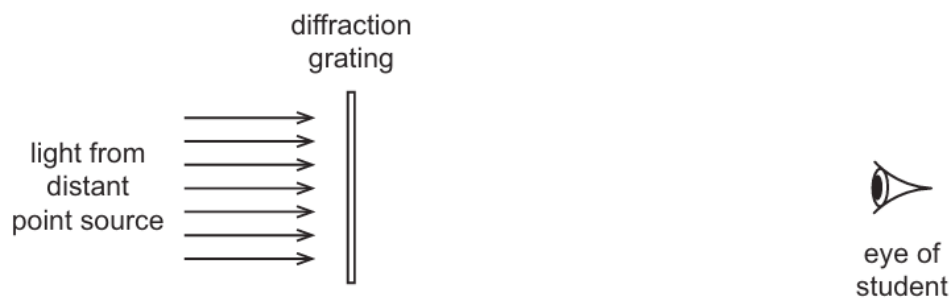
Water waves in a ripple tank are made to pass through a small gap as shown.

Which diagram shows the waves after they have passed through the gap?



171. 9702/13/M/J/20/Q30

Light of a single wavelength from a distant point source falls normally onto a diffraction grating positioned with its lines vertical.



NOT TO SCALE

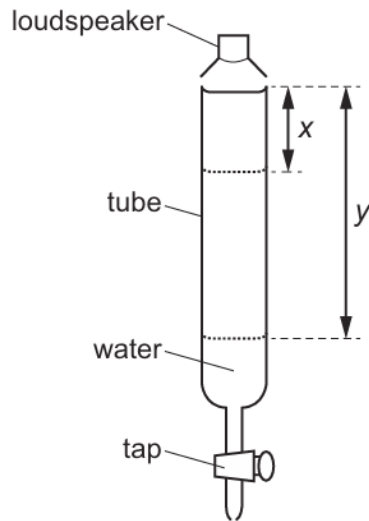
The plane of the diffraction grating is at right angles to the incident light. A student looks at the grating from a position near to the grating.

What could the student see?

- A** a central point source with a series of point source images on either side
- B** a central vertical line with a series of spectra on either side
- C** a series of fine vertical lines
- D** a single point source

172. 9702/11/O/N/20/Q23

A loudspeaker emits a sound wave into a tube initially full of water.



A tap at the bottom of the tube is opened so that water slowly leaves the tube. For some lengths of the air column in the tube, the sound heard is much louder.

The first loud sound is heard when the air column in the tube has length x .

The next time that a loud sound is heard is when the air column in the tube has length y .

What is the wavelength of the sound wave from the loudspeaker?

- A** $2x$ **B** $4y$ **C** $2(y - x)$ **D** $4(y - x)$

173. 9702/11/O/N/20/Q26

Which statement concerning a stationary wave is correct?

- A** All the particles between two adjacent nodes oscillate in phase.
- B** The amplitude of the stationary wave is equal to the amplitude of one of the waves creating it.
- C** The wavelength of the stationary wave is equal to the separation of two adjacent nodes.
- D** There is no displacement of a particle at an antinode at any time.

174. 9702/11/O/N/20/Q27

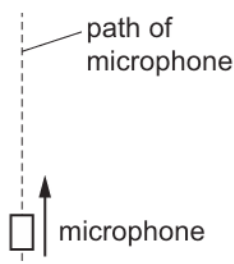
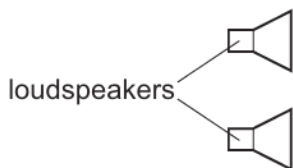
Which waves would best demonstrate diffraction through a doorway?

- A** sound waves
- B** ultraviolet waves
- C** visible light waves
- D** X-rays

175. 9702/11/O/N/20/Q28

Two loudspeakers are placed near to each other and facing in the same direction.

A microphone connected to an oscilloscope is moved along a line some distance away from the loudspeakers, as shown.



Which statement about the waves emitted by the loudspeakers is **not** a necessary condition for the microphone to detect a fixed point along the line where there is no sound?

- A The waves must be emitted in phase.
- B The waves must be emitted with a similar amplitude.
- C The waves must have the same frequency.
- D The waves must have the same wavelength.

176. 9702/11/O/N/20/Q29

A parallel beam of white light passes through a diffraction grating. Orange light of wavelength 600 nm in the fourth-order diffraction maximum coincides with blue light in the fifth-order diffraction maximum.

What is the wavelength of the blue light?

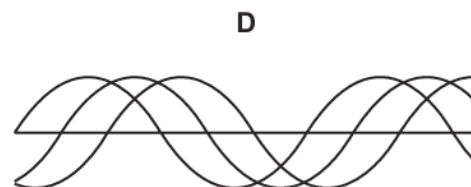
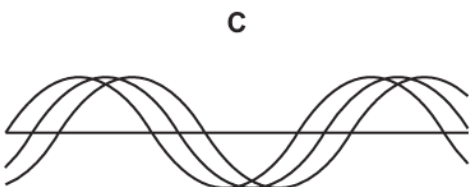
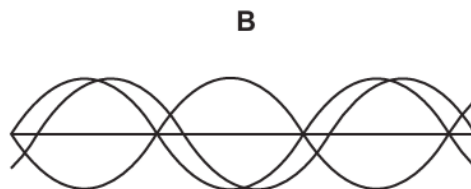
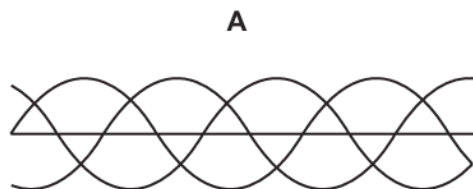
- A 450 nm B 480 nm C 500 nm D 750 nm

177. 9702/12/O/N/20/Q27

The three waves shown in each diagram have the same amplitude and frequency but different phase.

They are added together to give a resultant wave.

In which case is the resultant wave zero at this instant?



178. 9702/12/O/N/20/Q28

A transmitting mast sends out microwaves of wavelength 1.5 cm and radio waves of wavelength 1.5 km.



A receiving aerial behind a mountain can detect the radio waves but not the microwaves.

What is the reason for this?

- A** The radio waves are coherent but the microwaves are not.
- B** The radio waves are diffracted around the mountain but the microwaves are not.
- C** The radio waves are reflected by the mountain but the microwaves are not.
- D** The radio waves travel at the speed of light but the microwaves do not.

179. 9702/12/O/N/20/Q29

An experiment is carried out to demonstrate double-slit interference using light of wavelength 500 nm. The distance between bright fringes in the interference pattern is 5 mm.

What are possible values for the distance between the slits and the screen, and the slit separation?

	slit–screen distance	slit separation
A	50 cm	0.5 mm
B	50 cm	5 mm
C	5 m	0.5 mm
D	5 m	5 mm

180. 9702/12/O/N/20/Q30

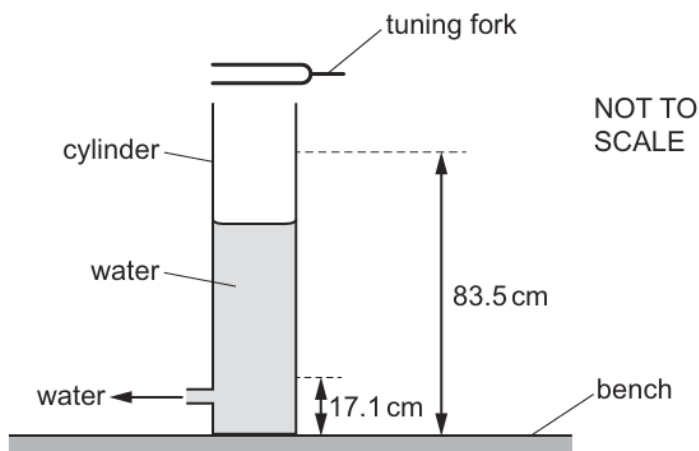
Light of a single frequency is incident on a diffraction grating. Seven bright spots are observed on a screen.

Which change will result in an increase in the number of bright spots observed?

- A** Increase the distance between the grating and the screen.
- B** Increase the frequency of the incident light.
- C** Increase the intensity of the incident light.
- D** Increase the number of lines per metre in the grating.

181. 9702/12/O/N/20/Q24

A vibrating tuning fork is held above a glass cylinder filled to the top with water. The water level is steadily lowered. A loud sound is first heard when the water level is 83.5 cm above the bench. The next loud sound is heard when the water level is 17.1 cm above the bench.



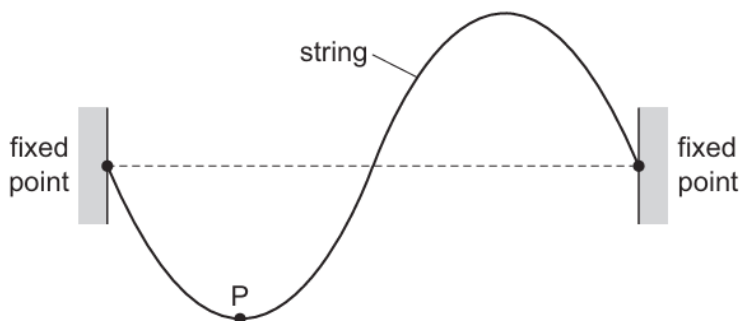
The speed of sound in air is 340 m s^{-1} .

What is the frequency of the tuning fork?

- A** 128 Hz **B** 256 Hz **C** 384 Hz **D** 512 Hz

182. 9702/13/O/N/20/Q27

A stationary wave is formed on a stretched string. The diagram illustrates the string at an instant of time when the displacement of the string is at its maximum.



The frequency of the wave is 250 Hz. Point P on the string has a vertical displacement of -1.0 mm .

What will be the vertical displacement of the point P after a time of 5.0 ms?

- A** -1.0 mm **B** zero **C** $+0.5 \text{ mm}$ **D** $+1.0 \text{ mm}$

183. 9702/13/O/N/20/Q28

What is meant by *diffraction*?

- A** the change in observed frequency when a wave source moves relative to an observer
B the formation of nodes and antinodes by two progressive waves travelling in opposite directions
C the spreading of a wave around the edge of an obstacle
D the superposition of two waves when they meet

184. 9702/13/O/N20/Q29

In a dark room, a small source of red light illuminates two slits that are 0.75 mm apart. A few metres beyond the slits, the light falls on a screen producing a series of equally spaced bright lines.

Which change would cause the distance between the bright lines on the screen to be reduced?

- A** Change the source for one emitting blue light.
- B** Reduce the distance between the light source and the slits.
- C** Reduce the distance between the slits to 0.55 mm.
- D** Reduce the intensity of the light source.

185. 9702/13/O/N20/Q30

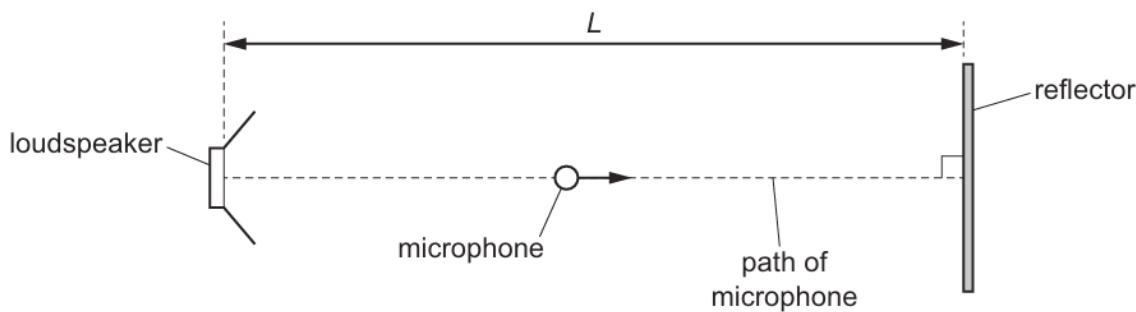
Light of wavelength 5.30×10^{-7} m is incident normally on a diffraction grating. The first-order maximum is observed at an angle of 15.4° to the direction of the incident light.

What is the angle between the first-order and second-order diffraction maxima?

- A** 7.7° **B** 15.4° **C** 16.7° **D** 32.1°

186. 9702/12/F/M/21/Q23

A loudspeaker emitting a sound wave of a single frequency is placed a distance L from a reflecting surface, as shown.



A stationary wave is formed with an antinode at the loudspeaker. A microphone is moved from the loudspeaker to the reflector.

Before the microphone reaches the reflector, it detects four points where the sound intensity is a minimum.

What is the wavelength of the sound wave?

- A** $\frac{2L}{9}$ **B** $\frac{2L}{8}$ **C** $\frac{4L}{9}$ **D** $\frac{4L}{8}$

187. 9702/12/F/M/21/Q26

Two waves, P and Q, meet at a point X and superpose.

Initially, the two waves meet at X in phase (zero phase difference) so that the resultant wave has an amplitude of 14.0 cm at that point.

The phase difference between the two waves is then changed so that they meet at X with a phase difference of 180° . The resultant wave now has an amplitude of 4.0 cm at X.

What is the amplitude of one of the waves at point X?

- A** 2.0 cm **B** 5.0 cm **C** 10 cm **D** 18 cm

188. 9702/12/F/M/21/Q27

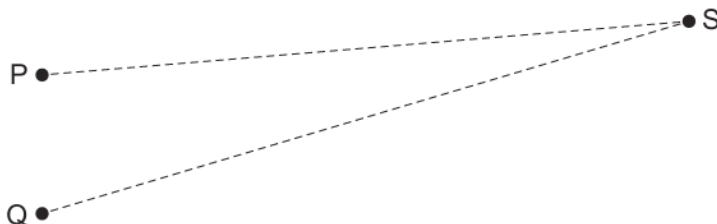
A water wave is diffracted as it passes through a gap between two barriers in a ripple tank. The wave is observed to 'spread out' as it moves through the gap.

Which two factors both affect the amount of diffraction observed?

- A** the amplitude and frequency of the incident wave
B the amplitude of the incident wave and the width of the gap
C the wavelength and amplitude of the incident wave
D the wavelength of the incident wave and the width of the gap

189. 9702/12/F/M/21/Q28

Two sources of microwaves P and Q produce coherent waves with a phase difference of 180° . The waves have the same wavelength λ .



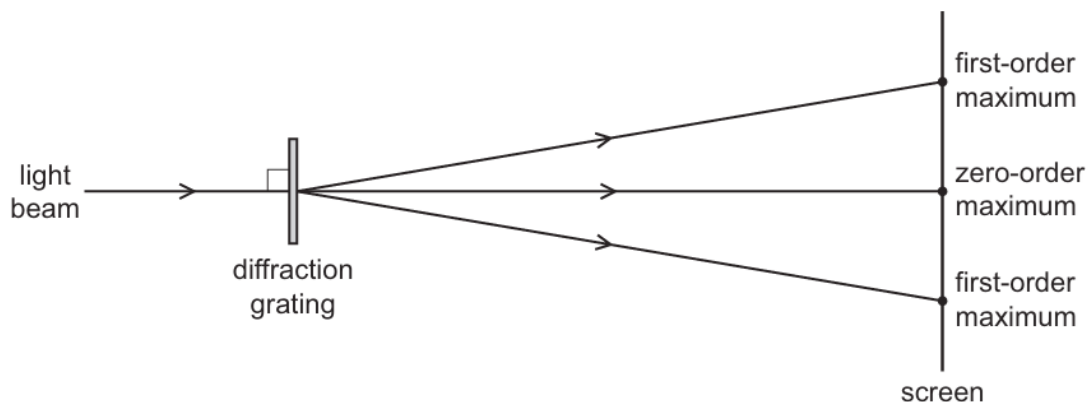
At the point S there is a minimum in the interference pattern produced by waves from the two sources. The distance $(QS - PS)$ is called the path difference.

Which expression could represent the path difference?

- A** $\frac{\lambda}{4}$ **B** $\frac{\lambda}{2}$ **C** λ **D** $\frac{3\lambda}{2}$

190. 9702/12/F/M/21/Q29

A beam of red laser light of wavelength 633 nm is incident normally on a diffraction grating with 600 lines per mm.



The beam of red light is now replaced by a beam of blue laser light of wavelength 445 nm. A replacement diffraction grating is used so that the first-order maximum of the blue light appears at the same position on the screen as the first-order maximum of the red light from the original laser.

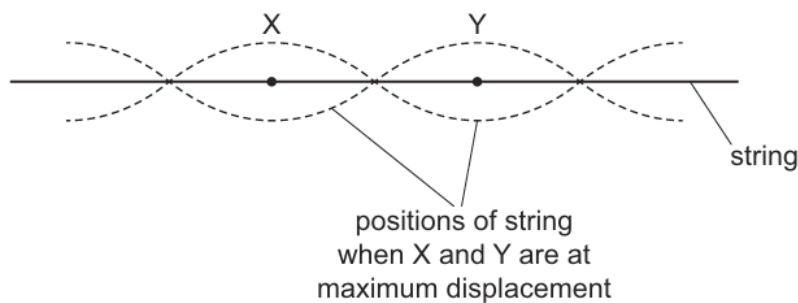
How many lines per mm are there in the replacement diffraction grating?

- A** 420 mm^{-1} **B** 470 mm^{-1} **C** 600 mm^{-1} **D** 850 mm^{-1}

191. 9702/11/M/J/21/Q26

The diagram shows part of a stationary wave on a string.

X and Y are points on the string. The vibrations at X and Y are 180° out of phase.

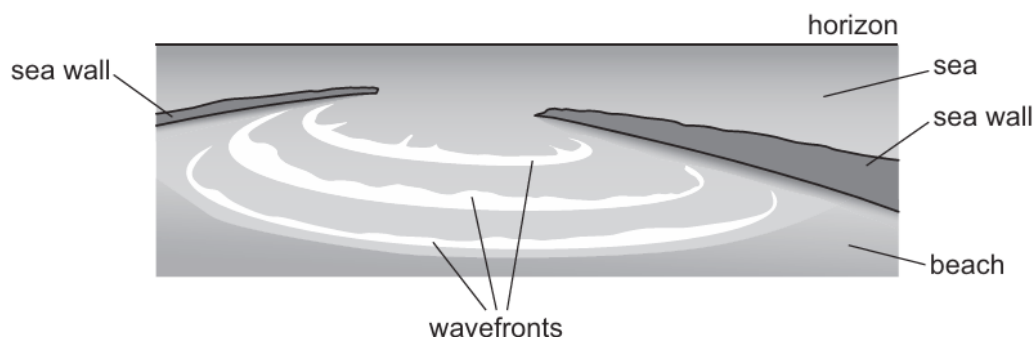


What is the distance between X and Y?

- A** one-quarter of a wavelength
B half a wavelength
C one wavelength
D two wavelengths

192. 9702/11/M/J/21/Q27

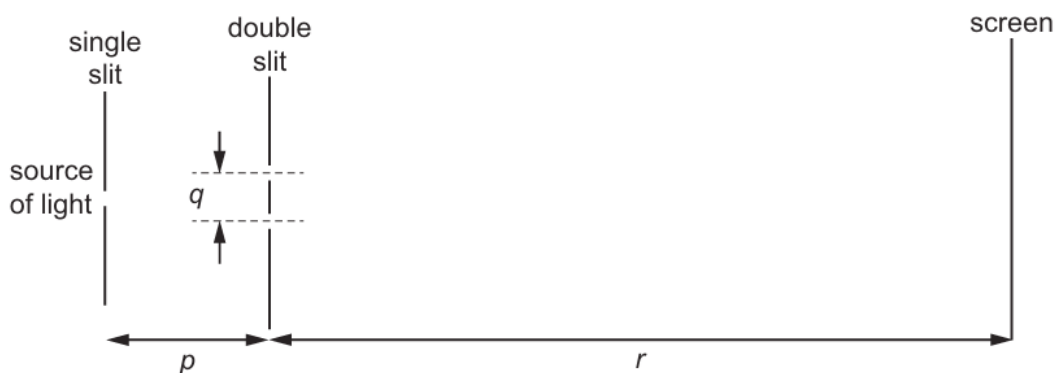
Which wave behaviour is shown in the diagram?



- A diffraction
- B Doppler shift
- C interference
- D superposition

193. 9702/11/M/J/21/Q28

A teacher sets up the apparatus shown to demonstrate a double-slit interference pattern on a screen.



Which change to the apparatus will increase the fringe spacing?

- A decrease the distance p
- B decrease the distance q
- C decrease the distance r
- D decrease the wavelength of the light

194. 9702/11/M/J/21/Q29

Light of a single unknown wavelength and blue light of a single wavelength are both incident normally on a diffraction grating. Two diffraction patterns are produced, one for each wavelength of light.

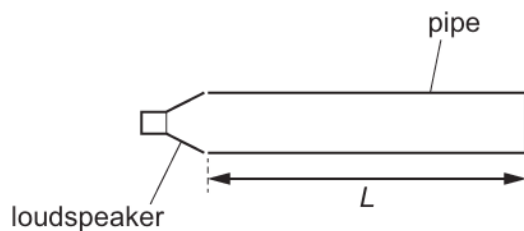
The third-order maximum for the blue light occurs at the same angle as the second-order maximum for the light of unknown wavelength. The wavelength of the blue light is 480 nm.

What is the unknown wavelength?

- A 320 nm
- B 720 nm
- C 960 nm
- D 1440 nm

195. 9702/12/M/J/21/Q26

A pipe of length L is open at one end and closed at the other end. A loudspeaker is at the open end and emits a sound wave into the pipe.



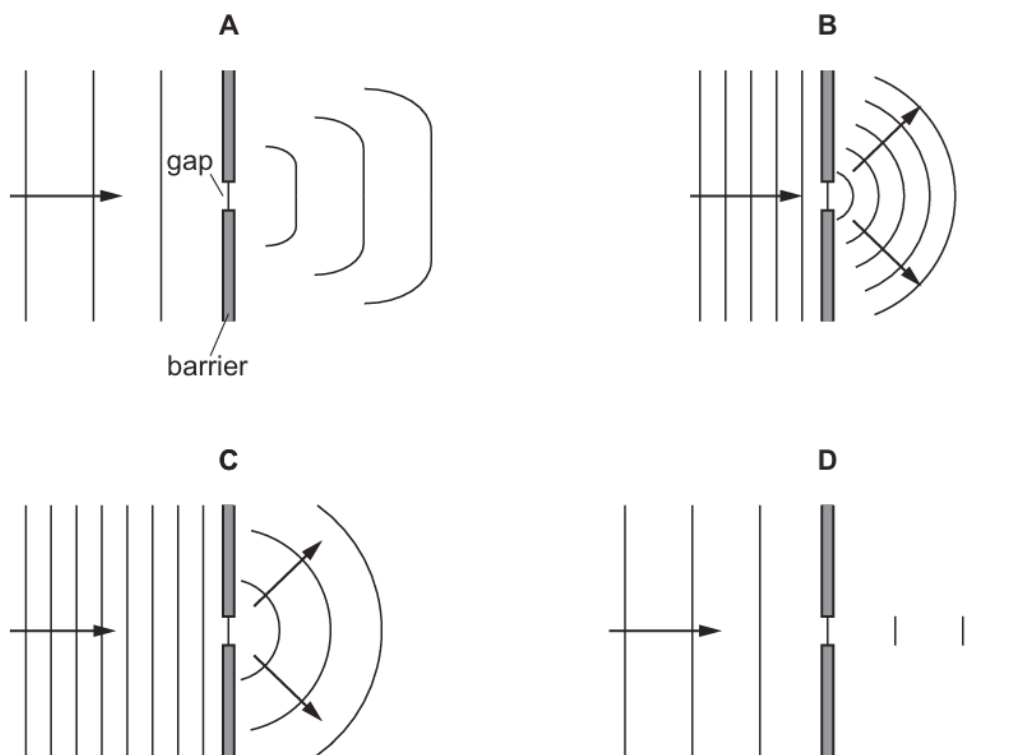
When a stationary wave is formed, there is an antinode at the open end of the pipe.

Which wavelength of sound could be used to produce a stationary wave?

- A** $\frac{2L}{3}$ **B** L **C** $\frac{4L}{3}$ **D** $2L$

196. 9702/12/M/J/21/Q27

Which diagram best shows how water waves diffract when they pass through a gap in a barrier?



197. 9702/12/M/J/21/Q28

In a two-source interference experiment, light of a single frequency is incident on a double slit.

The light waves emerging from the slits are coherent.

What is meant by *coherent*?

- A The waves are in phase.
- B The waves have a constant phase difference.
- C The waves have the same amplitude.
- D The waves interfere constructively wherever they overlap.

198. 9702/12/M/J/21/O29

A parallel beam of light consists of light of wavelength 420 nm and light of wavelength 630 nm.

The light is incident normally on a diffraction grating.

The diffraction maxima for the two wavelengths overlap only at an angle of 31° from the direction of the incident light beam.

What could be the line spacing of the diffraction grating?

- A $1.2\ \mu\text{m}$ B $1.6\ \mu\text{m}$ C $2.4\ \mu\text{m}$ D $3.7\ \mu\text{m}$

199. 9702/13/M/J/21/Q27

To produce a stationary wave, two waves must travel in opposite directions through the same space.

Which statement about the properties of the two waves **must** also be correct?

- A The waves must have equal frequencies, but different speeds and wavelengths.
- B The waves must have equal speeds, but different wavelengths and frequencies.
- C The waves must have equal speeds, frequencies and wavelengths.
- D The waves must have equal wavelengths, but different speeds and frequencies.

200. 9702/13/M/J/21/Q28

The speed of sound in air is $330\ \text{m s}^{-1}$.

Which size of architectural features in a large concert hall would best diffract sound waves of frequency 0.44 kHz?

- A 1.3 mm B 750 mm C 7.5 m D 17 m

201. 9702/13/M/J/21/Q29

A double-slit interference pattern using red light of wavelength $7.0 \times 10^{-7}\ \text{m}$ has a fringe spacing of 3.5 mm.

Which fringe spacing would be observed for the same arrangement of apparatus but using blue light of wavelength $4.5 \times 10^{-7}\ \text{m}$?

- A 2.3 mm B 3.5 mm C 5.4 mm D 9.0 mm

202. 9702/13/M/J/21/Q30

A beam of light of a single wavelength is incident normally on a diffraction grating.

The angle of diffraction θ is measured for each order of diffraction n . The distance between adjacent slits in the diffraction grating is d .

A graph is plotted to determine the wavelength of the light.

Which graph should be plotted and how is the wavelength determined from the graph?

	y-axis	x-axis	wavelength
A	n	$d \sin \theta$	gradient
B	n	$d \sin \theta$	1 / gradient
C	$\sin \theta$	d / n	gradient
D	$\sin \theta$	$d \times n$	1 / gradient

203. 9702/11/O/N/21/Q23

A glass tube is closed at one end and has a loudspeaker at the other end.



A stationary wave is formed with a node at the closed end of the tube when the sound has frequency f_0 . There are no other nodes.

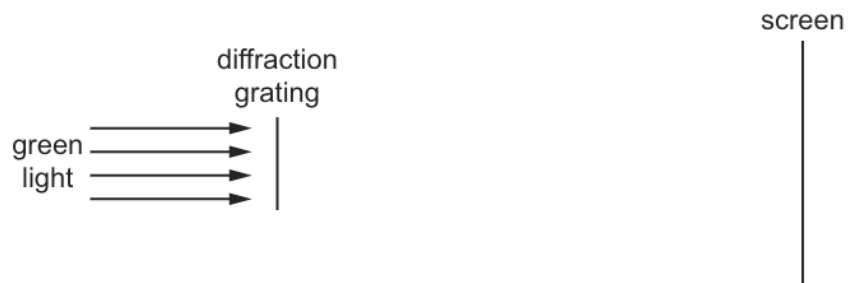
The frequency of the sound is then slowly increased.

What is the frequency of the sound that produces the next stationary wave?

- A** $1.25f_0$ **B** $1.50f_0$ **C** $2.00f_0$ **D** $3.00f_0$

204. 9702/11/O/N/21/Q29

Green light is incident normally on a diffraction grating and forms a diffraction pattern on a distant screen.

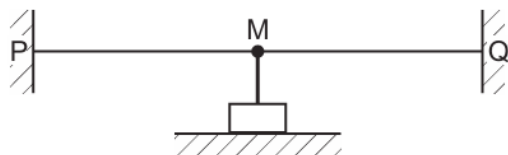


Which change, on its own, would **decrease** the separation of the diffraction maxima on the screen?

- A** Increase the distance between the screen and the diffraction grating.
B Replace the diffraction grating with a grating that has a smaller separation between the slits.
C Replace the diffraction grating with a grating that has fewer slits per unit length.
D Replace the green light with red light.

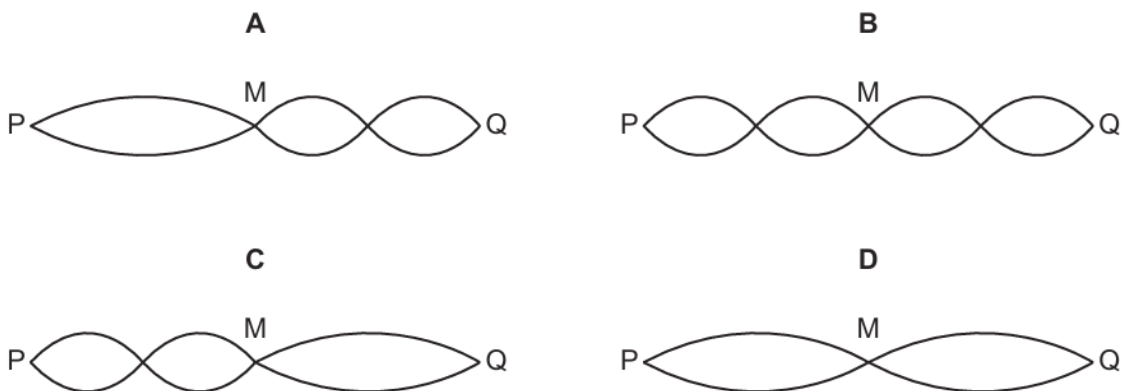
205. 9702/11/O/N/21/Q26

A string is fixed between point P and an oscillator M. Another string is fixed between M and point Q. M is midway between P and Q.



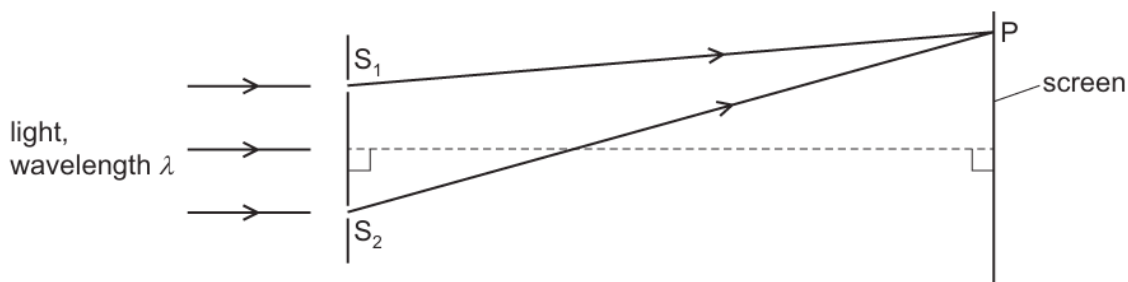
The frequency of the oscillator is adjusted until a stationary wave is formed on both strings. The speed of the wave between P and M is twice the speed of the wave between M and Q.

Which diagram could represent the stationary wave pattern?



206. 9702/11/O/N/21/Q28

Light of wavelength λ is incident normally on two narrow slits S_1 and S_2 , a small distance apart. Bright and dark fringes are observed on a screen a long distance away from the slits.



The n th **dark** fringe from the central bright fringe is observed at point P on the screen.

Which equation is correct for all positive values of n ?

- A** $S_2P - S_1P = \frac{n\lambda}{2}$
- B** $S_2P - S_1P = n\lambda$
- C** $S_2P - S_1P = (n - \frac{1}{2})\lambda$
- D** $S_2P - S_1P = (n + \frac{1}{2})\lambda$

207. 9702/11/O/N/21/Q27

A water wave in a ripple tank is diffracted as it passes through a gap in a barrier.

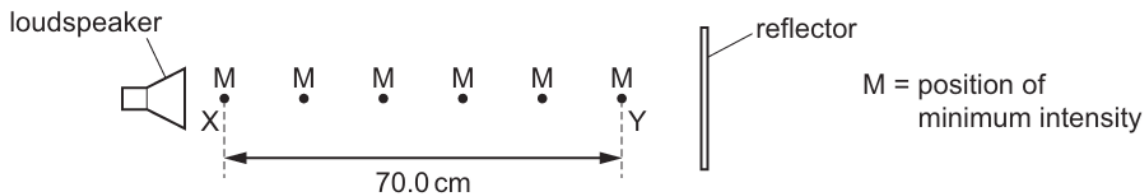
Which two factors affect the angle of diffraction of the wave?

- A the amplitude and frequency of the incident wave
- B the amplitude of the incident wave and the width of the gap
- C the wavelength and amplitude of the incident wave
- D the wavelength of the incident wave and the width of the gap

208. 9702/12/O/N/21/Q24

A sound wave from a loudspeaker is reflected back along its original path by a reflector.

A microphone is initially at point X where the sound intensity is a minimum, as shown.



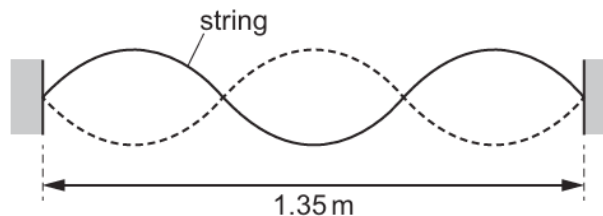
The microphone is moved towards the reflector and passes through four more intensity minima until reaching a fifth minimum at point Y. The distance XY is 70.0 cm.

What is the wavelength of the sound?

- A 11.7 cm
- B 14.0 cm
- C 23.3 cm
- D 28.0 cm

209. 9702/12/O/N/21/Q27

A stationary wave is produced on a string that is stretched between two fixed points that are a distance of 1.35 m apart, as shown.



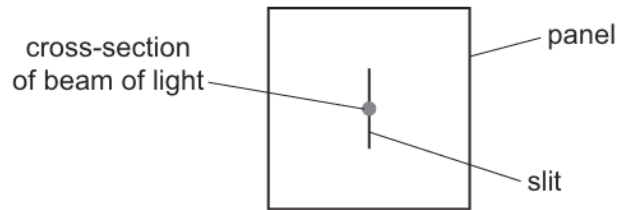
The speed of the waves on the string is 450 m s^{-1} .

What is the frequency of oscillation of the stationary wave?

- A 333 Hz
- B 405 Hz
- C 500 Hz
- D 1000 Hz

210. 9702/12/O/N/21/O28

A beam of laser light is directed towards a narrow slit.



After emerging from the other side of the slit, the diffracted light then falls on a screen.

What is the pattern of light seen on the screen?



211. 9702/12/O/N/21/Q29

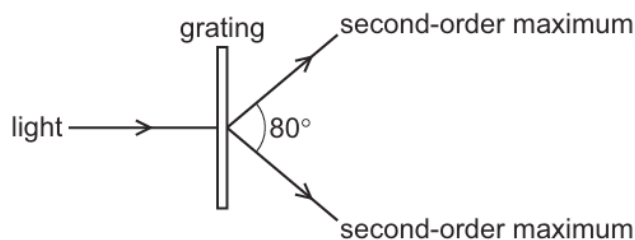
Two waves, each with a constant amplitude, interfere and produce an interference pattern. The pattern has minima at fixed points where the displacement is zero at all times.

Which statement describes the two waves?

- A** They must be coherent and of the same amplitude.
- B** They must be coherent but not necessarily of the same amplitude.
- C** They must be of the same amplitude but not necessarily coherent.
- D** They must not be coherent or of the same amplitude.

212. 9702/12/O/N/21/Q30

Light of wavelength 5.5×10^{-7} m is incident normally on a diffraction grating.



The angle between the second-order diffraction maxima is 80° , as shown.

What is the number of lines per metre of the diffraction grating?

- A** 5.8×10^5 lines per metre
- B** 9.0×10^5 lines per metre
- C** 1.2×10^6 lines per metre
- D** 2.3×10^6 lines per metre

213. 9702/13/O/N/21/Q24

A stationary sound wave is formed in a pipe that is closed at one end and open at the other end. The wave has two antinodes. One of these antinodes is at the open end of the pipe.

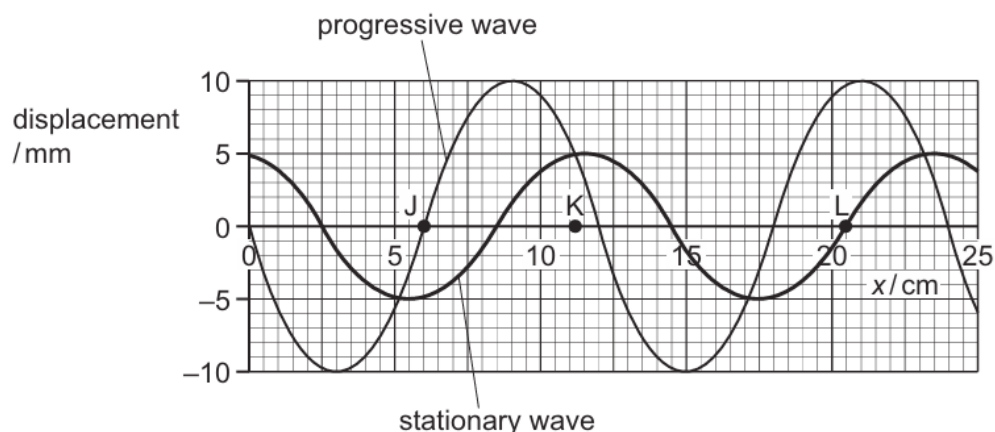
The length of the pipe is 0.600 m. The speed of sound in the air column in the pipe is 340 ms^{-1} .

What is the frequency of the sound wave?

- A** 425 Hz **B** 850 Hz **C** 1130 Hz **D** 2270 Hz

214. 9702/13/O/N/21/Q27

Two progressive waves travel in opposite directions and form a stationary wave. The graph shows the variation with distance x of the displacement of the stationary wave and of one of the two progressive waves at the same instant in time.



What are the approximate displacements of the **other** progressive wave at the positions J, K and L?

	displacement / mm		
	J	K	L
A	-5	0	-10
B	-5	+5	0
C	0	+5	+10
D	+5	-5	0

215. 9702/13/O/N/21/Q29

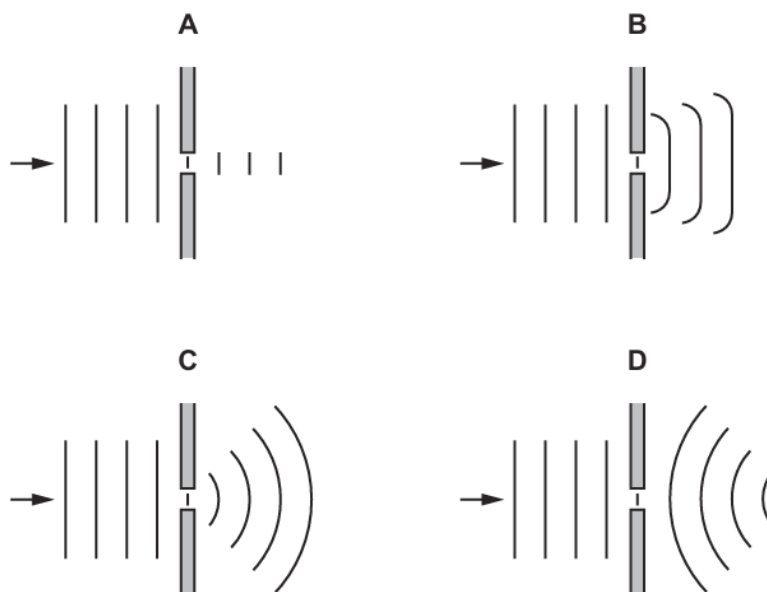
Interference fringes are produced on a screen by double-slit interference using light of wavelength 600 nm. The fringe separation is 4.0 mm and the separation of the slits is 0.60 mm.

What is the distance between the double slit and the screen?

- A** 0.25 m **B** 0.40 m **C** 2.5 m **D** 4.0 m

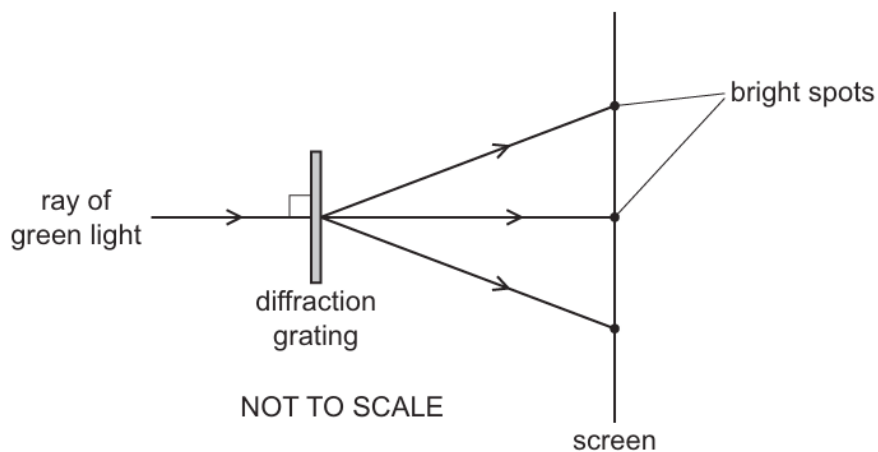
216. 9702/13/O/N/21/Q28

Which diagram shows the diffraction of water waves in a ripple tank?



217. 9702/13/O/N/21/Q30

A ray of green light is incident normally on a diffraction grating. Several bright spots are produced on a screen on the other side of the grating, as shown.



Which pair of changes could result in bright spots at exactly the same angles as previously?

- A** Use blue light and increase the distance between the grating and the screen.
- B** Use blue light and increase the number of lines per unit length in the grating.
- C** Use red light and increase the distance between the grating and the screen.
- D** Use red light and increase the number of lines per unit length in the grating.

218. 9702/12/F/M/22/Q24

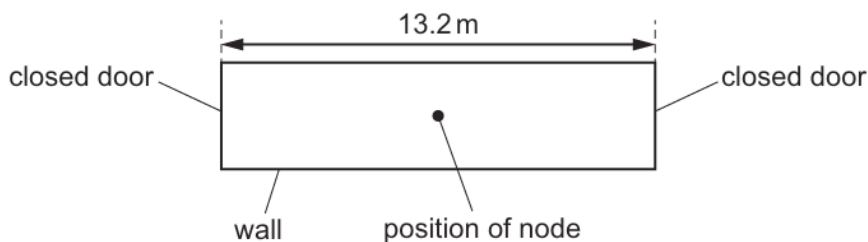
Two coherent progressive waves from different sources meet at a point.

Which condition **must** be satisfied for there to be zero resultant amplitude at the point where the waves meet?

- A** The two waves must be emitted from their sources with the same intensity.
- B** The two waves must be in phase with each other at the point.
- C** The two waves must be travelling in opposite directions.
- D** The two waves must have the same amplitude at the point.

219. 9702/12/F/M/22/Q25

A corridor is 13.2m long and has closed doors that reflect sound at both ends. The speed of sound in the air in the corridor is 330 m s^{-1} .



What is the lowest frequency of sound that could create a stationary wave in the corridor with a node halfway along it?

- A** 0.040 Hz
- B** 13 Hz
- C** 25 Hz
- D** 50 Hz

220. 9702/12/F/M/22/Q26

Water waves of wavelength λ are formed in a ripple tank. The waves are diffracted as they pass through a narrow gap of width d (d is greater than λ).

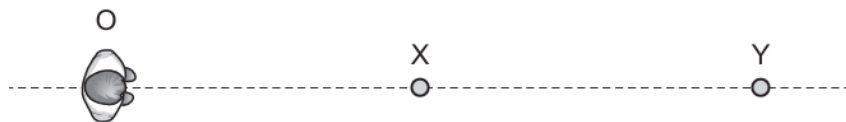
Which gap width and which wavelength will cause the largest decrease in the amount of diffraction?

	gap width	wavelength
A	$\frac{1}{2} d$	$\frac{1}{2} \lambda$
B	$\frac{1}{2} d$	2λ
C	$2d$	$\frac{1}{2} \lambda$
D	$2d$	2λ

221. 9702/12/F/M/22/Q27

Two loudspeakers X and Y emit sound waves that are in phase and of wavelength 0.75 m.

An observer O is able to stand anywhere on a straight line that passes through X and Y, as shown. The observer stands at a point where the sound waves from X and Y meet in phase.



What could be the distances OY and XY?

	distance OY / m	distance XY / m
A	1.25	3.50
B	2.00	2.75
C	2.75	2.00
D	3.25	1.50

222. 9702/12/F/M/22/Q28

Light of a single wavelength is incident normally on a diffraction grating.

The resulting diffraction pattern is displayed on a screen.

Which change makes the first orders of intensity maxima further apart from each other on the screen?

- A** placing the screen closer to the diffraction grating
- B** using a diffraction grating with less separation between adjacent slits
- C** using a diffraction grating with more slits but keeping the same separation between adjacent slits
- D** using light with a shorter wavelength

223. 9702/11/M/J/22/Q26

A stationary wave is set up on a string that is stretched between two fixed points that are 48 cm apart.

At one instant, the appearance of the string is as shown.

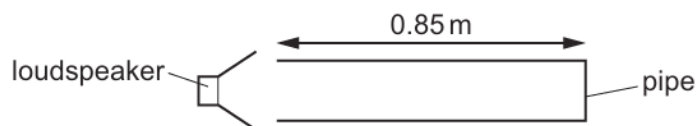


What is the wavelength of the stationary wave?

- A** 16 cm **B** 32 cm **C** 48 cm **D** 72 cm

224. 9702/11/M/J/22/Q27

A pipe, closed at one end, has a loudspeaker at the open end. For some frequencies of sound from the loudspeaker, a stationary sound wave is formed in the air within the pipe with an antinode at the open end of the pipe.



The length of the pipe is 0.85 m.

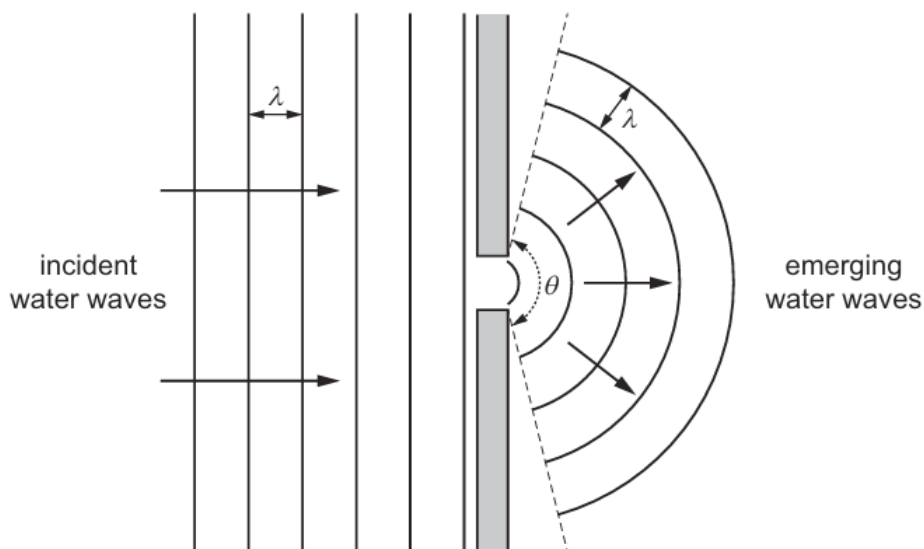
The speed of sound in air is 340 m s^{-1} .

Which frequency of sound from the loudspeaker would **not** produce a stationary wave?

- A** 100 Hz **B** 200 Hz **C** 300 Hz **D** 500 Hz

225. 9702/11/M/J/22/Q28

Water waves of wavelength λ are incident normally on an obstacle with a narrow gap. The width of the gap is equal to λ . The waves from the gap emerge over an angle θ , as shown.



The gap is slowly widened.

Which changes, if any, occur to θ and to the wavelength of the emerging waves?

	θ	wavelength
A	decreases	remains the same
B	increases	remains the same
C	remains the same	decreases
D	remains the same	increases

226. 9702/11/M/J/22/Q29

Light of a single frequency passes through two narrow slits and produces an interference pattern on a screen some distance away. The interference fringes are very close together.

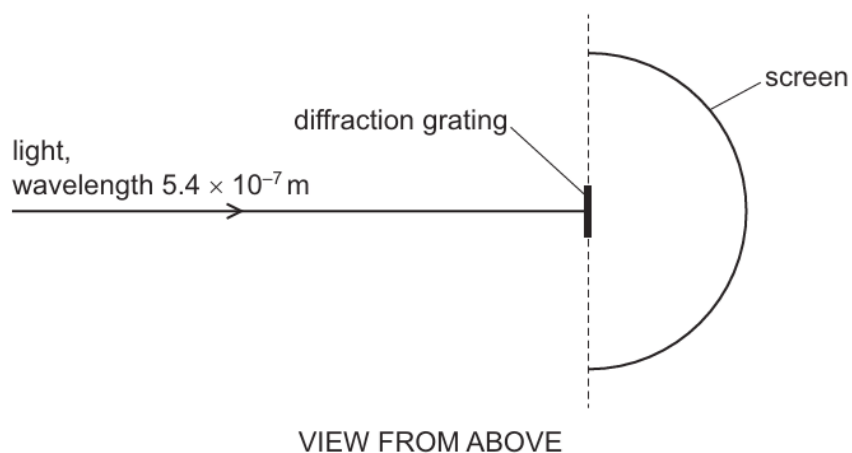
Which change would **increase** the distance between the fringes?

- A** Increase the brightness of the light source.
- B** Increase the distance between the slits and the screen.
- C** Increase the distance between the two slits.
- D** Increase the frequency of the light used.

227. 9702/11/M/J/22/Q30

Light of wavelength $5.4 \times 10^{-7} \text{ m}$ is incident normally on a diffraction grating.

The separation between adjacent lines in the grating is $2.0 \times 10^{-6} \text{ m}$. The light that emerges from the grating falls on a semicircular screen, as shown in the view from above.



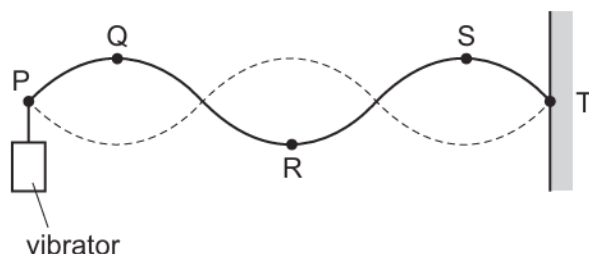
The grating is at the centre of the semicircle, and the lines of the grating are vertical.

How many bright dots are formed on the screen?

- A** 3
- B** 4
- C** 6
- D** 7

228. 9702/12/M/J/22/Q27

A stationary wave on a stretched string is set up between two points P and T.

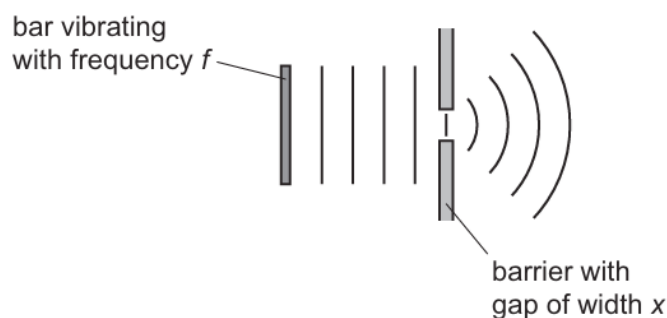


Which statement about the stationary wave is correct?

- A** Point R is at a node.
- B** Points Q and S vibrate in phase.
- C** The distance between P and T is three wavelengths.
- D** The wave transfers energy from P to T.

229. 9702/12/M/J/22/Q28

A bar vibrates with frequency f to produce water waves in a ripple tank.



The waves pass through a gap of width x in a barrier so that diffraction occurs.

Which combination of vibration frequency and gap width will produce the smallest angle of diffraction?

	vibration frequency	gap width
A	$\frac{f}{2}$	$\frac{x}{2}$
B	$\frac{f}{2}$	$2x$
C	$2f$	$\frac{x}{2}$
D	$2f$	$2x$

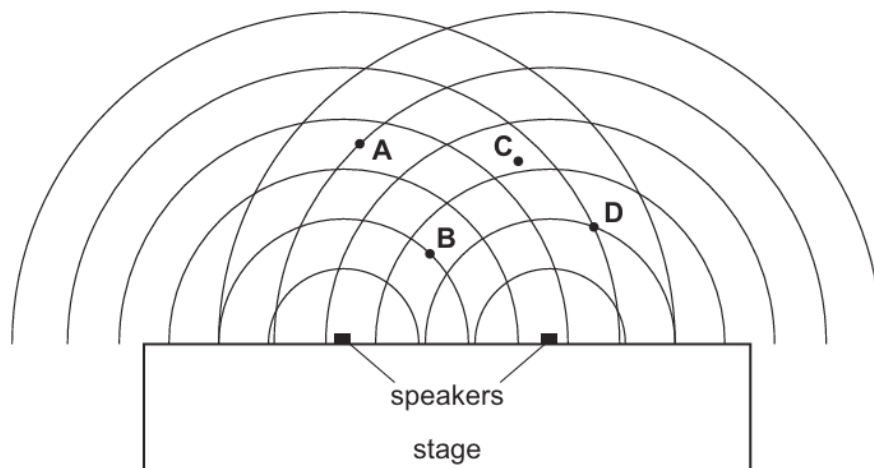
230. 9702/12/M/J/22/Q29

An outdoor concert has two large speakers beside the stage for broadcasting music.

In order to test the speakers, they are made to emit sound of the same wavelength and the same amplitude.

The curved lines in the diagram represent wavefronts.

Where is the loudest sound heard?

**231. 9702/12/M/J/22/Q30**

The equation

$$\lambda = \frac{d \sin \theta}{n}$$

is used to calculate the wavelength λ of light in an experiment that uses a diffraction grating. The light from the diffraction grating is displayed on a screen.

What do the symbols n and d represent?

	n	d
A	number of slits in the grating	distance between adjacent slits in the grating
B	number of slits in the grating	distance from grating to screen
C	order of intensity maximum	distance between adjacent slits in the grating
D	order of intensity maximum	distance from grating to screen

232. 9702/13/M/J/22/Q27

The principle of superposition states that a certain quantity is added when two or more waves meet at a point.

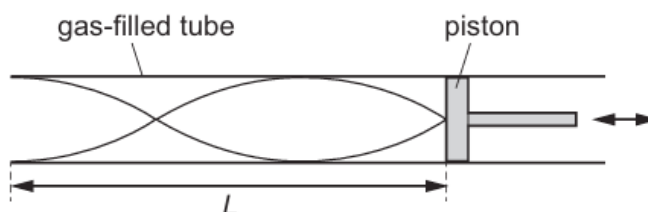
What is this quantity?

- A** amplitude
- B** displacement
- C** intensity
- D** wavelength

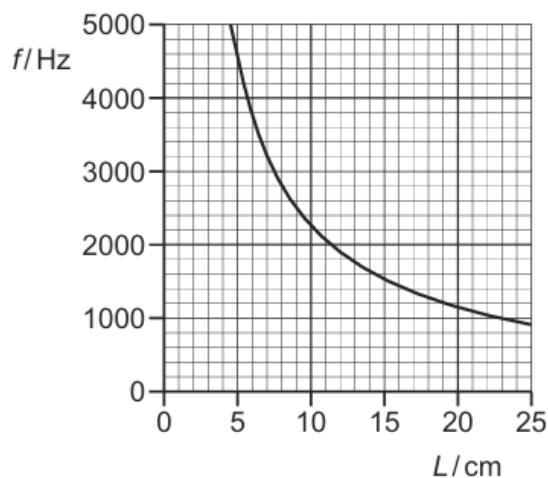
233. 9702/13/M/J/22/Q28

A stationary sound wave is formed in a gas-filled tube of length L , which is closed at one end by a piston. The length of the tube can be altered by moving the piston.

The length of the tube and the frequency of the sound are varied so that the stationary wave always has two antinodes and two nodes, as shown.



The graph shows the variation of the frequency f of the stationary sound wave with the length L of the tube.



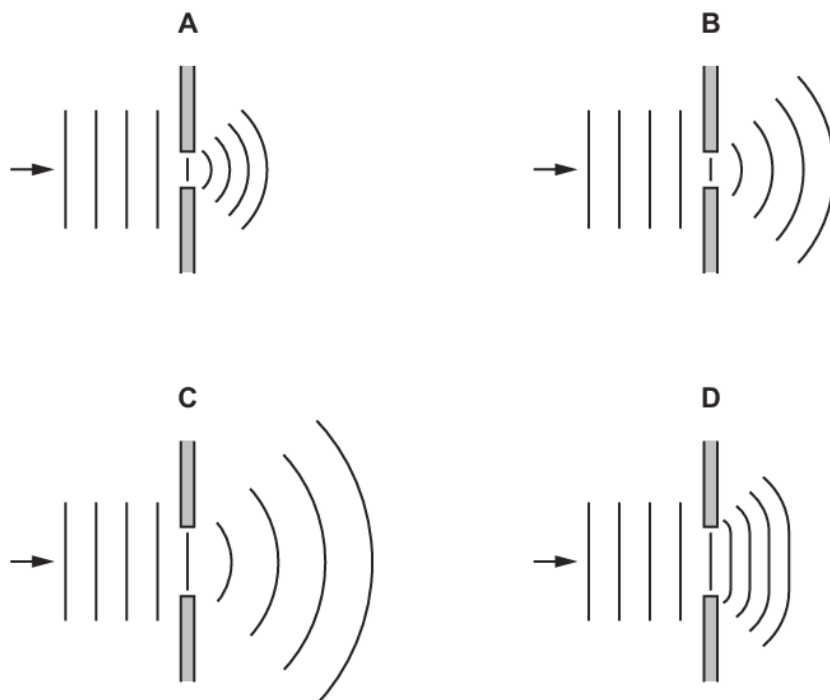
What is the speed of sound in the gas in the tube?

- A** 150 ms^{-1}
- B** 230 ms^{-1}
- C** 300 ms^{-1}
- D** 340 ms^{-1}

234. 9702/13/M/J/22/Q29

The diagrams show the diffraction of water waves in a ripple tank as they pass through a gap between two barriers.

Which diagram is correct?



235. 9702/13/M/J/22/Q30

A beam of light from a laser is incident normally on a double slit. Interference fringes are seen on a screen placed parallel to the double slit.

The separation of the two slits is a . The distance between the slits and the screen is D . The distance between the centres of two adjacent bright fringes is x .

D and a are both halved.

What is the distance between the centres of two adjacent bright fringes after these changes?

A $\frac{x}{2}$

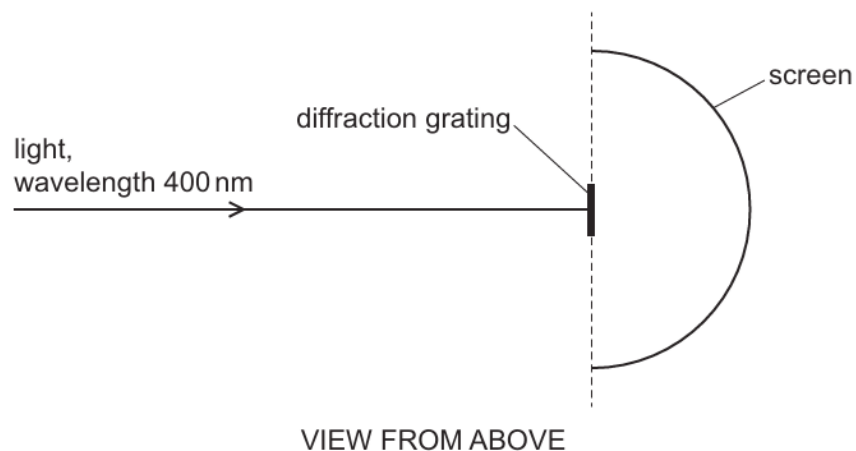
B x

C $2x$

D $4x$

236. 9702/13/M/J/22/Q31

A beam of light of wavelength 400 nm is incident normally on a diffraction grating that has 300 lines per millimetre. The light passes through the grating and produces a series of maxima which are observed on a semicircular screen, as shown.



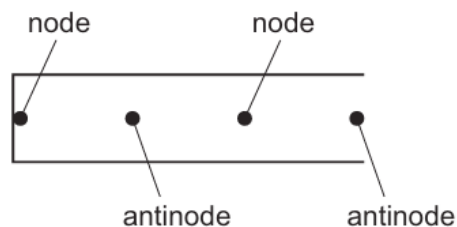
What is the total number of maxima observed on the screen?

- A** 8 **B** 9 **C** 16 **D** 17

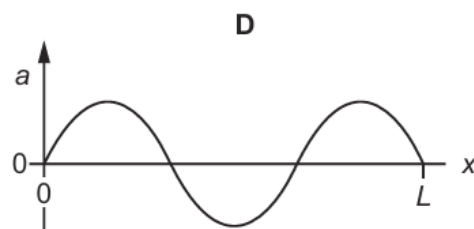
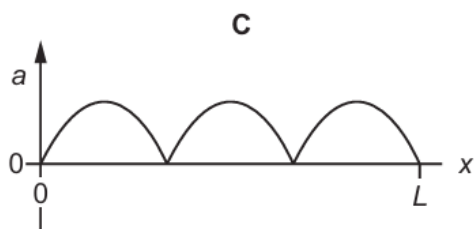
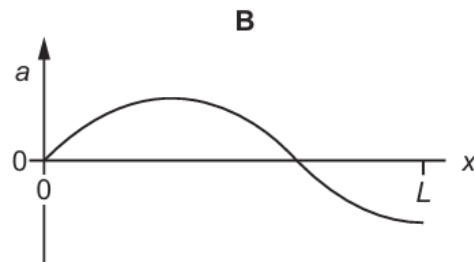
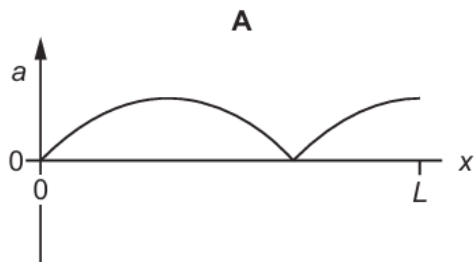
237. 9702/11/O/N/22/Q27

A stationary sound wave is formed in a tube of length L that is closed at one end.

The diagram shows the positions of the nodes and antinodes of the stationary wave.



Which graph shows the variation of the amplitude a of the wave with distance x measured from the closed end of the tube?



238. 9702/11/O/N/22/Q26

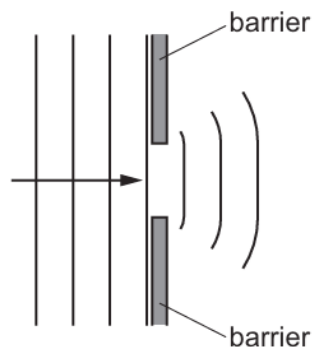
Two waves of the same type overlap.

When does the principle of superposition apply?

- A always
- B only when the waves have the same amplitude
- C only when the waves have the same frequency
- D only when the waves travel in opposite directions

239. 9702/11/O/N/22/Q28

A wave on the surface of water passes through a gap between two barriers and is diffracted, as shown.



What happens when the frequency of the wave is halved?

- A Less diffraction is observed.
- B More diffraction is observed.
- C There is no diffraction.
- D The same amount of diffraction is observed.

240. 9702/11/O/N/22/Q29

Observable interference fringes are produced using light from a double slit. The intensity of the light emerging from each slit is initially the same.

The intensity of the light emerging from one of the slits is now reduced.

How does this affect the interference pattern?

- A The bright fringes and the dark fringes all become brighter.
- B The bright fringes and the dark fringes all become darker.
- C The bright fringes become brighter and the dark fringes become darker.
- D The bright fringes become darker and the dark fringes become brighter.

241. 9702/11/O/N/22/Q30

A diffraction grating has 4.00×10^5 lines per metre. A beam of light of wavelength 589×10^{-9} m is incident normally on the diffraction grating.

What is the angle between the second-order maximum and the direction of the incident beam of light?

- A** 13.6° **B** 27.3° **C** 28.1° **D** 56.2°

242. 9702/12/O/N/22/Q26

The wavelength of sound in air may be determined by using stationary waves.

In one experiment, a loudspeaker produces a sound wave of constant frequency which is reflected directly back along its original path by a metal plate approximately 1 m away. A microphone connected to a cathode-ray oscilloscope (CRO) is moved between the loudspeaker and plate to identify regions of high sound intensity ('loud' spots) and low sound intensity ('quiet' spots).

The wavelength of the sound is determined using the **least** possible number of measured quantities.

Which row shows the quantities that are needed?

	frequency of sound	mean separation of 'quiet' spots	speed of sound in air
A	✓	✓	✗
B	✓	✗	✓
C	✓	✗	✗
D	✗	✓	✗

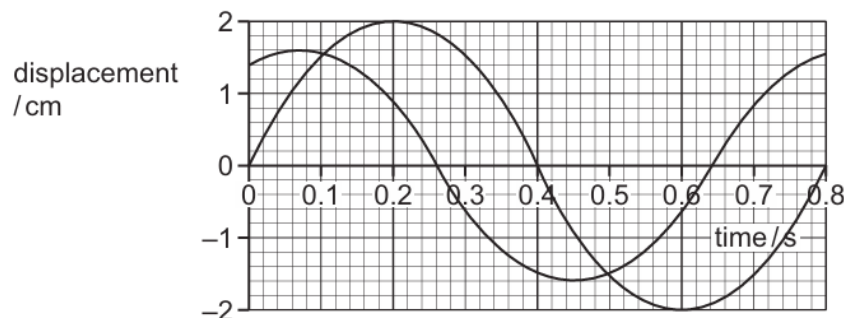
key

✓ = needed

✗ = not needed

243. 9702/12/O/N/22/Q27

Two progressive waves meet at a fixed point P. The variation with time of the displacement of each wave at point P is shown.



The two waves superpose at point P.

What is the resultant displacement at time 0.38 s?

- A** +1.0 cm **B** -1.0 cm **C** +1.8 cm **D** -1.8 cm

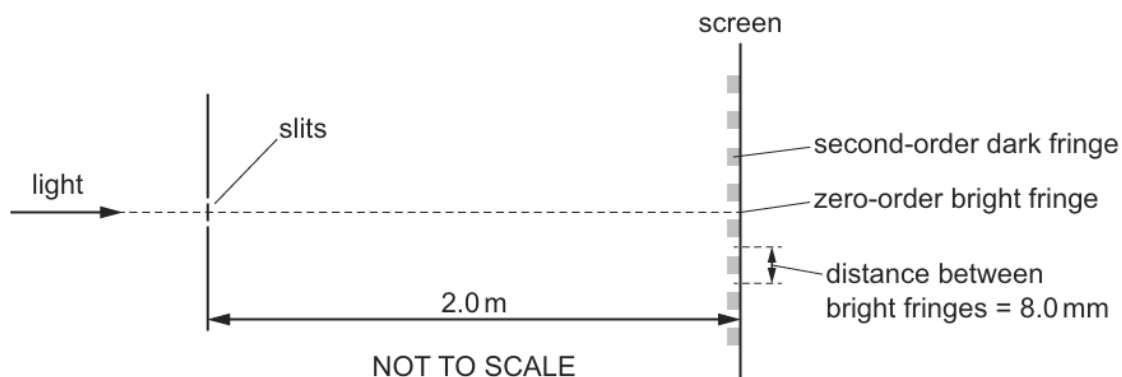
244. 9702/12/O/N/22/Q28

In which situation does diffraction occur?

- A** A wave bounces back from a surface.
- B** A wave passes from one medium into another.
- C** A wave passes through a gap in a barrier.
- D** Waves from two identical sources are superposed.

245. 9702/12/O/N/22/Q29

Light of a single frequency is incident on a pair of narrow slits that are a distance of 0.10 mm apart. A series of bright and dark fringes is observed on a screen a distance of 2.0 m away. The distance between adjacent bright fringes is 8.0 mm.



What is the path difference of the light waves from the two slits that meet at the second-order **dark** fringe?

- A** $2.0 \times 10^{-7} \text{ m}$
- B** $4.0 \times 10^{-7} \text{ m}$
- C** $6.0 \times 10^{-7} \text{ m}$
- D** $8.0 \times 10^{-7} \text{ m}$

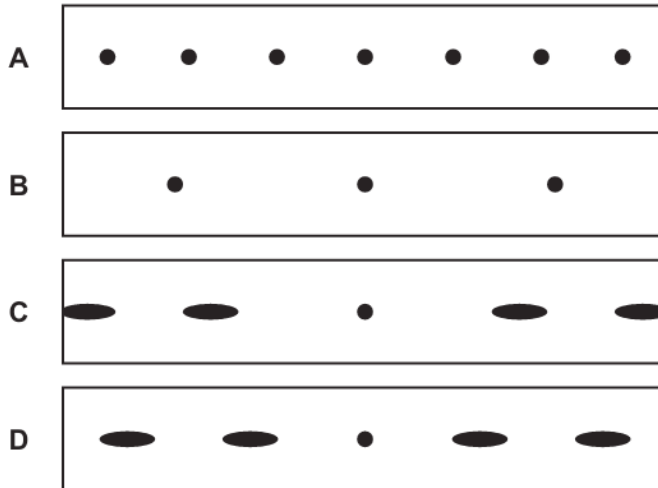
246. 9702/12/O/N/22/Q30

Red light of a single wavelength passes through a diffraction grating. Bright dots are formed on a screen, as shown.



The red light is replaced with white light.

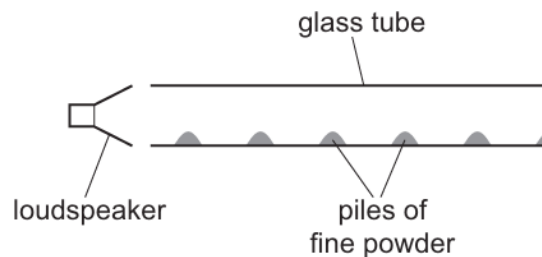
Which diagram, drawn to the same scale, shows a possible pattern of bright light on the screen?



247. 9702/13/O/N/22/Q26

In an experiment to produce a stationary sound wave in air, a fine powder is initially evenly distributed along the length of a horizontal glass tube which is closed at one end.

At the open end of the tube, a loudspeaker emits a sound wave of a constant wavelength. A stationary wave is formed and the powder accumulates in regularly spaced piles, as shown.



Which statement explains the positions of the piles of powder within the tube?

- A** The piles are where the air molecules vibrate with maximum amplitude.
- B** The piles are where the air molecules vibrate with minimum amplitude.
- C** The piles are where the air molecules vibrate with the highest frequency.
- D** The piles are where the air molecules vibrate vertically.

248. 9702/13/O/N/22/Q25

Two progressive waves overlap.

What is an essential condition for the two waves to form a stationary wave?

- A The waves are longitudinal.
- B The waves are polarised.
- C The waves travel in opposite directions.
- D The waves travel in the same direction.

249. 9702/13/O/N/22/Q27

A sound wave of wavelength 0.50 m passes through a doorway of width 1.0 m between two rooms.

Which change increases the amount of diffraction that takes place?

- A Double the amplitude of the sound wave.
- B Double the width of the doorway.
- C Halve the frequency of the sound wave.
- D Halve the period of the sound wave.

250. 9702/13/O/N/22/Q29

Which property of a light wave can be determined using a diffraction grating?

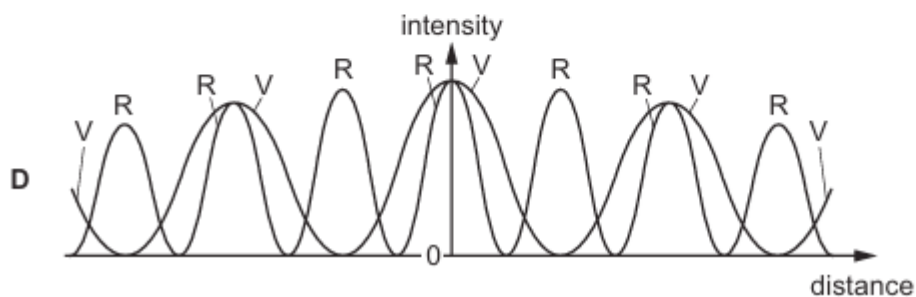
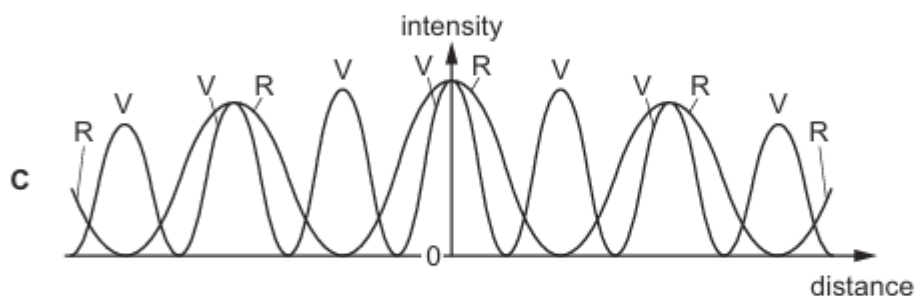
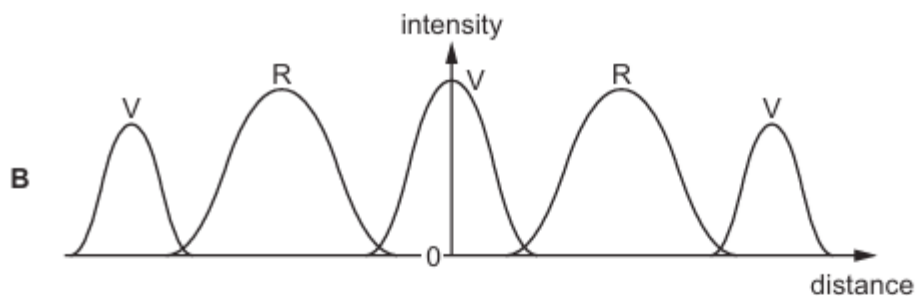
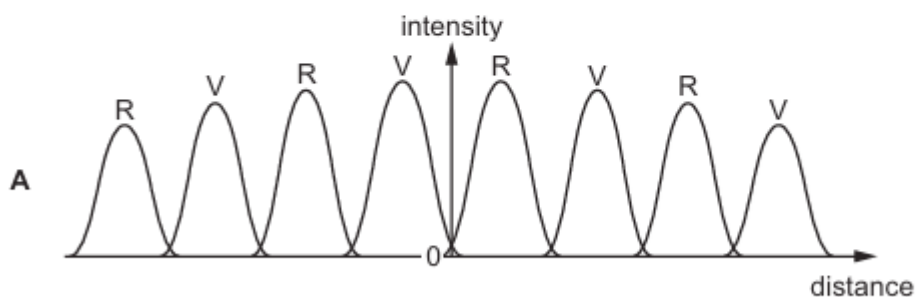
- A amplitude
- B intensity
- C speed
- D wavelength

251. 9702/13/O/N/22/Q28

A light source consists of a vertical slit illuminated by red light (R) and violet light (V). The wavelength of R is approximately twice the wavelength of V. A parallel vertical double slit is placed nearby. A white screen is placed so that fringes are formed on it.



Which graph best represents the interference fringes formed on the screen?



252. 9702/12/F/M/23/Q26

Two waves of equal frequency and amplitude are travelling in opposite directions along a stretched string. When they meet, they form a stationary wave with three nodes and two antinodes.

The frequency of both waves is doubled and a new stationary wave is formed.

How many antinodes are there in the new stationary wave?

- A** 1 **B** 2 **C** 3 **D** 4

253. 9702/12/F/M/23/Q27

A transmitting mast sends out microwaves of wavelength 1.5 cm and radio waves of wavelength 1.5 km.



A receiving aerial behind a mountain can detect the radio waves but not the microwaves.

What is the reason for this?

- A** The radio waves are coherent but the microwaves are not.
B The radio waves are diffracted around the mountain but the microwaves are not.
C The radio waves are reflected by the mountain but the microwaves are not.
D The radio waves travel at the speed of light but the microwaves do not.

254. 9702/12/F/M/23/Q28

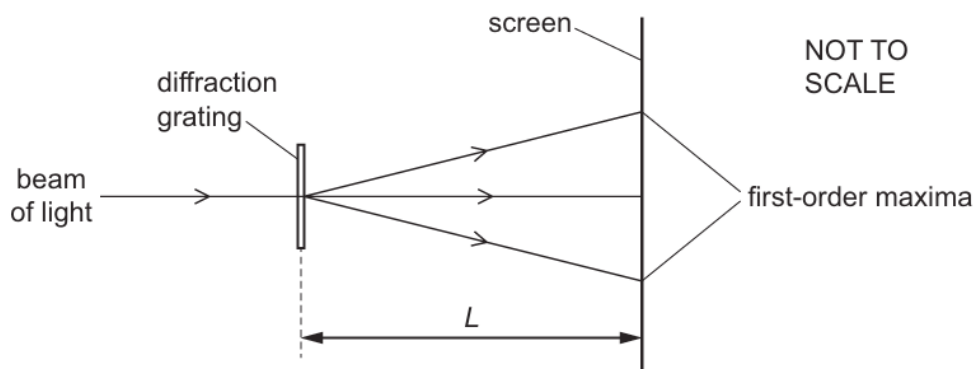
Waves are emitted from two coherent sources.

Which statement about the waves must be correct?

- A** They are in phase.
B They are transverse waves.
C They have a constant phase difference.
D They have the same amplitude.

255. 9702/12/F/M/23/Q29

The diagram shows a screen that is a distance L from a diffraction grating. The grating has a total number of N lines. Any two adjacent lines are a distance d apart. A beam of parallel light of wavelength λ is incident normally on the grating.



Which quantities affect the distance between the first-order diffraction maxima on the screen?

	d	λ	L	N
A	✓	✓	✓	✗
B	✓	✓	✗	✗
C	✓	✗	✓	✓
D	✗	✓	✗	✓

key

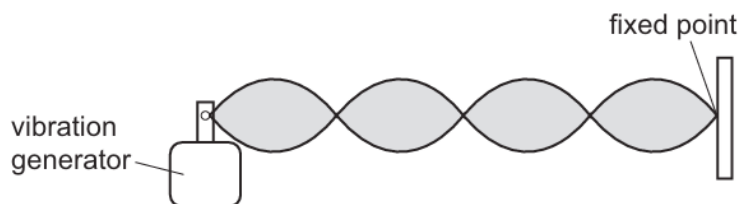
✓ = affects the distance

✗ = does not affect the distance

256. 9702/11/M/J/23/Q26

A string is stretched between a vibration generator and a fixed point.

When the vibration generator is vibrating at a frequency f , a stationary wave with five nodes is created on the stretched string, as shown. There is a node at the end of the string that is attached to the vibration generator.



The frequency of vibration of the vibration generator is slowly increased.

What is the next frequency that produces a stationary wave on the string?

A $1.25f$

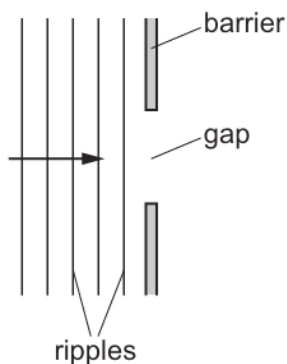
B $1.50f$

C $1.75f$

D $2.00f$

257. 9702/11/M/J/23/Q27

A ripple tank and a barrier with a single gap are used to demonstrate the diffraction of surface ripples on water. Initially, the wavelength of the ripples is five times smaller than the gap in the barrier.



Which change increases the amount of diffraction observed?

- A** double the amplitude of the ripples
- B** double the width of the gap
- C** halve the frequency of the ripples
- D** halve the wavelength of the ripples

258. 9702/11/M/J/23/Q28

A laser produces a beam of light of wavelength 650 nm. The beam is incident normally on two slits that are a distance of 0.12 mm apart.

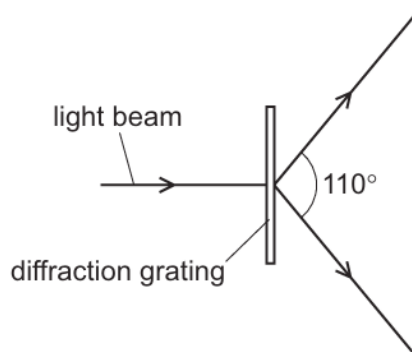
A screen is placed parallel to the slits. The bright interference fringes on the screen have a separation of 7.5 cm.

What is the distance between the screen and the two slits?

- A** 1.4 m
- B** 2.8 m
- C** 7.0 m
- D** 14 m

259. 9702/11/M/J/23/Q29

A beam of light from a laser is incident normally on a diffraction grating.



The diagram shows only the **second-order** maxima that are produced.

The grating has a line spacing of 1.0×10^{-6} m. The angle between the two second-order maxima is 110° .

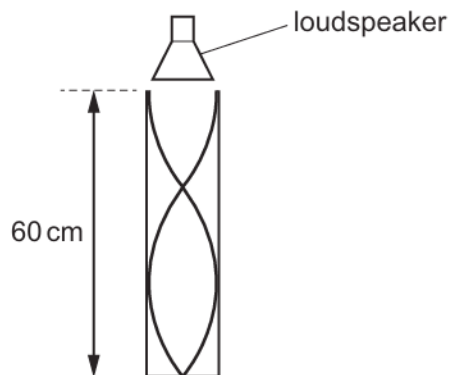
What is the wavelength of the light?

- A** 4.1×10^{-7} m
- B** 4.7×10^{-7} m
- C** 8.2×10^{-7} m
- D** 9.4×10^{-7} m

260. 9702/12/M/J/23/Q26

The sound from a loudspeaker placed above a tube causes resonance of the air in the tube.

A stationary wave is formed with two nodes and two antinodes, as shown.



The tube has height 60 cm.

The speed of sound in the air is 340 m s^{-1} .

What is the frequency of the sound?

- A** 430 Hz
- B** 570 Hz
- C** 850 Hz
- D** 1700 Hz

261. 9702/12/M/J/23/Q25

A stationary wave is formed by two progressive waves travelling in opposite directions along the same line of travel.

Which statement about the two progressive waves is **not** correct?

- A** They must have a constant phase difference.
- B** They must have the same frequency.
- C** They must have the same wavelength.
- D** They must travel at the same speed.

262. 9702/12/M/J/23/Q27

A continuous progressive water wave in a ripple tank passes through a gap in a barrier and diffracts.

The width of the gap is greater than the wavelength of the wave.

Which change causes the wave to spread over a larger angle as it passes through the gap?

- A** a small decrease in the wavelength of the wave
- B** a small decrease in the width of the gap
- C** a small increase in the frequency of the wave
- D** a small increase in the height of the barrier

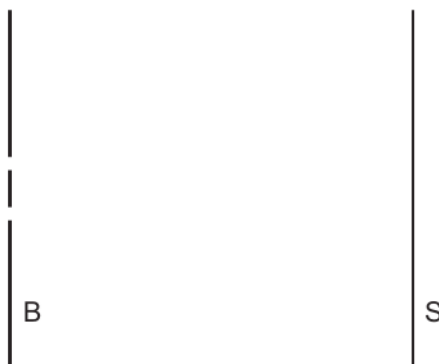
263. 9702/12/M/J/23/Q28

The diagram shows a view from above of a double-slit interference demonstration.

L is a monochromatic light source with a vertical filament. B is a barrier with two narrow vertical slits and S is a screen upon which interference fringes form.

NOT TO
SCALE

*
L



The intensity is I at the point on the screen where the centre of the fringe pattern forms.

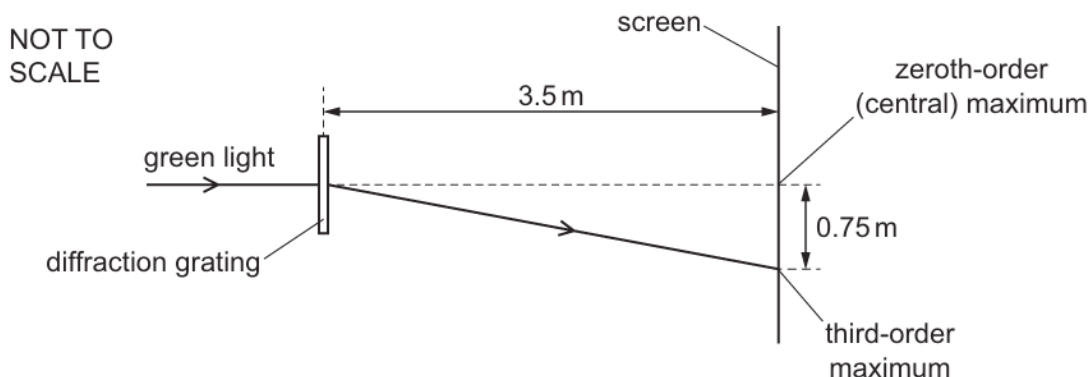
When one of the slits is covered, what is the intensity at the same point on the screen?

- A** $\frac{I}{\sqrt{2}}$
- B** $\frac{I}{2}$
- C** $\frac{I}{2\sqrt{2}}$
- D** $\frac{I}{4}$

264. 9702/12/M/J/23/Q29

Green light of wavelength 550 nm is incident normally on a diffraction grating and produces a diffraction pattern on a screen placed 3.5 m from the diffraction grating.

The **third**-order maximum on the screen is a distance of 0.75 m from the zeroth-order (central) maximum.

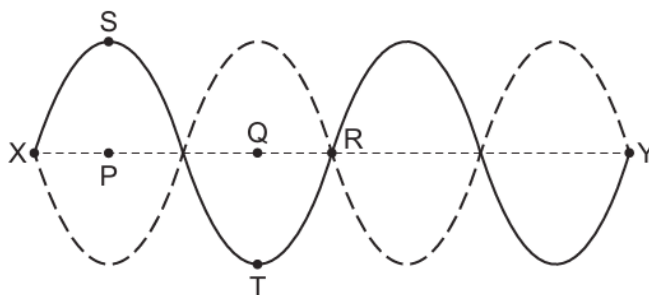


What is the distance between two adjacent slits in the diffraction grating?

- A $2.6 \times 10^{-6} \text{ m}$
- B $7.7 \times 10^{-6} \text{ m}$
- C $7.9 \times 10^{-6} \text{ m}$
- D $1.0 \times 10^{-5} \text{ m}$

265. 9702/13/M/J/23/Q25

The diagram shows a string stretched between fixed points X and Y. There is a stationary wave on the string.



The solid curve shows the string at a position of maximum displacement at time t_0 . The dashed curve shows the other position of maximum displacement. The straight central dashed line shows the mean position of the string.

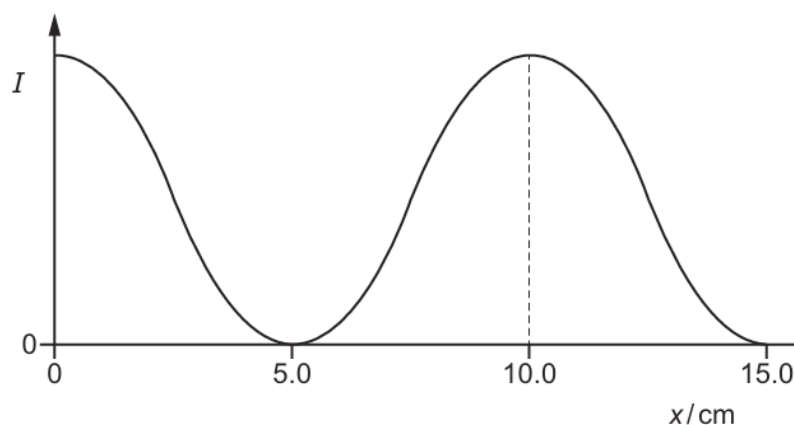
Point S on the string is directly above point P. Point T on the string is directly below point Q.

Which statement is correct?

- A A short time after t_0 , point R on the string will be displaced.
- B A short time after t_0 , points S and T on the string move in opposite directions.
- C The distance between P and Q is one wavelength.
- D Two moving points on the string that are equal distances from point R vibrate in phase.

266. 9702/13/M/J/23/Q26

The variation with distance x of the intensity I along a stationary sound wave in air is shown.



The speed of sound in air is 340 m s^{-1} .

What is the frequency of the sound wave?

- A** 1700 Hz **B** 2300 Hz **C** 3400 Hz **D** 6800 Hz

267. 9702/13/M/J/23/Q27

An experiment is set up to demonstrate the diffraction of water waves in a ripple tank.

The waves pass through a gap of width w and some diffraction of the waves is observed.

The wavelength of the waves is now doubled.

What is the new gap width needed to cause the same amount of diffraction as before?

- A** $0.5w$ **B** w **C** $2w$ **D** $4w$

268. 9702/13/M/J/23/Q28

Two coherent electromagnetic waves are travelling in a vacuum. The two waves meet at a point. At this point, the two waves have different intensities.

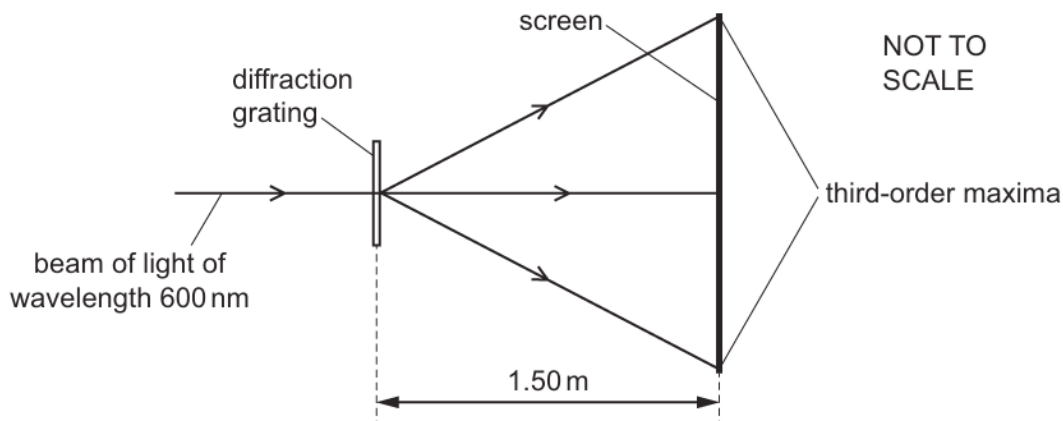
Which statement about the waves is **not** correct?

- A** They have a constant phase difference at the point.
B They have the same amplitude at the point.
C They have the same frequency.
D They travel at the same speed.

269. 9702/13/M/J/23/Q29

A parallel beam of light of wavelength 600 nm is incident normally on a diffraction grating.

The distance between adjacent slits in the grating is 2.0×10^{-6} m. A screen is placed parallel to the grating, at a distance of 1.50 m from the grating. **Third-order** diffraction maxima are observed at the two ends of the screen, as shown.



What is the distance between the two ends of the screen?

- A** 1.4 m **B** 2.7 m **C** 3.1 m **D** 6.2 m

270. 9702/11/O/N/23/Q25

In an experiment, a stationary wave is formed on a string stretched horizontally between two fixed points.

Which statement about the experiment is correct?

- A** At certain times, the string between two nodes is horizontal with all points having zero displacement.
B Each point on the string between two antinodes has an oscillation of the same amplitude.
C The number of nodes is equal to the number of antinodes.
D Two adjacent antinodes oscillate in phase.

271. 9702/11/O/N/23/Q26

A musical organ produces notes by blowing air into a set of pipes that are open at one end and closed at the other.

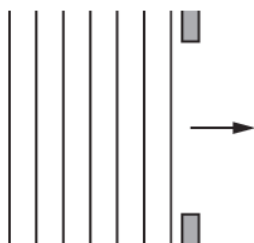
The speed of sound in the air in the pipes is 320 m s^{-1} .

What is the lowest frequency of sound produced by a pipe of length 10 m?

- A** 4 Hz **B** 8 Hz **C** 16 Hz **D** 32 Hz

272. 9702/11/O/N/23/Q27

In an experiment, water waves in a ripple tank are incident on a gap, as shown.



Some diffraction of the water waves is observed.

Which change to the experiment would provide a better demonstration of diffraction?

- A** Increase the amplitude of the waves.
- B** Increase the frequency of the waves.
- C** Increase the wavelength of the waves.
- D** Increase the width of the gap.

273. 9702/11/O/N/23/Q28

Light of wavelength λ is emitted from two point sources R and S and falls onto a distant screen.



At point P on the screen, the light intensity is zero.

What could explain the zero intensity at P?

- A** Light from the two sources is emitted 180° out of phase and the path difference to P is $\frac{1}{2}\lambda$.
- B** Light from the two sources is emitted in phase and the path difference to P is λ .
- C** Light from the two sources is emitted 90° out of phase and the path difference to P is λ .
- D** Light from the two sources is emitted in phase and the path difference to P is $\frac{1}{2}\lambda$.

274. 9702/11/O/N/23/Q29

A beam of red light of wavelength 720 nm is incident normally on a diffraction grating and produces a diffraction pattern on a screen placed parallel to the grating.

The beam of red light is replaced with a beam of electromagnetic radiation of wavelength X, which is incident normally on the same diffraction grating.

The third-order maximum for the electromagnetic radiation of wavelength X is at the same position on the screen as the second-order maximum for the red light.

What is wavelength X?

- A** 480 nm
- B** 540 nm
- C** 960 nm
- D** 1100 nm

275. 9702/12/O/N/23/Q25

What can explain how stationary waves are formed from progressive waves?

- A diffraction
- B polarisation
- C superposition
- D the Doppler effect

276. 9702/12/O/N/23/Q26

A pipe has a length of 2.0 m. It is open at one end and closed at the other end.

A stationary sound wave is set up within the pipe. There are four nodes (N) and four antinodes (A) within the length of the pipe.



What is the wavelength of the sound wave?

- A 0.57 m
- B 1.1 m
- C 1.3 m
- D 1.6 m

277. 9702/12/O/N/23/Q27

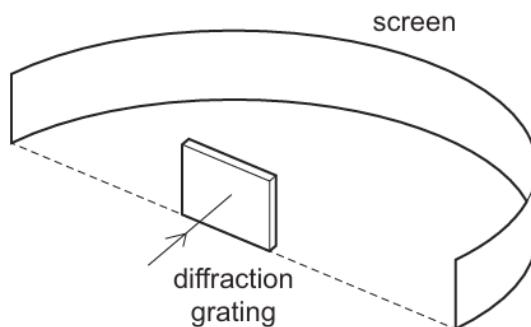
A teacher is explaining diffraction to a group of students.

Which piece of apparatus is most appropriate for the teacher to use to demonstrate diffraction?

- A a long spring
- B a ripple tank
- C a rope
- D a stretched string

278. 9702/12/O/N/23/Q29

Light of wavelength 690 nm passes through a diffraction grating with 300 lines per mm, producing a series of bright spots (maxima) on a screen.



What is the total number of bright spots that are produced?

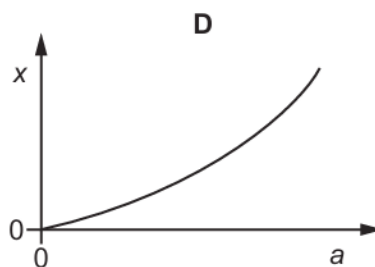
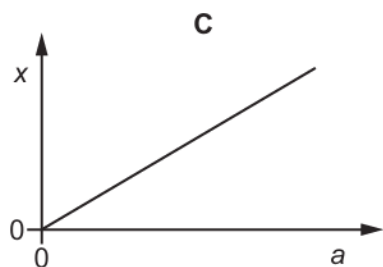
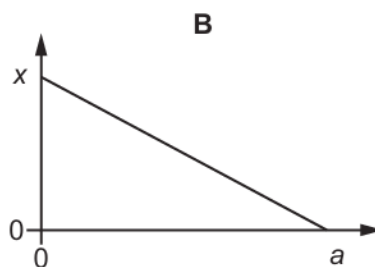
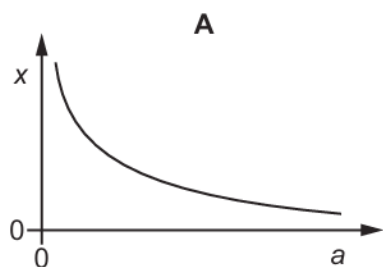
- A 4
- B 5
- C 8
- D 9

279. 9702/12/O/N/23/Q28

Coherent light of constant wavelength is incident normally on a double slit. Interference fringes are formed on a screen that is a fixed distance from the double slit. The screen is parallel to the double slit.

The separation of the slits is varied.

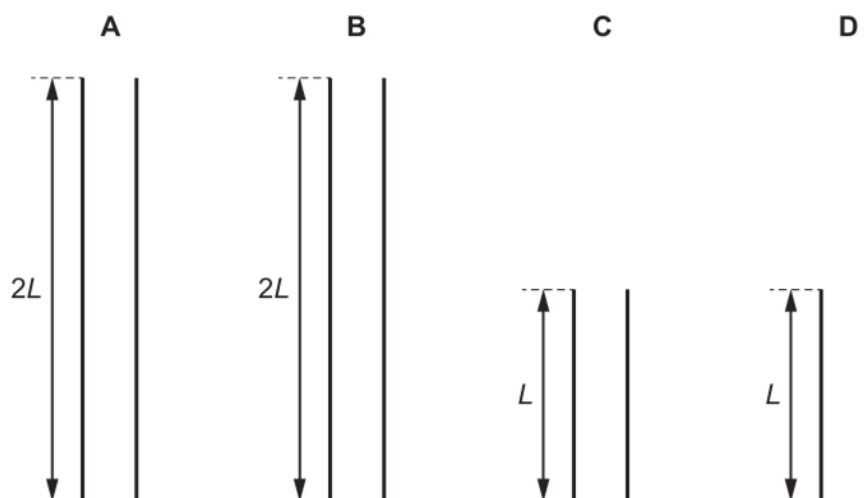
Which graph best shows the variation with slit separation a of the spacing x of the interference fringes?



280. 9702/13/O/N/23/Q26

Stationary sound waves can be formed in the air columns of pipes. One type of pipe is closed at one end and open at the other end. Another type of pipe is open at both ends.

Which pipe can form a stationary sound wave with the lowest frequency?



281. 9702/13/O/N/23/Q25

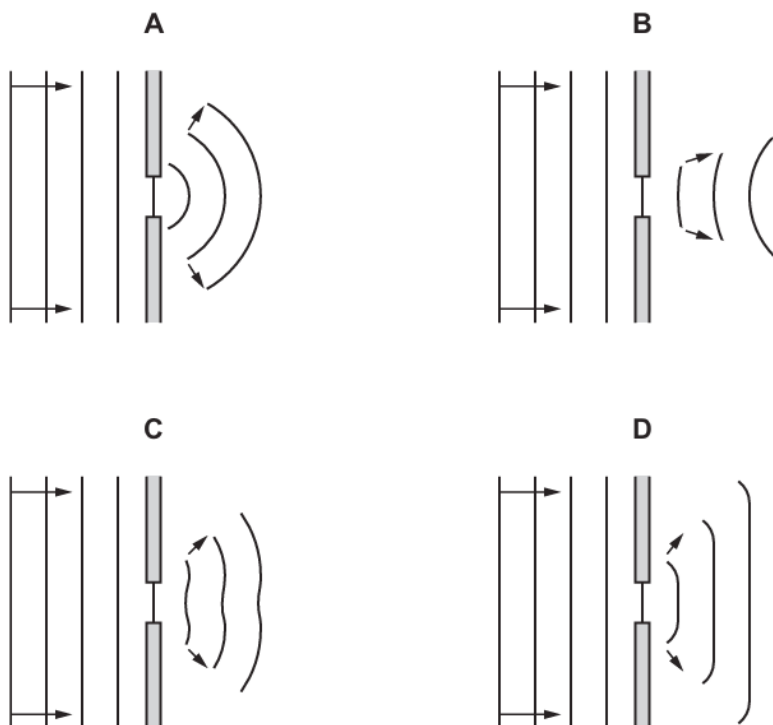
Which statement concerning a stationary wave is correct?

- A** All the particles between two adjacent nodes oscillate in phase.
- B** The amplitude of the stationary wave is equal to the amplitude of one of the waves creating it.
- C** The wavelength of the stationary wave is equal to the separation of two adjacent nodes.
- D** There is no displacement of a particle at an antinode at any time.

282. 9702/13/O/N/23/Q27

Water waves in a ripple tank are made to pass through a small gap, as shown.

Which diagram shows the waves after they have passed through the gap?



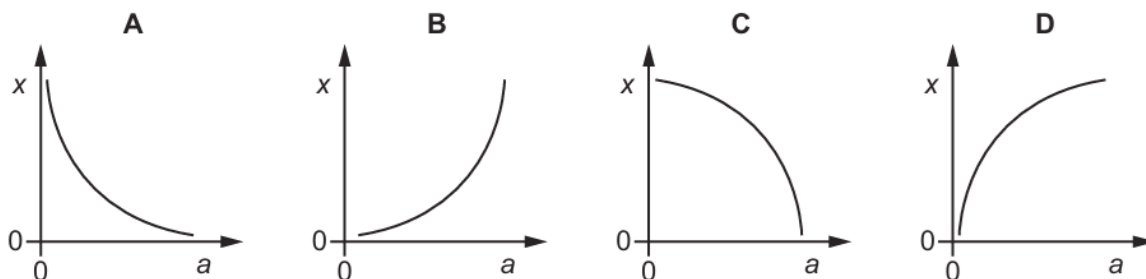
283. 9702/13/O/N/23/Q28

Light of a single wavelength is incident normally on a double slit. The slit separation can be varied.

A screen is placed a fixed distance away from the double slit. The screen and double slit are parallel. A pattern of bright interference fringes is observed on a screen.

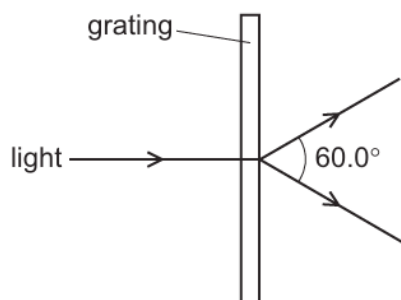


Which graph best shows the variation of the separation x of the bright interference fringes with the slit separation a ?

**284. 9702/13/O/N/23/Q29**

A diffraction grating is used to measure the wavelength of light.

The spacing of the slits in the grating is $1.15 \times 10^{-6} \text{ m}$. The angle between the first-order diffraction maxima is 60.0° , as shown.



What is the wavelength of the light?

- A** 288 nm **B** 498 nm **C** 575 nm **D** 996 nm

ANSWERS

1. C	48. D	95. A	139. C	183. C
2. D	49. A	96. B	140. A	184. A
3. B	50. C	97. C	141. A	185. C
4. D	51. C	98. C	142. B	186. C
5. B	52. A	99. D	143. C	187. B
6. C	53. C	100. A	144. C	188. D
7. C	54. C	101. D	145. D	189. C
8. D	55. B	102. C	146. B	190. D
9. D	56. D	103. B	147. B	191. B
10. C	57. D	104. C	148. A	192. A
11. D	58. D	105. D	149. C	193. B
12. C	59. B	106. C	150. D	194. B
13. B	60. B	107. A	151. C	195. C
14. B	61. A	108. C	152. C	196. B
15. C	62. A	109. C	153. C	197. B
16. A	63. D	110. C	154. B	198. C
17. C	64. B	111. A	155. C	199. C
18. B	65. A	112. D	156. D	200. B
19. C	66. A	113. A	157. C	201. A
20. D	67. B	114. B	158. A	202. B
21. D	68. A	115. A	159. B	203. D
22. A	69. D	116. C	160. B	204. C
23. D	70. D	117. D	161. B	205. A
24. B	71. B	118. C	162. D	206. C
25. B	72. B	119. C	163. B	207. D
26. A	73. D	120. B	164. B	208. D
27. C	74. D	121. C	165. A	209. C
28. D	75. B	122. D	166. A	210. B
29. D	76. B	123. B	167. B	211. A
30. B	77. D	124. D	168. C	212. A
31. A	78. B	125. C	169. D	213. A
32. B	79. C	126. A	170. A	214. A
33. D	80. B	127. D	171. C	215. D
34. C	81. C	128. C	172. C	216. C
35. B	82. B	129. B	173. A	217. B
36. D	83. C	130. A	174. A	218. D
37. B	84. A	131. B	175. A	219. C
38. D	85. B	132. C	176. B	220. C
39. B	86. B	133. D	177. A	221. D
40. C	87. A	134. B	178. B	222. B
41. D	88. A	135. D	179. C	223. B
42. C	89. C	136. B	180. B	224. B
43. A	90. C	137. A	181. B	225. A
44. D	91. D	138. D	182. B	226. B
45. C	92. B			227. D
46. B	93. C			228. B
47. C	94. B			

229. D	276. B
230. D	277. B
231. C	278. D
232. B	279. A
233. C	280. B
234. B	281. A
235. B	282. A
236. D	283. A
237. A	284. C
238. A	
239. B	
240. D	
241. C	
242. D	
243. B	
244. C	
245. C	
246. D	
247. B	
248. C	
249. C	
250. D	
251. C	
252. D	
253. B	
254. C	
255. A	
256. A	
257. C	
258. D	
259. A	
260. A	
261. A	
262. B	
263. D	
264. C	
265. B	
266. A	
267. C	
268. B	
269. D	
270. A	
271. B	
272. C	
273. D	
274. A	
275. C	

- 1 A double-slit interference experiment is set up using coherent red light as illustrated in Fig. 5.1.

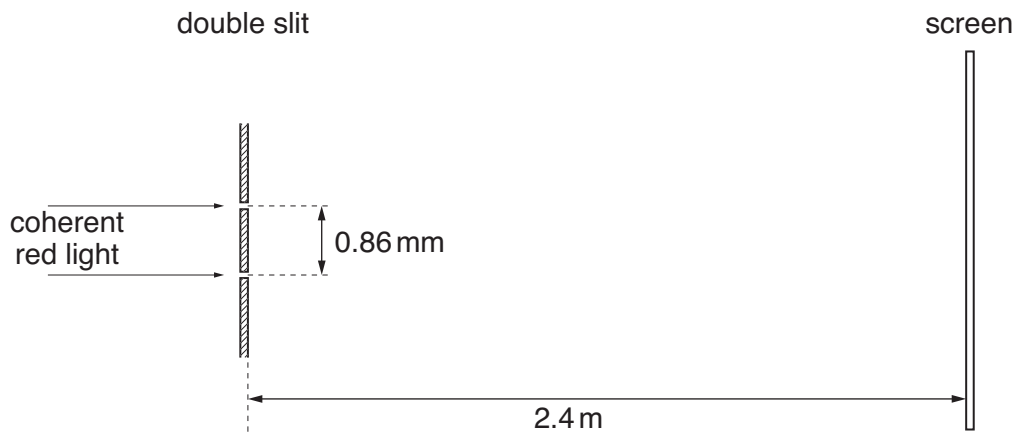


Fig. 5.1 (not to scale)

The separation of the slits is 0.86 mm.
The distance of the screen from the double slit is 2.4 m.
A series of light and dark fringes is observed on the screen.

- (a) State what is meant by *coherent* light.

.....
..... [1]

- (b) Estimate the separation of the dark fringes on the screen.

separation = mm [3]

- (c) Initially, the light passing through each slit has the same intensity.
The intensity of light passing through one slit is now reduced.
Suggest and explain the effect, if any, on the dark fringes observed on the screen.

.....
.....
..... [2]

- 2 (a) State what is meant by the *diffraction* of a wave.

.....

 [2]

- (b) A laser produces a narrow beam of coherent light of wavelength 632 nm. The beam is incident normally on a diffraction grating, as shown in Fig. 4.1.

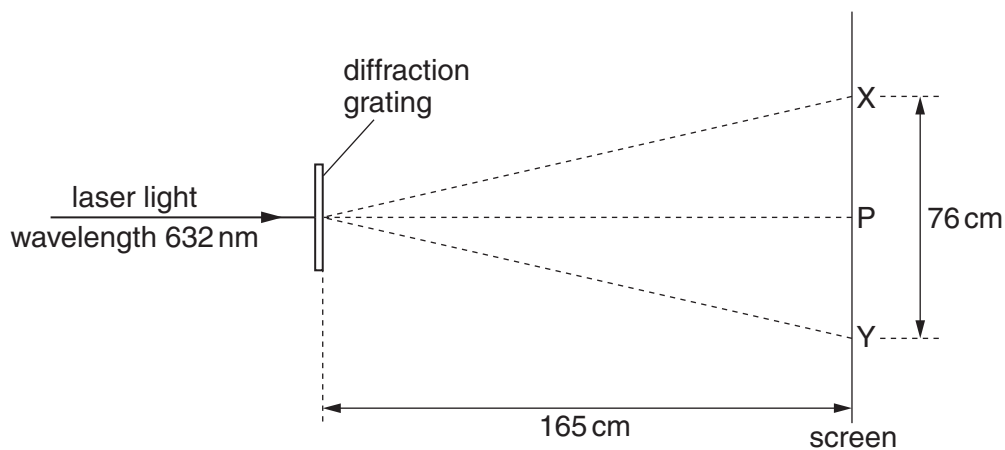


Fig. 4.1

Spots of light are observed on a screen placed parallel to the grating. The distance between the grating and the screen is 165 cm.

The brightest spot is P. The spots formed closest to P and on each side of P are X and Y.

X and Y are separated by a distance of 76 cm.

Calculate the number of lines per metre on the grating.

number per metre = [4]

- (c) The grating in (b) is now rotated about an axis parallel to the incident laser beam, as shown in Fig. 4.2.

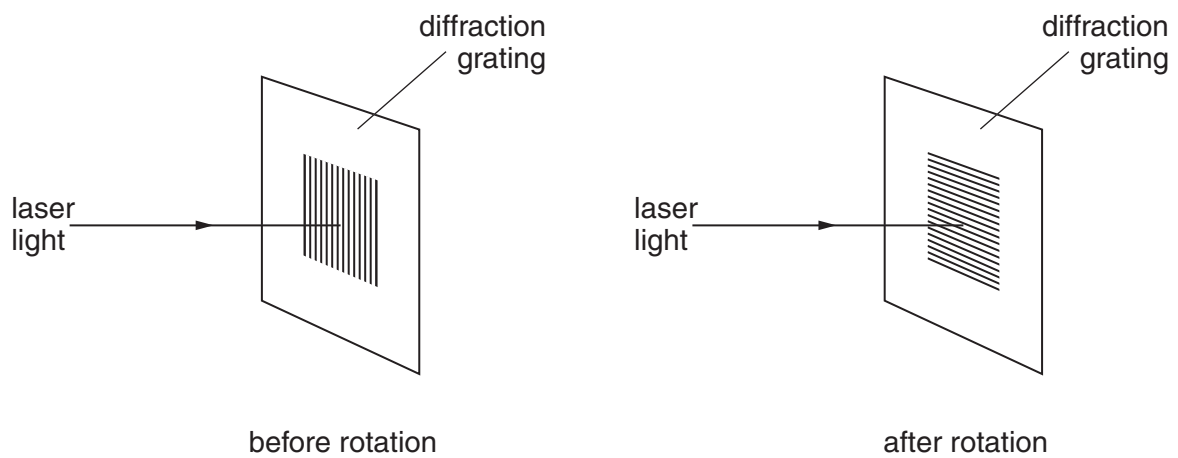


Fig. 4.2

State what effect, if any, this rotation will have on the positions of the spots P, X and Y.

.....

 [2]

- (d) In another experiment using the apparatus in (b), a student notices that the distances XP and PY, as shown in Fig. 4.1, are not equal. Suggest a reason for this difference.

.....
 [1]

- 3 (a) State what is meant by the *diffraction* of a wave.

.....

.....

..... [2]

- (b) Plane wavefronts are incident on a slit, as shown in Fig. 5.1.

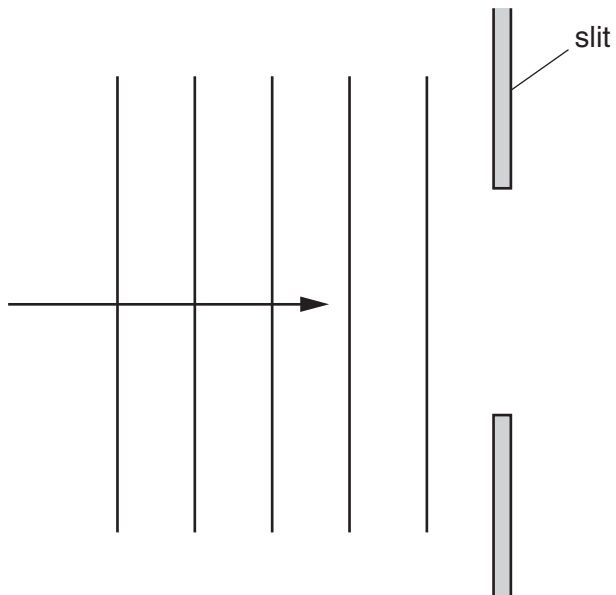


Fig. 5.1

Complete Fig. 5.1 to show four wavefronts that have emerged from the slit.

[2]

- (c) Monochromatic light is incident normally on a diffraction grating having 650 lines per millimetre, as shown in Fig. 5.2.

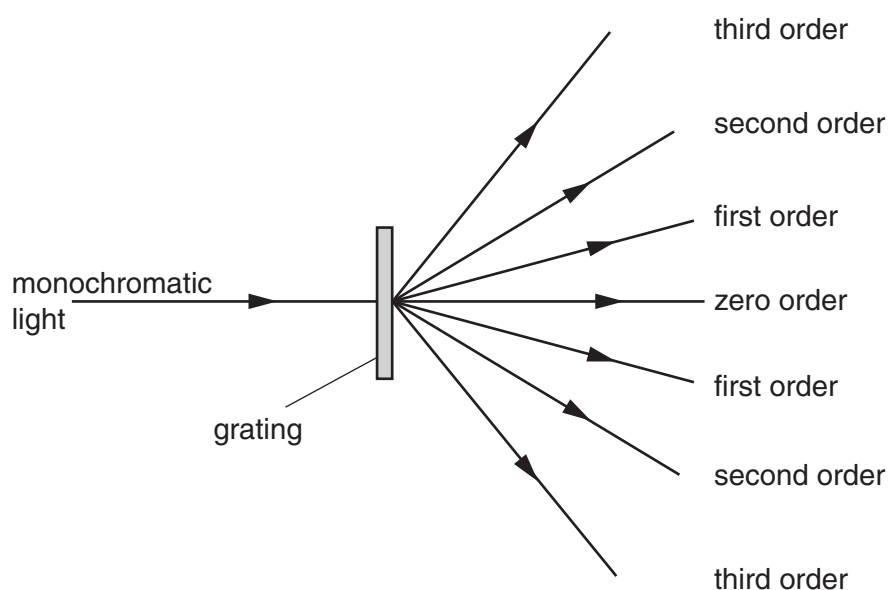


Fig. 5.2

An image (the zero order) is observed for light that has an angle of diffraction equal to zero.

For incident light of wavelength 590 nm, determine the number of orders of diffracted light that can be observed on each side of the zero order.

number = [3]

- (d) The images in Fig. 5.2 are viewed, starting with the zero order and then with increasing order number.
State how the appearance of the images changes as the order number increases.

.....
..... [1]

4 (a) Explain the term *interference*.

9702/21/M/J/11/Q7

.....
.....
..... [1]

(b) A ripple tank is used to demonstrate interference between water waves.

Describe

(i) the apparatus used to produce two sources of coherent waves that have circular wavefronts,

.....
.....
.....
..... [2]

(ii) how the pattern of interfering waves may be observed.

.....
.....
.....
..... [2]

(c) A wave pattern produced in (b) is shown in Fig. 7.1.

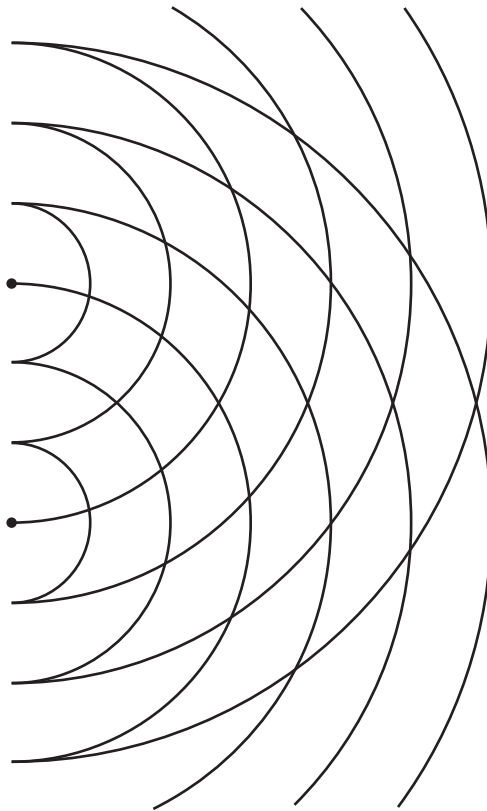


Fig. 7.1

Solid lines on Fig. 7.1 represent crests.

On Fig. 7.1,

- (i) draw two lines to show where maxima would be seen (label each of these lines with the letter X),

[1]

- (ii) draw one line to show where minima would be seen (label this line with the letter N).

[1]

- 5 (a) Apparatus used to produce interference fringes is shown in Fig. 6.1. The apparatus is not drawn to scale.

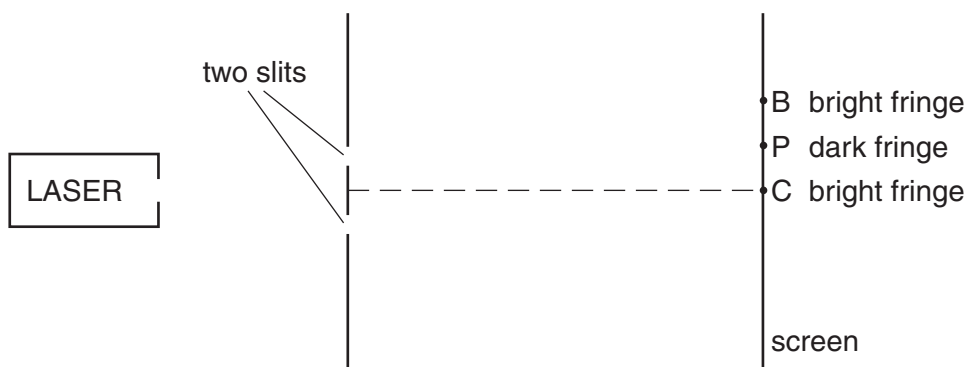


Fig. 6.1 (not to scale)

Laser light is incident on two slits. The laser provides light of a single wavelength. The light from the two slits produces a fringe pattern on the screen. A bright fringe is produced at C and the next bright fringe is at B. A dark fringe is produced at P.

- (i) Explain why one laser and two slits are used, instead of two lasers, to produce a visible fringe pattern on the screen.

.....
 [1]

- (ii) State the phase difference between the waves that meet at

1. B [1]

2. P [1]

- (iii) 1. State the *principle of superposition*.

.....

 [2]

2. Use the principle of superposition to explain the dark fringe at P.

.....
 [1]

- (b) In Fig. 6.1 the distance from the two slits to the screen is 1.8 m. The distance CP is 2.3 mm and the distance between the slits is 0.25 mm. Calculate the wavelength of the light provided by the laser.

wavelength = nm [3]

6 (a) State the *principle of superposition*.

9702/22/O/N/11/Q6

.....
.....
.....[2]

(b) An arrangement that can be used to determine the speed of sound in air is shown in Fig. 6.1.

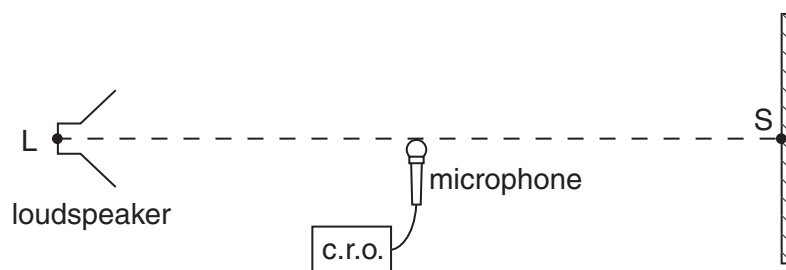


Fig. 6.1

Sound waves of constant frequency are emitted from the loudspeaker L and are reflected from a point S on a hard surface.

The loudspeaker is moved away from S until a stationary wave is produced.

Explain how sound waves from L give rise to a stationary wave between L and S.

.....
.....
.....[2]

(c) A microphone connected to a cathode ray oscilloscope (c.r.o.) is positioned between L and S as shown in Fig. 6.1. The trace obtained on the c.r.o. is shown in Fig. 6.2.

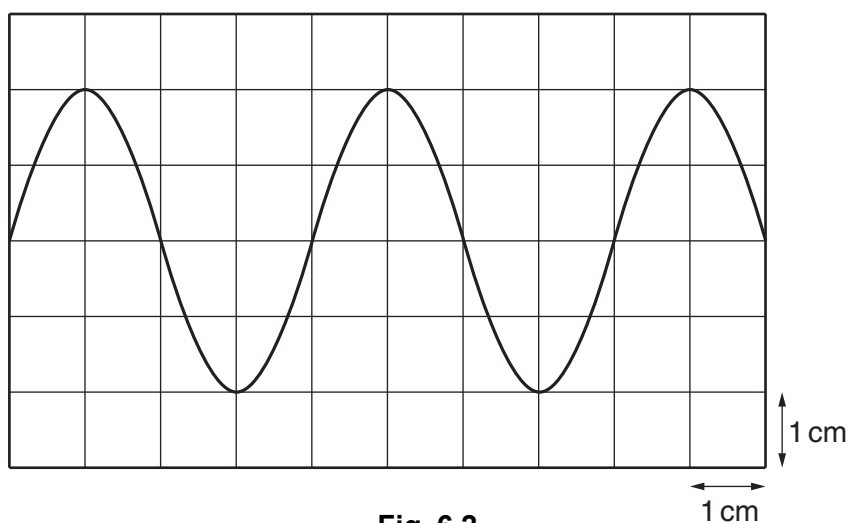


Fig. 6.2

The time-base setting on the c.r.o. is 0.10 ms cm^{-1} .

- (i) Calculate the frequency of the sound wave.

frequency = Hz [2]

- (ii) The microphone is now moved towards S along the line LS. When the microphone is moved 6.7 cm, the trace seen on the c.r.o. varies from a maximum amplitude to a minimum and then back to a maximum.

1. Use the properties of stationary waves to explain these changes in amplitude.

.....

.....

..... [1]

2. Calculate the speed of sound.

speed of sound = ms^{-1} [3]

- 7 (a) A laser is used to produce an interference pattern on a screen, as shown in Fig. 6.1.

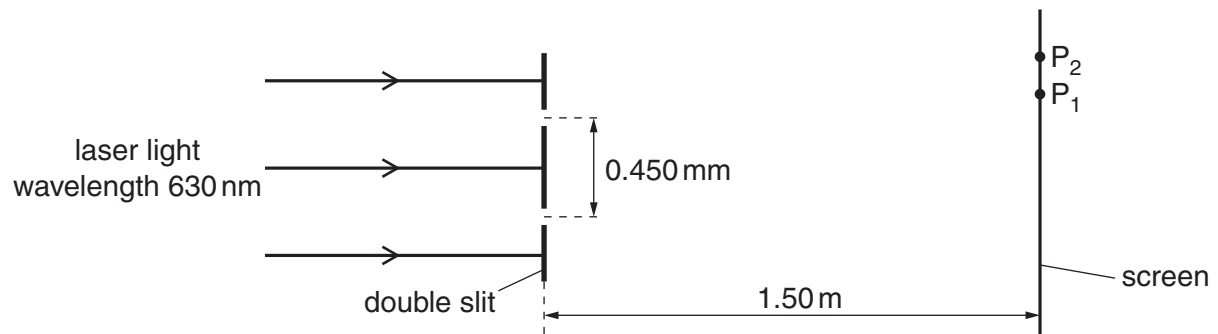


Fig. 6.1 (not to scale)

The laser emits light of wavelength 630 nm. The slit separation is 0.450 mm. The distance between the slits and the screen is 1.50 m. A maximum is formed at P₁ and a minimum is formed at P₂.

Interference fringes are observed only when the light from the slits is coherent.

- (i) Explain what is meant by *coherence*.

.....

 [2]

- (ii) Explain how an interference maximum is formed at P₁.

.....
 [1]

- (iii) Explain how an interference minimum is formed at P₂.

.....
 [1]

- (iv) Calculate the fringe separation.

fringe separation = m [3]

- (b) State the effects, if any, on the fringes when the amplitude of the waves incident on the double slits is increased.

.....
.....
.....
.....[3]

9702/23/M/J/12/Q6

- 8 (a) Monochromatic light is diffracted by a diffraction grating. By reference to this, explain what is meant by

- (i) diffraction,

.....
.....
.....[2]

- (ii) coherence,

.....
.....[1]

- (iii) superposition.

.....
.....[1]

- (b) A parallel beam of red light of wavelength 630 nm is incident normally on a diffraction grating of 450 lines per millimetre.

Calculate the number of diffraction orders produced.

number of orders = [3]

- (c) The red light in (b) is replaced with blue light. State and explain the effect on the diffraction pattern.

.....
.....
.....
.....[3]

- 9 (a) Describe the diffraction of monochromatic light as it passes through a diffraction grating.

.....
 [2]

- (b) White light is incident on a diffraction grating, as shown in Fig. 4.1.

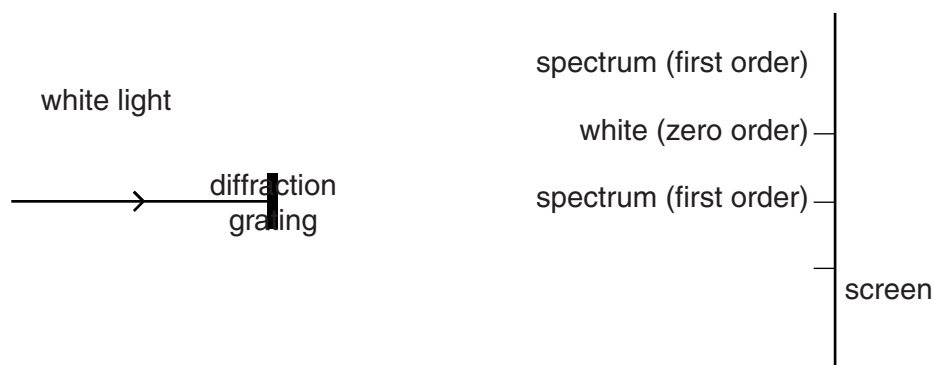


Fig. 4.1 (not to scale)

The diffraction pattern formed on the screen has white light, called zero order, and coloured spectra in other orders.

- (i) Describe how the principle of superposition is used to explain

1. white light at the zero order,

.....
 [2]

2. the difference in position of red and blue light in the first-order spectrum.

.....
 [2]

- (ii) Light of wavelength 625 nm produces a second-order maximum at an angle of 61.0° to the incident direction.

Determine the number of lines per metre of the diffraction grating.

number of lines = m^{-1} [2]

- (iii) Calculate the wavelength of another part of the visible spectrum that gives a maximum for a different order at the same angle as in (ii).

wavelength =nm [2]

9702/21/M/J/13/Q5

- 10 (a)** State three conditions required for maxima to be formed in an interference pattern produced by two sources of microwaves.

1.

.....

2.

.....

3.

.....

[3]

- (b)** A microwave source M emits microwaves of frequency 12 GHz. Show that the wavelength of the microwaves is 0.025 m.

[3]

- (c) Two slits S_1 and S_2 are placed in front of the microwave source M described in (b), as shown in Fig 5.1.

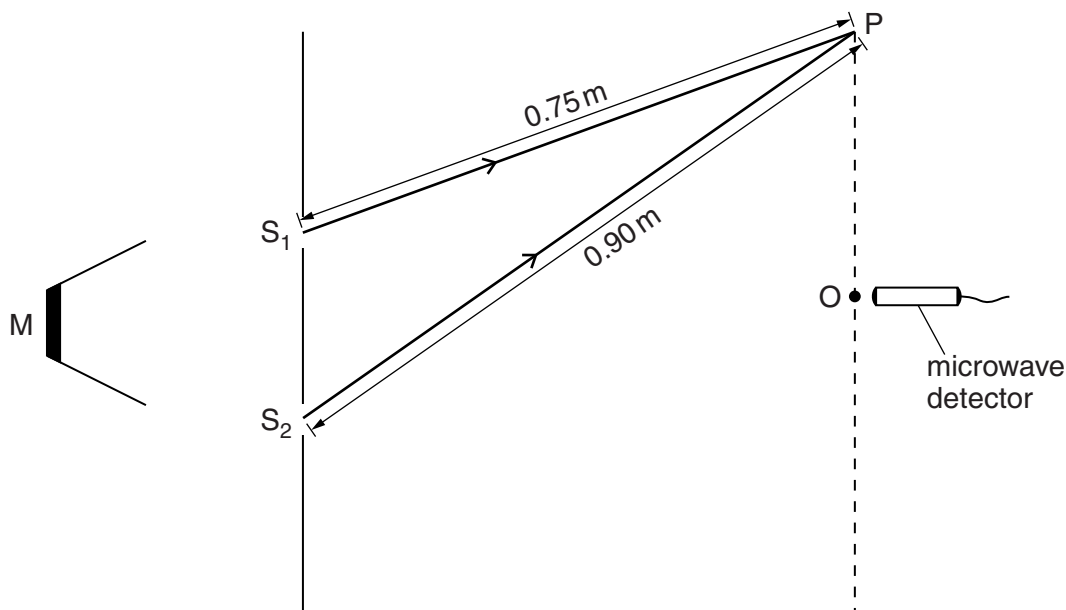


Fig. 5.1 (not to scale)

The distances S_1O and S_2O are equal. A microwave detector is moved from O to P. The distance S_1P is 0.75 m and the distance S_2P is 0.90 m.

The microwave detector gives a maximum reading at O.

State the variation in the readings on the microwave detector as it is moved slowly along the line from O to P.

.....

.....

.....

.....

[3]

.....

- (d) The microwave source M is replaced by a source of coherent light.

State two changes that must be made to the slits in Fig. 5.1 in order to observe an interference pattern.

1.
 2.
- [2]

- 11 (a) Explain the principle of superposition.

.....

 [2]

- (b) Sound waves travel from a source S to a point X along two paths SX and SPX, as shown in Fig. 5.1.

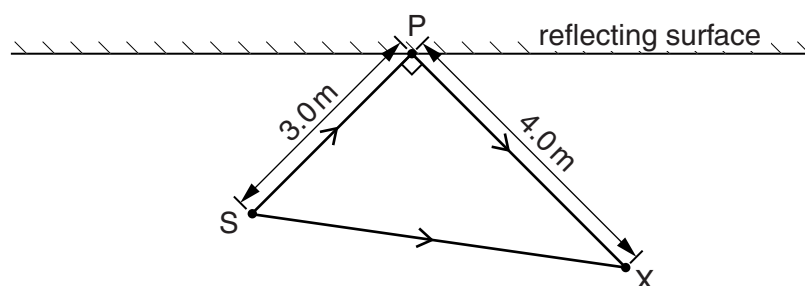


Fig. 5.1

- (i) State the phase difference between these waves at X for this to be the position of
1. a minimum,
 phase difference = unit [1]
 2. a maximum.
 phase difference = unit [1]
- (ii) The frequency of the sound from S is 400 Hz and the speed of sound is 320 m s^{-1} . Calculate the wavelength of the sound waves.

wavelength = m [2]

- (iii) The distance SP is 3.0 m and the distance PX is 4.0 m. The angle SPX is 90° . Suggest whether a maximum or a minimum is detected at point X. Explain your answer.

.....
 [2]

12 (a) A laser is placed in front of a double slit, as shown in Fig. 7.1.

9702/22/M/J/14/Q7

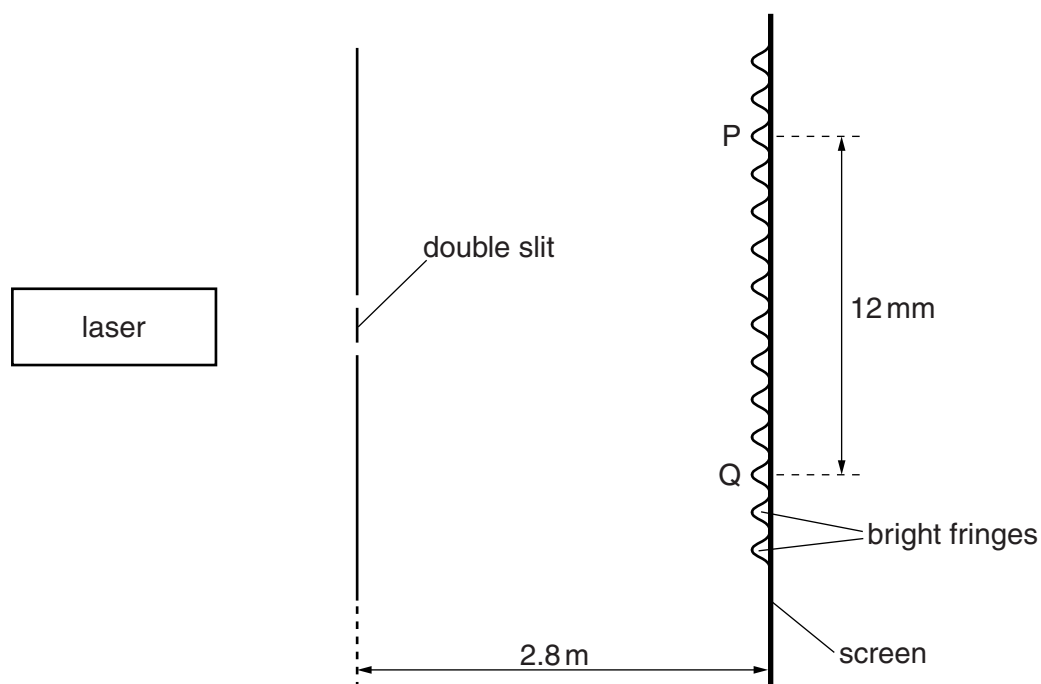


Fig. 7.1 (not to scale)

The laser emits light of frequency 670 THz. Interference fringes are observed on the screen.

(a) Explain how the interference fringes are formed.

.....

.....

.....

.....

.....

..... [3]

(b) Show that the wavelength of the light is 450 nm.

- (c) The separation of the maxima P and Q observed on the screen is 12mm.
The distance between the double slit and the screen is 2.8m.

Calculate the separation of the two slits.

separation = m [3]

- (d) The laser is replaced by a laser emitting red light. State and explain the effect on the interference fringes seen on the screen.

.....

.....

.....

..... [2]

- 13 (a) State one difference and one similarity between longitudinal and transverse waves.

difference:

.....

similarity:

.....

[2]

- (b) A laser is placed in front of two slits as shown in Fig. 6.1.

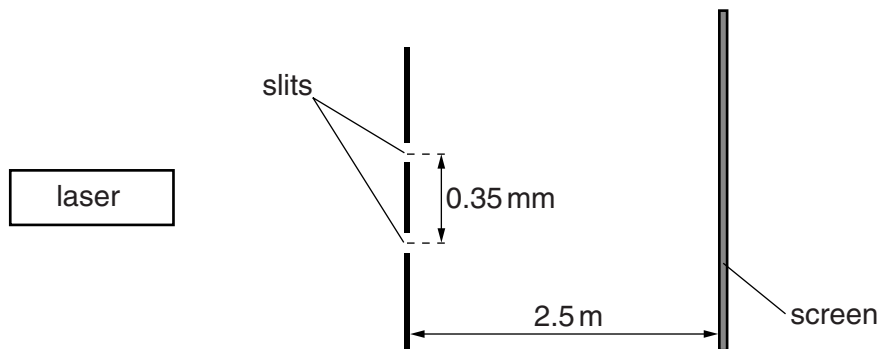


Fig. 6.1 (not to scale)

The laser emits light of wavelength 6.3×10^{-7} m.

The distance from the slits to the screen is 2.5 m. The separation of the slits is 0.35 mm.

An interference pattern of maxima and minima is observed on the screen.

- (i) Explain why an interference pattern is observed on the screen.

.....

.....

.....

.....[2]

- (ii) Calculate the distance between adjacent maxima.

distance =m [2]

- (c) State and explain the effect, if any, on the distance between adjacent maxima when the laser is replaced by another laser emitting ultra-violet radiation.

.....

.....[1]

14 (a) State what is meant by *diffraction* and by *interference*.

9702/21/M/J/15/Q6

diffraction:

.....

interference:

.....[3]

(b) Light from a source S_1 is incident on a diffraction grating, as illustrated in Fig. 6.1.

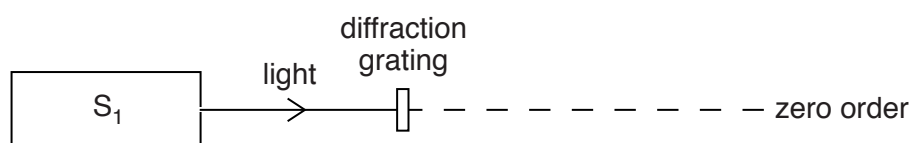


Fig. 6.1 (not to scale)

The light has a single frequency of 7.06×10^{14} Hz. The diffraction grating has 650 lines per millimetre.

Calculate the number of orders of diffracted light produced by the grating. Do not include the zero order.
Show your working.

number = [3]

(c) A second source S_2 is used in place of S_1 . The light from S_2 has a single frequency lower than that of the light from S_1 .

State and explain whether more orders are seen with the light from S_2 .

.....

.....[1]

- 15 (a) Two overlapping waves of the same type travel in the same direction. The variation with distance x of the displacement y of each wave is shown in Fig. 6.1.

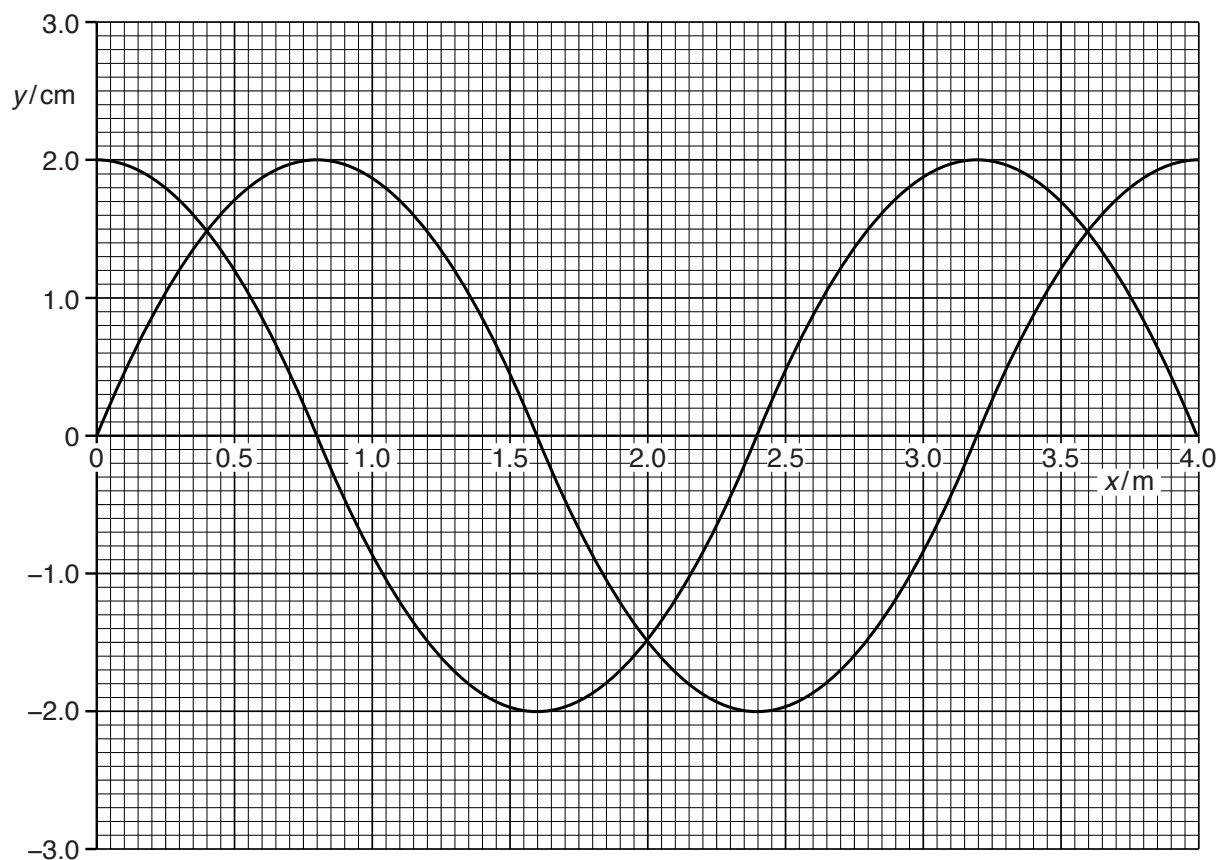


Fig. 6.1

The speed of the waves is 240 m s^{-1} . The waves are coherent and produce an interference pattern.

- (i) Explain the meaning of *coherence* and *interference*.

coherence:

.....

interference:

.....

[2]

- (ii) Use Fig. 6.1 to determine the frequency of the waves.

frequency = Hz [2]

(iii) State the phase difference between the waves.

phase difference = ° [1]

(iv) Use the principle of superposition to sketch, on Fig. 6.1, the resultant wave. [2]

(b) An interference pattern is produced with the arrangement shown in Fig. 6.2.

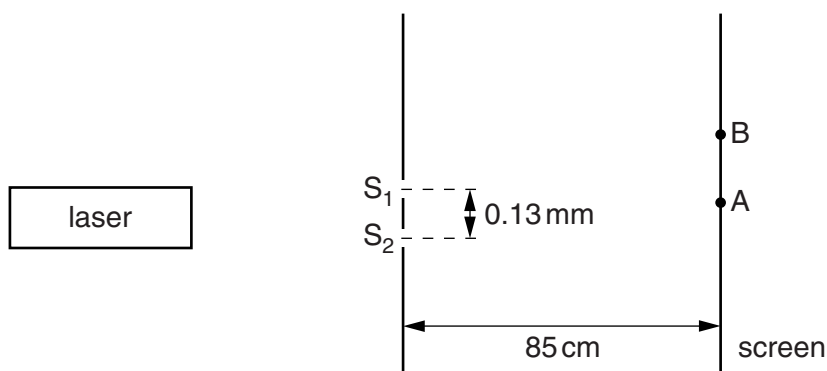


Fig. 6.2 (not to scale)

Laser light of wavelength λ of 546 nm is incident on the slits S_1 and S_2 . The slits are a distance 0.13 mm apart. The distance between the slits and the screen is 85 cm .

Two points on the screen are labelled A and B. The path difference between S_1A and S_2A is zero. The path difference between S_1B and S_2B is 2.5λ . Maxima and minima of intensity of light are produced on the screen.

(i) Calculate the distance AB.

distance = m [3]

(ii) The laser is replaced by a laser emitting blue light. State and explain the change in the distance between the maxima observed on the screen.

.....

 [1]

- 16 (a) (i) By reference to the direction of propagation of energy, state what is meant by a *transverse* wave.

.....
 [1]

- (ii) State the principle of superposition.

.....

 [2]

- (b) Circular water waves may be produced by vibrating dippers at points P and Q, as illustrated in Fig. 4.1.

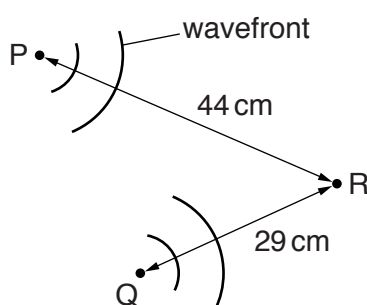


Fig. 4.1 (not to scale)

The waves from P alone have the same amplitude at point R as the waves from Q alone. Distance PR is 44 cm and distance QR is 29 cm.

The dippers vibrate in phase with a period of 1.5 s to produce waves of speed 4.0 cm s^{-1} .

- (i) Determine the wavelength of the waves.

wavelength = cm [2]

- 17 A double-slit interference experiment is used to determine the wavelength of light emitted from a laser, as shown in Fig. 5.2.

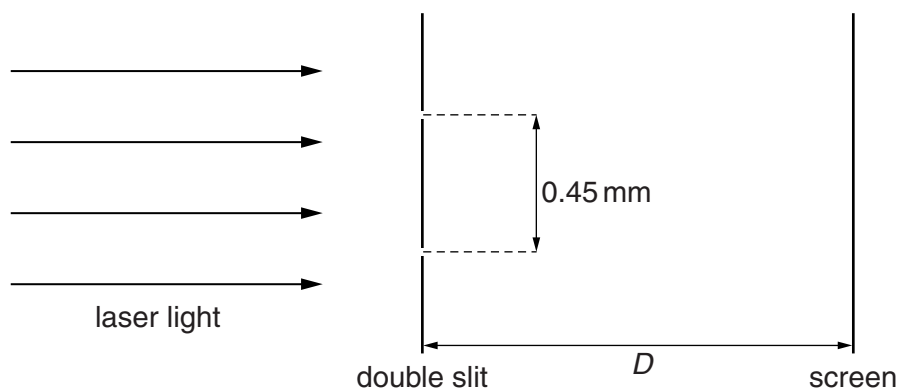


Fig. 5.2 (not to scale)

The separation of the slits is 0.45 mm. The fringes are viewed on a screen at a distance D from the double slit.

The fringe width x is measured for different distances D . The variation with D of x is shown in Fig. 5.3.

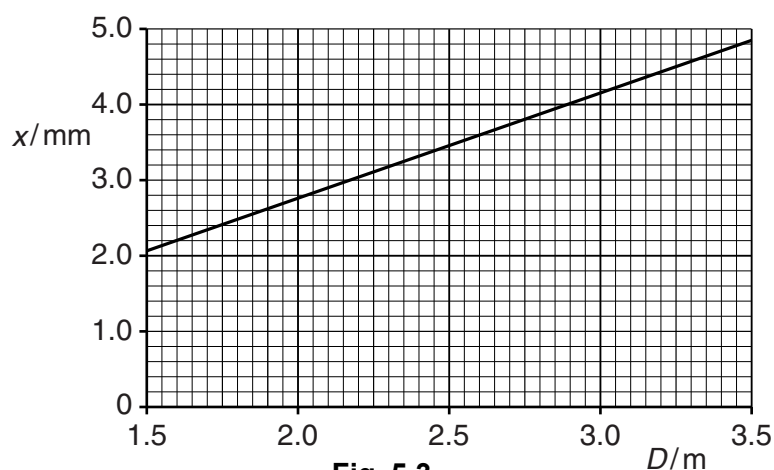


Fig. 5.3

- (i) Use the gradient of the line in Fig. 5.3 to determine the wavelength, in nm, of the laser light.

wavelength = nm [4]

- (ii) The separation of the slits is increased. State and explain the effects, if any, on the graph of Fig. 5.3.

.....
 [2]

- 18 (a) Light of a single wavelength is incident on a diffraction grating. Explain the part played by *diffraction* and *interference* in the production of the first order maximum by the diffraction grating.

diffraction:

.....

interference:

.....

.....

.....

[3]

- (b) The diffraction grating illustrated in Fig. 5.1 is used with light of wavelength 486 nm.

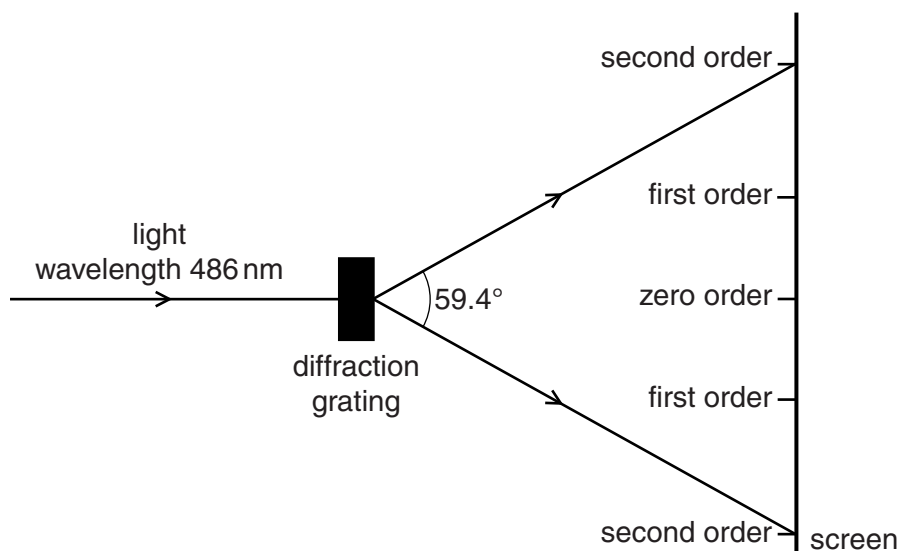


Fig. 5.1 (not to scale)

The orders of the maxima produced are shown on the screen in Fig. 5.1. The angle between the two second order maxima is 59.4° .

Calculate the number of lines per millimetre of the grating.

number of lines per millimetre = mm^{-1} [3]

- 19 (a) State what is meant by the *diffraction* of a wave.

.....
.....[2]

- (b) Laser light of wavelength 500 nm is incident normally on a diffraction grating. The resulting diffraction pattern has diffraction maxima up to and including the fourth-order maximum.

Calculate, for the diffraction grating, the minimum possible line spacing.

line spacing = m [3]

- (c) The light in (b) is now replaced with red light. State and explain whether this is likely to result in the formation of a fifth-order diffraction maximum.

.....
.....
.....
.....[2]

20 (a) State what is meant by the *diffraction* of a wave.

9702/22/O/N/16/Q4

.....
.....[2]

(b) An arrangement for demonstrating the interference of light is shown in Fig. 4.1.

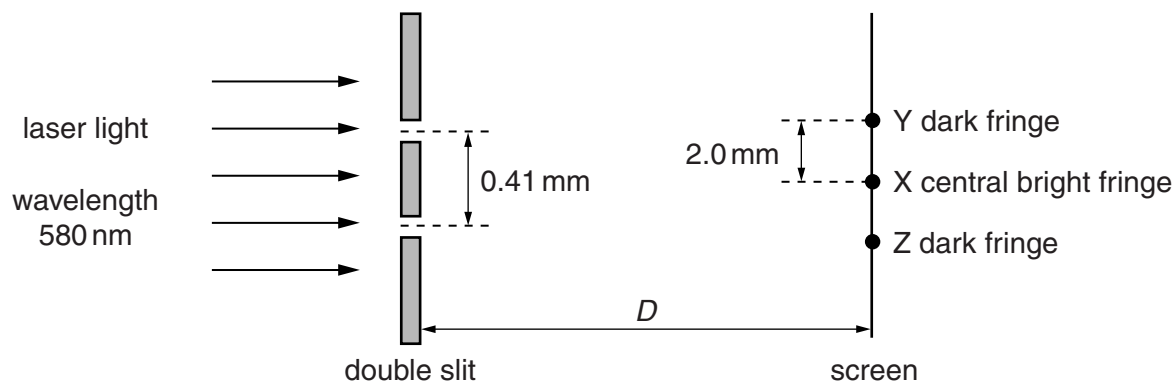


Fig. 4.1 (not to scale)

The wavelength of the light from the laser is 580 nm. The separation of the slits is 0.41 mm. The perpendicular distance between the double slit and the screen is D .

Coherent light emerges from the slits and an interference pattern is observed on the screen. The central bright fringe is produced at point X. The closest dark fringes to point X are produced at points Y and Z. The distance XY is 2.0 mm.

(i) Explain why a bright fringe is produced at point X.

.....
.....
.....
.....[2]

(ii) State the difference in the distances, in nm, from each slit to point Y.

distance = nm [1]

(iii) Calculate the distance D .

$D =$ m [3]

- (iv) The intensity of the light passing through the two slits was initially the same. The intensity of the light through **one** of the slits is now reduced. Compare the appearance of the fringes before and after the change of intensity.

.....

.....

.....

.....[2]

- 21 (a) Interference fringes may be observed using a light-emitting laser to illuminate a double slit. The double slit acts as two sources of light.

Explain

- (i) the part played by diffraction in the production of the fringes,

.....
[2]

- (ii) the reason why a double slit is used rather than two separate sources of light.

.....
[1]

- (b) A laser emitting light of a single wavelength is used to illuminate slits S_1 and S_2 , as shown in Fig. 6.1.

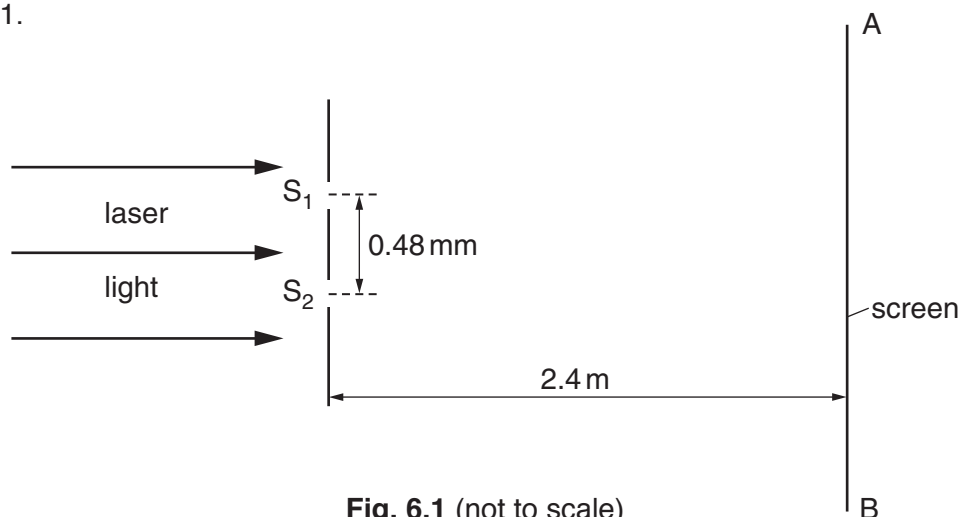


Fig. 6.1 (not to scale)

An interference pattern is observed on the screen AB. The separation of the slits is 0.48 mm . The slits are 2.4 m from AB. The distance on the screen across 16 fringes is 36 mm , as illustrated in Fig. 6.2.

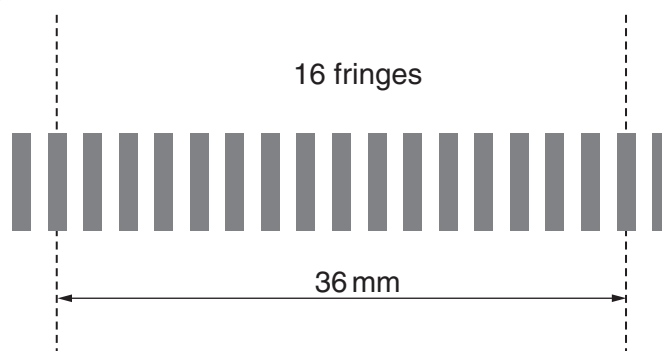


Fig. 6.2

Calculate the wavelength of the light emitted by the laser.

wavelength =m [3]

- (c) Two dippers D_1 and D_2 are used to produce identical waves on the surface of water, as illustrated in Fig. 6.3.

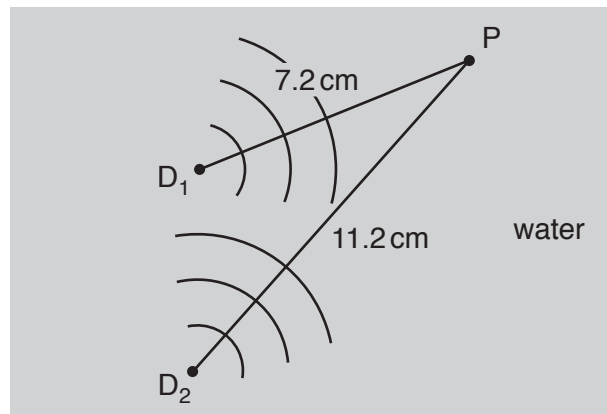


Fig. 6.3 (not to scale)

Point P is 7.2 cm from D_1 and 11.2 cm from D_2 .

The wavelength of the waves is 1.6 cm. The phase difference between the waves produced at D_1 and D_2 is zero.

- (i) State and explain what is observed at P.

.....

[2]

- (ii) State and explain the effect on the answer to (c)(i) if the apparatus is changed so that, separately,

1. the phase difference between the waves at D_1 and at D_2 is 180° ,

.....

2. the intensity of the wave from D_1 is less than the intensity of that from D_2 .

.....

- 22 (a)** When monochromatic light is incident normally on a diffraction grating, the emergent light waves have been diffracted and are coherent.

Explain what is meant by

- (i)** *diffracted waves*,

.....
[1]

- (ii)** *coherent waves*.

.....
[1]

- (b)** Light consisting of only two wavelengths λ_1 and λ_2 is incident normally on a diffraction grating.

The third order diffraction maximum of the light of wavelength λ_1 and the fourth order diffraction maximum of the light of wavelength λ_2 are at the same angle θ to the direction of the incident light.

- (i)** Show that the ratio $\frac{\lambda_2}{\lambda_1}$ is 0.75. Explain your working.

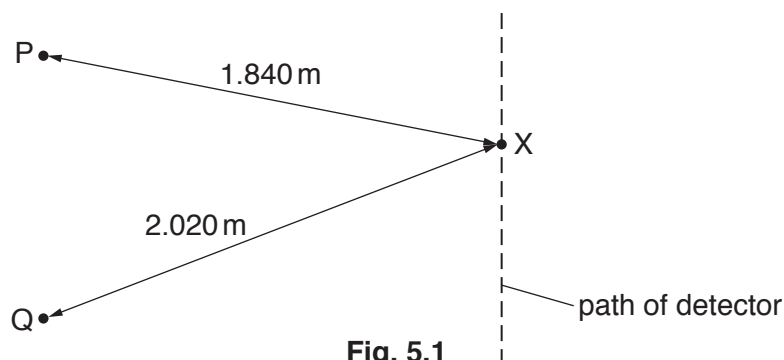
- (ii)** The difference between the two wavelengths is 170 nm.
 Determine wavelength λ_1 .

$\lambda_1 =$ nm [1]

23 (a) State the relationship between the intensity and the amplitude of a wave. 9702/23/M/J/18/Q5

.....
[1]

- (b) Microwaves of the same amplitude and wavelength are emitted in phase from two sources P and Q. The sources are arranged as shown in Fig. 5.1.



A microwave detector is moved along a path that is parallel to the line joining P and Q. A series of intensity maxima and intensity minima are detected.

When the detector is at a point X, the distance PX is 1.840 m and the distance QX is 2.020 m. The microwaves have a wavelength of 6.0 cm.

- (i) Calculate the frequency of the microwaves.

frequency = Hz [2]

- (ii) Describe and explain the intensity of the microwaves detected at X.

.....

[3]

- (iii) Describe the effect on the interference pattern along the path of the detector due to each of the following separate changes.

1. The wavelength of the microwaves decreases.

.....

2. The phase difference between the microwaves emitted from the sources changes to 180° .

.....
[2]

24 (a) State the principle of superposition.

9702/21/O/N/18/Q4

.....
.....
..... [2]

(b) An arrangement for demonstrating the interference of light is shown in Fig. 4.1.

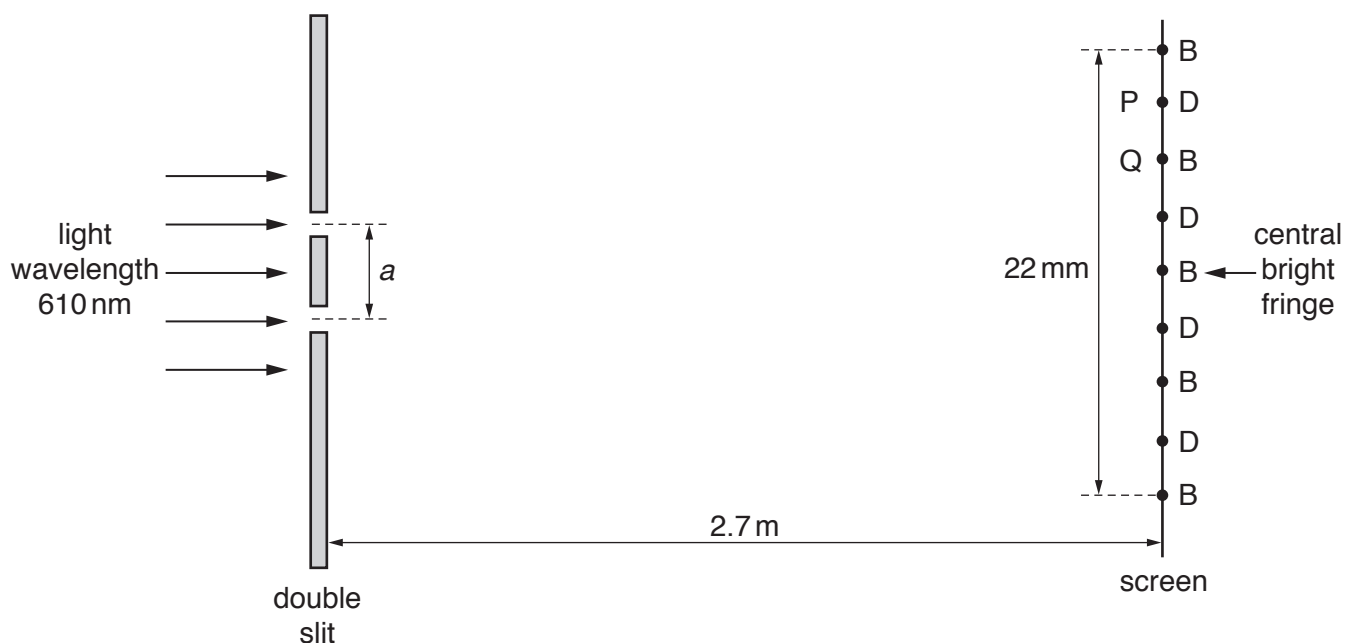


Fig. 4.1 (not to scale)

The wavelength of the light is 610 nm. The distance between the double slit and the screen is 2.7 m.

An interference pattern of bright fringes and dark fringes is observed on the screen. The centres of the bright fringes are labelled B and centres of the dark fringes are labelled D. Point P is the centre of a particular dark fringe and point Q is the centre of a particular bright fringe, as shown in Fig. 4.1. The distance across five bright fringes is 22 mm.

(i) The light waves leaving the two slits are coherent.

State what is meant by *coherent*.

.....
..... [1]

- (ii) 1. State the phase difference between the waves meeting at Q.

phase difference = °

2. Calculate the path difference, in nm, of the waves meeting at P.

path difference = nm
[2]

- (iii) Determine the distance a between the two slits.

a = m [3]

- (iv) A higher frequency of visible light is now used. State and explain the change to the separation of the fringes.

.....
..... [1]

- (v) The intensity of the light incident on the double slit is now increased without altering its frequency. Compare the appearance of the fringes after this change with their appearance before this change.

.....
.....
.....
..... [2]

- 25 Red light of wavelength 640 nm is incident normally on a diffraction grating having a line spacing of 1.7×10^{-6} m, as shown in Fig. 5.1.

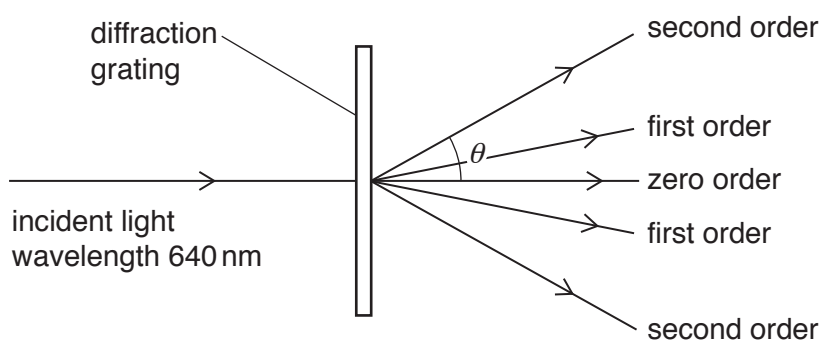


Fig. 5.1 (not to scale)

The second order diffraction maximum of the light is at an angle θ to the direction of the incident light.

- (a) Show that angle θ is 49° .

[3]

- (b) Determine a different wavelength of **visible** light that will also produce a diffraction maximum at an angle of 49° .

wavelength = m [2]

- 26 (a) On Fig. 4.1, complete the two graphs to illustrate what is meant by the amplitude A , the wavelength λ and the period T of a progressive wave.

Ensure that you label the axes of each graph.

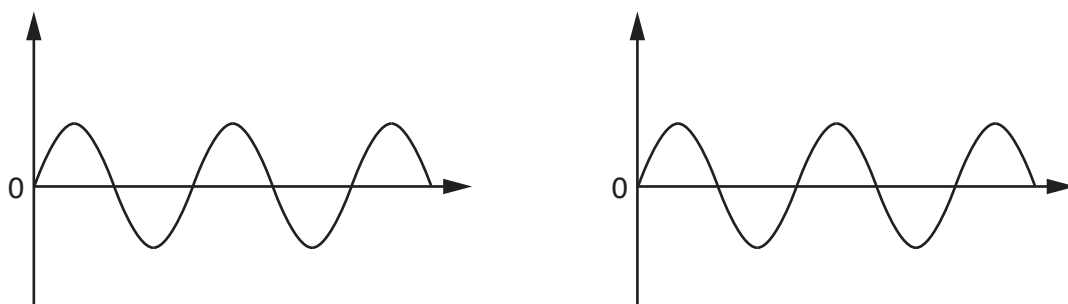


Fig. 4.1

[3]

- (b) A horizontal string is stretched between two fixed points X and Y. A vibrator is used to oscillate the string and produce a stationary wave. Fig. 4.2 shows the string at one instant in time.



Fig. 4.2

The speed of a progressive wave along the string is 30 ms^{-1} . The stationary wave has a period of 40 ms.

- (i) Explain how the stationary wave is formed on the string.

.....

.....

.....

.....[2]

- (ii) A particle on the string oscillates with an amplitude of 13 mm. At time t , the particle has zero displacement.

Calculate

1. the displacement of the particle at time $(t + 100 \text{ ms})$,

displacement = mm

2. the total distance moved by the particle from time t to time $(t + 100 \text{ ms})$.

distance = mm
[3]

- (iii) Determine

1. the frequency of the wave,

frequency = Hz [1]

2. the horizontal distance from X to Y.

distance = m [3]

- 27 (a) A loudspeaker oscillates with frequency f to produce sound waves of wavelength λ . The loudspeaker makes N oscillations in time t .

(i) State expressions, in terms of some or all of the symbols f , λ and N , for:

1. the distance moved by a wavefront in time t

distance =

2. time t .

time t =

[2]

(ii) Use your answers in (i) to deduce the equation relating the speed v of the sound wave to f and λ .

[1]

- (b) The waveform of a sound wave is displayed on the screen of a cathode-ray oscilloscope (c.r.o.), as shown in Fig. 5.1.

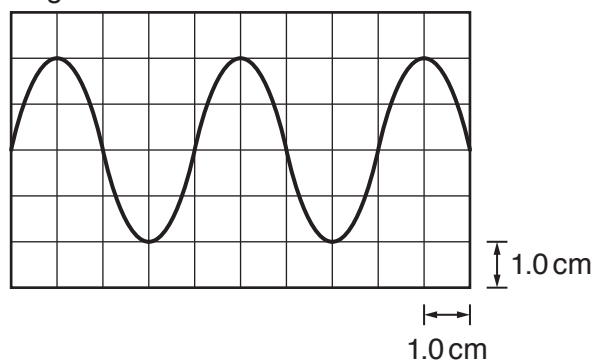


Fig. 5.1

The time-base setting is 0.20 ms cm^{-1} .

Determine the frequency of the sound wave.

frequency = Hz [2]

- (c) Two sources S_1 and S_2 of sound waves are positioned as shown in Fig. 5.2.

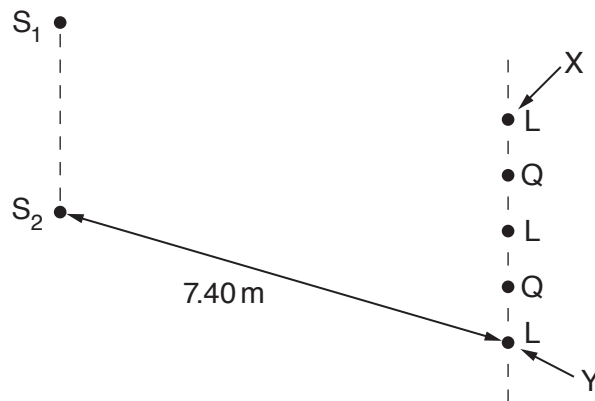


Fig. 5.2 (not to scale)

The sources emit coherent sound waves of wavelength 0.85 m. A sound detector is moved parallel to the line S_1S_2 from a point X to a point Y. Alternate positions of maximum loudness L and minimum loudness Q are detected, as illustrated in Fig. 5.2.

Distance S_1X is equal to distance S_2X . Distance S_2Y is 7.40 m.

- (i) Explain what is meant by *coherent* waves.

.....
[1]

- (ii) State the phase difference between the two waves arriving at the position of minimum loudness Q that is closest to point X.

phase difference = ° [1]

- (iii) Determine the distance S_1Y .

distance = m [2]

28 (a) State Newton's second law of motion.

.....
[1]

(b) A car of mass 850 kg tows a trailer in a straight line along a horizontal road, as shown in Fig. 2.1.

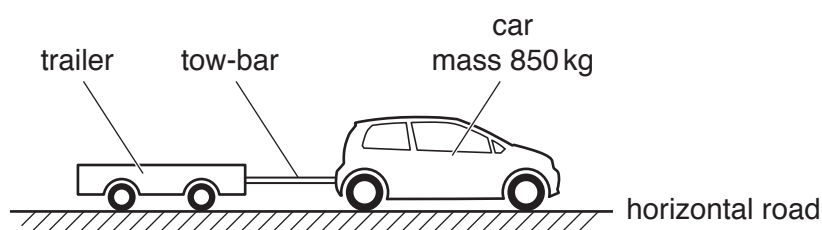


Fig. 2.1

The car and the trailer are connected by a horizontal tow-bar.

The variation with time t of the velocity v of the car for a part of its journey is shown in Fig. 2.2.

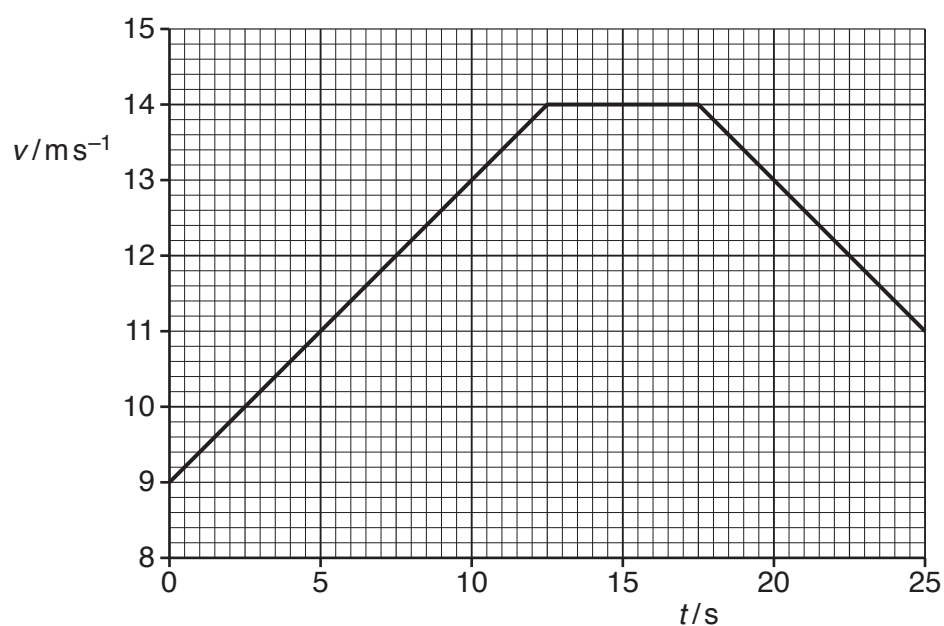


Fig. 2.2

- (i) Calculate the distance travelled by the car from time $t = 0$ to $t = 10$ s.

distance = m [2]

- (ii) At time $t = 10$ s, the resistive force acting on the car due to air resistance and friction is 510 N. The tension in the tow-bar is 440 N.

For the car at time $t = 10$ s:

1. use Fig. 2.2 to calculate the acceleration

acceleration = m s^{-2} [2]

2. use your answer to calculate the resultant force acting on the car

resultant force = N [1]

3. show that a horizontal force of 1300 N is exerted on the car by its engine

[1]

4. determine the useful output power of the engine.

output power = W [2]

- (c) A short time later, the car in (b) is travelling at a constant speed and the tension in the tow-bar is 480 N.

The tow-bar is a solid metal rod that obeys Hooke's law. Some data for the tow-bar are listed below.

Young modulus of metal = 2.2×10^{11} Pa

original length of tow-bar = 0.48 m

cross-sectional area of tow-bar = 3.0×10^{-4} m²

Determine the extension of the tow-bar.

extension = m [3]

- (d) The driver of the car in (b) sees a pedestrian standing directly ahead in the distance. The driver operates the horn of the car from time $t = 15$ s to $t = 17$ s. The frequency of the sound heard by the pedestrian is 480 Hz. The speed of the sound in the air is 340 m s^{-1} .

Use Fig. 2.2 to calculate the frequency of the sound emitted by the horn.

frequency = Hz [2]

29 (a) For a progressive water wave, state what is meant by:

(i) *displacement*

.....
[1]

(ii) *amplitude*.

.....
[1]

(b) Two coherent waves X and Y meet at a point and superpose. The phase difference between the waves at the point is 180° . Wave X has an amplitude of 1.2 cm and intensity I . Wave Y has an amplitude of 3.6 cm.

Calculate, in terms of I , the resultant intensity at the meeting point.

intensity = [2]

(c) (i) Monochromatic light is incident on a diffraction grating. Describe the diffraction of the light waves as they pass through the grating.

.....

[2]

- (ii) A parallel beam of light consists of two wavelengths 540 nm and 630 nm. The light is incident normally on a diffraction grating. Third-order diffraction maxima are produced for each of the two wavelengths. No higher orders are produced for either wavelength.

Determine the smallest possible line spacing d of the diffraction grating.

$$d = \dots\dots\dots \text{ m [3]}$$

- (iii) The beam of light in (c)(ii) is replaced by a beam of blue light incident on the same diffraction grating.

State and explain whether a third-order diffraction maximum is produced for this blue light.

.....
.....
.....[2]

- 30 A vertical tube of length 0.60 m is open at both ends, as shown in Fig. 5.1.

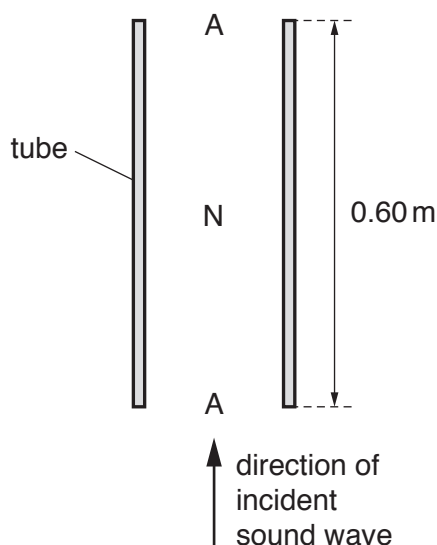


Fig. 5.1

An incident sinusoidal sound wave of a single frequency travels up the tube. A stationary wave is then formed in the air column in the tube with antinodes A at both ends and a node N at the midpoint.

- (a) Explain how the stationary wave is formed from the incident sound wave.

.....

.....

.....

.....[2]

- (b) On Fig. 5.2, sketch a graph to show the variation of the amplitude of the stationary wave with height h above the bottom of the tube.

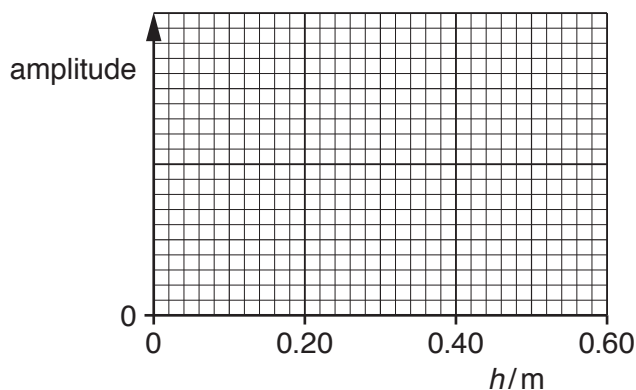


Fig. 5.2

(c) For the stationary wave, state:

- (i)** the direction of the oscillations of an air particle at a height of 0.15 m above the bottom of the tube

.....[1]

- (ii)** the phase difference between the oscillations of a particle at a height of 0.10 m and a particle at a height of 0.20 m above the bottom of the tube.

phase difference = ° [1]

(d) The speed of the sound wave is 340 m s^{-1} .

Calculate the frequency of the sound wave.

frequency = Hz [2]

(e) The frequency of the sound wave is gradually increased.

Determine the frequency of the wave when a stationary wave is next formed.

frequency = Hz [1]

- 31** A ripple tank is used to demonstrate the interference of water waves. Two dippers D1 and D2 produce coherent waves that have circular wavefronts, as illustrated in Fig. 5.1.

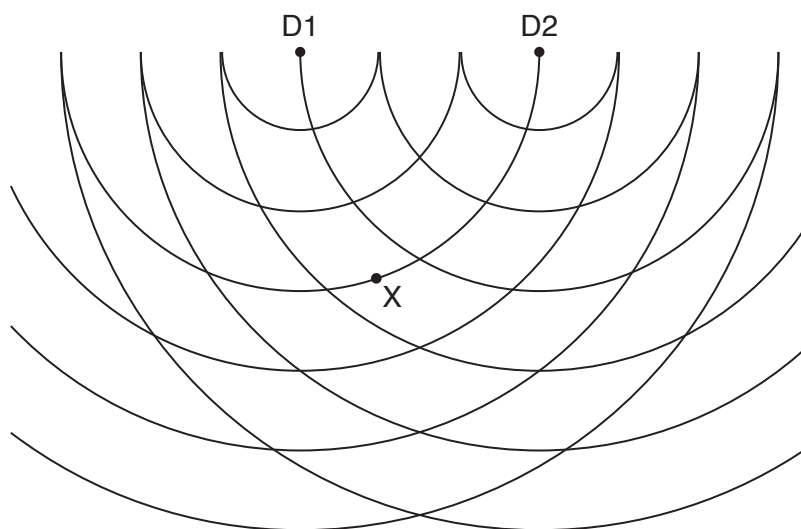


Fig. 5.1

The lines in the diagram represent crests. The waves have a wavelength of 6.0 cm.

- (a)** One condition that is required for an observable interference pattern is that the waves must be coherent.

- (i)** Describe how the apparatus is arranged to ensure that the waves from the dippers are coherent.

.....
 [1]

- (ii)** State one other condition that must be satisfied by the waves in order for the interference pattern to be observable.

.....
 [1]

- (b)** Light from a lamp above the ripple tank shines through the water onto a screen below the tank. Describe one way of seeing the illuminated pattern more clearly.

.....
 [1]

- (c) The speed of the waves is 0.40 m s^{-1} . Calculate the period of the waves.

period = s [2]

- (d) Fig. 5.1 shows a point X that lies on a crest of the wave from D1 and midway between two adjacent crests of the wave from D2.

For the waves at point X, state:

- (i) the path difference, in cm

path difference = cm [1]

- (ii) the phase difference.

phase difference = ° [1]

- (e) On Fig. 5.1, draw **one** line, at least 4 cm long, which joins points where only maxima of the interference pattern are observed. [1]
-

- 32 (a) Light waves emerging from the slits of a diffraction grating are coherent and produce an interference pattern.

Explain what is meant by:

- (i) *coherence*

.....
 [1]

- (ii) *interference*.

.....
 [1]

- (b) A narrow beam of light from a laser is incident normally on a diffraction grating, as shown in Fig. 5.1.

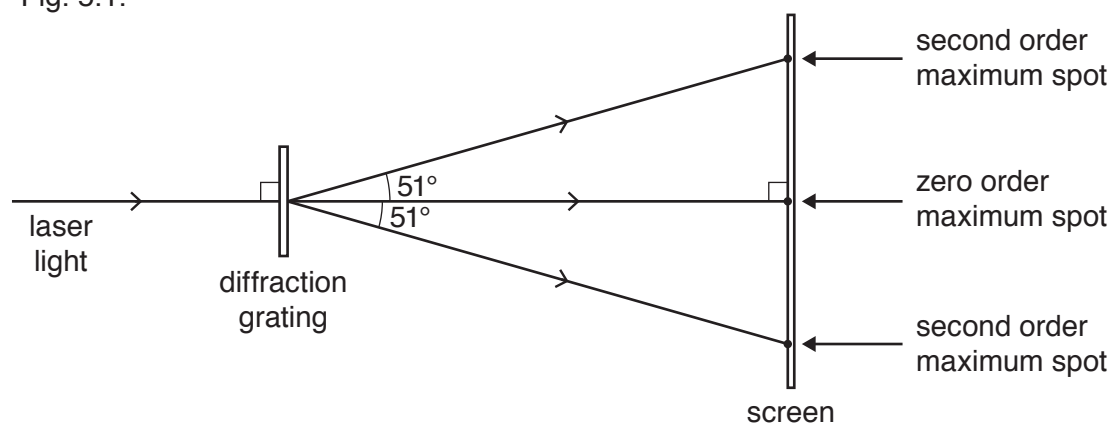


Fig. 5.1 (not to scale)

Spots of light are seen on a screen positioned parallel to the grating. The angle corresponding to each of the **second** order maxima is 51° . The number of lines per unit length on the diffraction grating is $6.7 \times 10^5 \text{ m}^{-1}$.

- (i) Determine the wavelength of the light.

wavelength = m [2]

- (ii) State and explain the change, if any, to the distance between the second order maximum spots on the screen when the light from the laser is replaced by light of a shorter wavelength.

.....
.....
.....[1]

9702/21/M/J/20/Q4

- 33 (a) (i) By reference to the direction of propagation of energy, state what is meant by a *longitudinal wave*.

.....
..... [1]

- (ii) State the principle of superposition.

.....
.....
..... [2]

- (b) The wavelength of light from a laser is determined using the apparatus shown in Fig. 4.1.

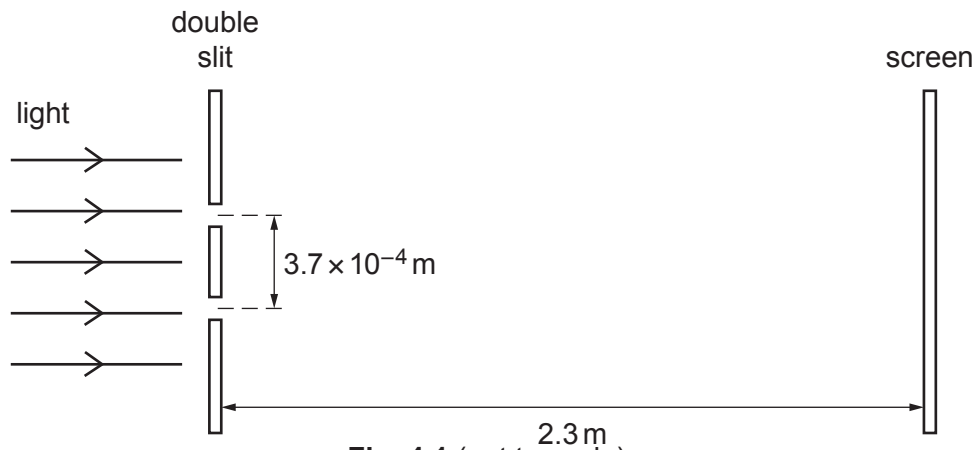


Fig. 4.1 (not to scale)

The light from the laser is incident normally on the plane of the double slit.

The separation of the two slits is $3.7 \times 10^{-4} \text{ m}$. The screen is parallel to the plane of the double slit. The distance between the screen and the double slit is 2.3 m.

A pattern of bright fringes and dark fringes is seen on the screen. The separation of adjacent bright fringes on the screen is $4.3 \times 10^{-3} \text{ m}$.

- (i) Calculate the wavelength, in nm, of the light.

wavelength = nm [3]

- (ii) The intensity of the light passing through each slit was initially the same. The intensity of the light through **one** of the slits is now reduced.

Compare the appearance of the fringes before and after the change of intensity.

.....

 [2]

- 34 (a) State the difference between progressive waves and stationary waves in terms of the transfer of energy along the wave.

.....
 [1]

- (b) A progressive wave travels from left to right along a stretched string. Fig. 4.1 shows part of the string at one instant.

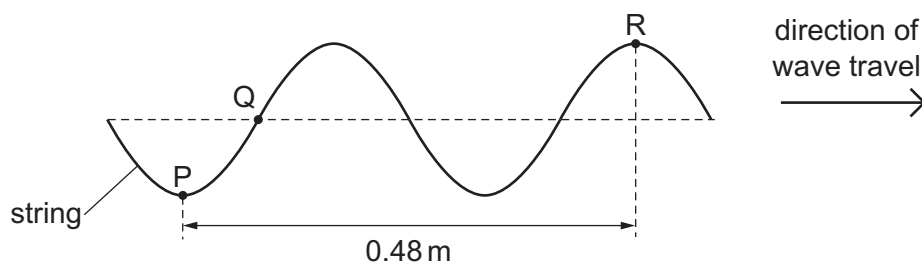


Fig. 4.1

P, Q and R are three different points on the string. The distance between P and R is 0.48 m. The wave has a period of 0.020 s.

- (i) Use Fig. 4.1 to determine the wavelength of the wave.

wavelength = m [1]

- (ii) Calculate the speed of the wave.

speed = ms^{-1} [2]

- (iii) Determine the phase difference between points Q and R.

phase difference = $^{\circ}$ [1]

- (iv) Fig. 4.1 shows the position of the string at time $t = 0$. Describe how the displacement of point Q on the string varies with time from $t = 0$ to $t = 0.010$ s.

.....

.....

.....

..... [2]

- (c) A stationary wave is formed on a different string that is stretched between two fixed points X and Y. Fig. 4.2 shows the position of the string when each point is at its maximum displacement.

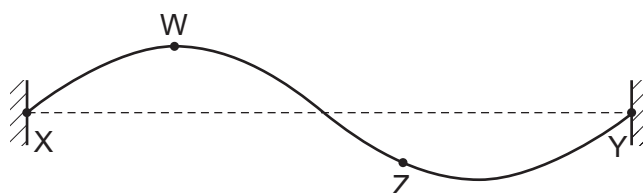


Fig. 4.2

- (i) Explain what is meant by a *node* of a stationary wave.

..... [1]

- (ii) State the number of antinodes of the wave shown in Fig. 4.2.

number = [1]

- (iii) State the phase difference between points W and Z on the string.

phase difference =° [1]

- (iv) A new stationary wave is now formed on the string. The new wave has a frequency that is half of the frequency of the wave shown in Fig. 4.2. The speed of the wave is unchanged.

On Fig. 4.3, draw a position of the string, for this new wave, when each point is at its maximum displacement.

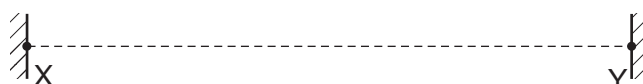


Fig. 4.3

[1]

35. 9702/21O/N/20/Q6

- (a) Describe the conditions required for two waves to be able to form a stationary wave.

.....
.....
.....
..... [2]

- (b) A stationary wave on a string has nodes and antinodes. The distance between a node and an adjacent antinode is 6.0 cm.

- (i) State what is meant by a *node*.

..... [1]

- (ii) Calculate the wavelength of the two waves forming the stationary wave.

wavelength = cm [1]

- (iii) State the phase difference between the particles at two adjacent antinodes of the stationary wave.

phase difference = ° [1]

[Total: 5]

36. 9702/22/O/N/20/Q5

Microwaves with the same wavelength and amplitude are emitted in phase from two sources X and Y, as shown in Fig. 5.1.

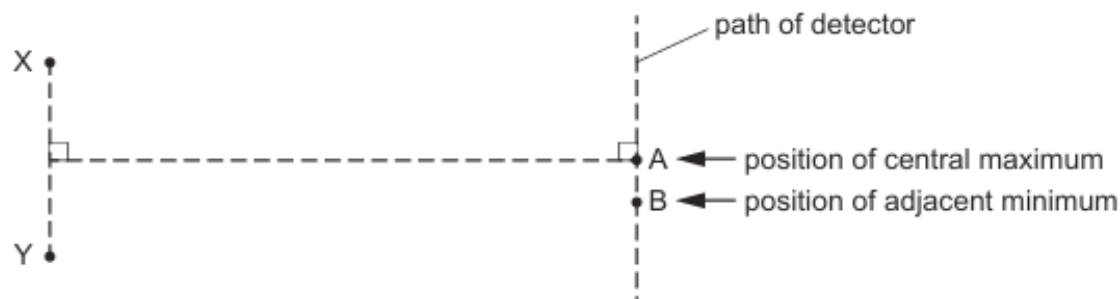


Fig. 5.1 (not to scale)

A microwave detector is moved along a path parallel to the line joining X and Y. An interference pattern is detected. A central intensity maximum is located at point A and there is an adjacent intensity minimum at point B. The microwaves have a wavelength of 0.040 m.

(a) Calculate the frequency, in GHz, of the microwaves.

frequency = GHz [3]

(b) For the waves arriving at point B, determine:

(i) the path difference

path difference = m [1]

(ii) the phase difference.

phase difference =° [1]

- (c) The amplitudes of the waves from the sources are changed. This causes a change in the amplitude of the waves arriving at point A. At this point, the amplitude of the wave arriving from source X is doubled and the amplitude of the wave arriving from source Y is also doubled.

Describe the effect, if any, on the intensity of the central maximum at point A.

.....
.....
..... [2]

- (d) Describe the effect, if any, on the positions of the central intensity maximum and the adjacent intensity minimum due to the following separate changes.

- (i) The separation of the sources X and Y is increased.

.....
..... [1]

- (ii) The phase difference between the microwaves emitted by the sources X and Y changes to 180° .

.....
..... [1]

[Total: 9]

37. 9702/23/O/N/20/Q5

- (a) A sound wave is detected by a microphone that is connected to a cathode-ray oscilloscope (CRO). The trace on the screen of the CRO is shown in Fig. 5.1.

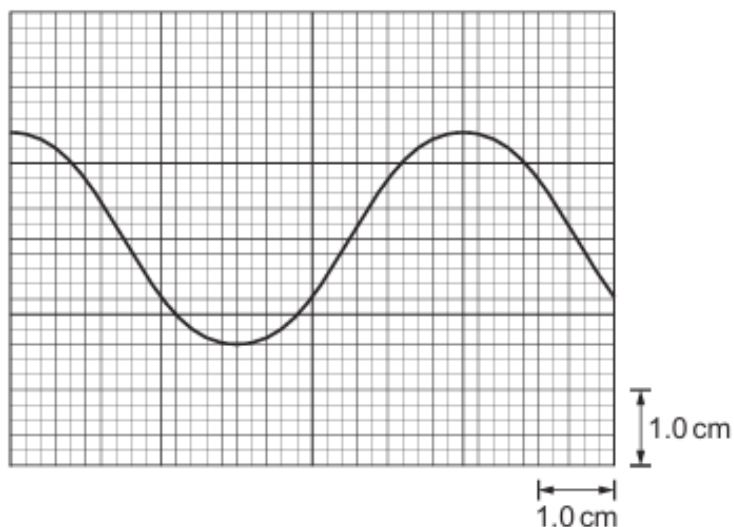


Fig. 5.1

The time-base setting of the CRO is $2.0 \times 10^{-5} \text{ s cm}^{-1}$.

- (i) Determine the frequency of the sound wave.

frequency = Hz [2]

- (ii) The intensity of the sound wave is now doubled. The frequency is unchanged. Assume that the amplitude of the trace is proportional to the amplitude of the sound wave.

On Fig. 5.1, sketch the new trace shown on the screen. [2]

- (iii) The time-base is now switched off.

Describe the trace seen on the screen.

.....
 [1]

- (b) A beam of light of a single wavelength is incident normally on a diffraction grating, as illustrated in Fig. 5.2.

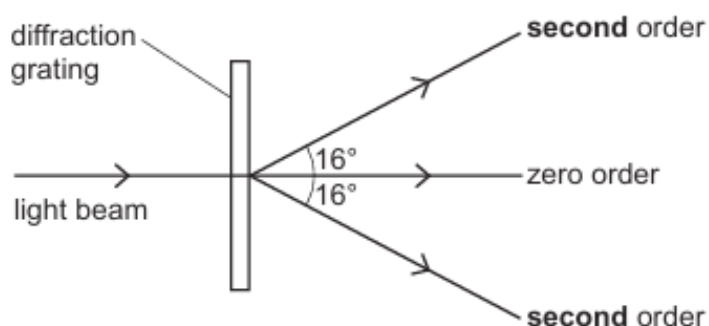


Fig. 5.2 (not to scale)

Fig. 5.2 does not show all of the emerging beams from the grating. The angle between the **second**-order emerging beam and the central zero-order beam is 16° . The grating has a line spacing of $3.4 \times 10^{-6} \text{ m}$.

- (i) Calculate the wavelength of the light.

wavelength = m [2]

- (ii) Determine the highest order of emerging beam from the grating.

highest order = [2]

[Total: 9]

38. 9702/22/F/M/21/Q4

(a) State the principle of superposition.

.....

.....

..... [2]

(b) A transmitter produces microwaves that travel in air towards a metal plate, as shown in Fig. 4.1.

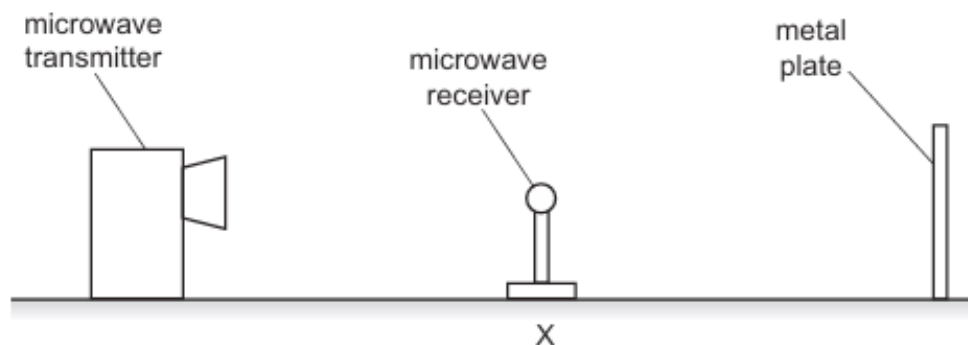


Fig. 4.1

The microwaves have a wavelength of 0.040 m. A stationary wave is formed between the transmitter and the plate.

(i) Explain the function of the metal plate.

.....

..... [1]

(ii) Calculate the frequency, in GHz, of the microwaves.

frequency = GHz [3]

(iii) A microwave receiver is initially placed at position X where it detects an intensity minimum. The receiver is then slowly moved away from X directly towards the plate.

1. Determine the shortest distance from X of the receiver when it detects another intensity minimum.

distance = m

2. Determine the number of intensity maxima that are detected by the receiver as it moves from X to a position that is 9.1 cm away from X.

number =
[2]

[Total: 8]

39. 9702/21/M/J/21/Q4

- (a) For a progressive wave, state what is meant by *wavelength*.

.....
 [1]

- (b) A light wave from a laser has a wavelength of 460 nm in a vacuum.

Calculate the period of the wave.

period = s [3]

- (c) The light from the laser is incident normally on a diffraction grating.

Describe the diffraction of the light waves at the grating.

.....

 [2]

- (d) A diffraction grating is used with different wavelengths of visible light. The angle θ of the **fourth-order** maximum from the zero-order (central) maximum is measured for each wavelength. The variation with wavelength λ of $\sin \theta$ is shown in Fig. 4.1.

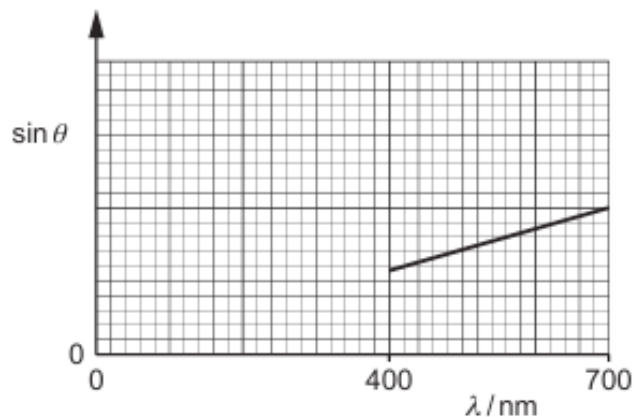


Fig. 4.1

- (i) The gradient of the graph is G .

Determine an expression, in terms of G , for the distance d between the centres of two adjacent slits in the diffraction grating.

$$d = \dots\dots\dots [2]$$

- (ii) On Fig. 4.1, sketch a graph to show the results that would be obtained for the **second-order** maxima. [2]

[Total: 10]

40. 9702/22/M/J/21/Q4

- (a) For a progressive wave, state what is meant by its *period*.

.....
 [1]

- (b) State the principle of superposition.

.....

 [2]

- (c) Electromagnetic waves of wavelength 0.040 m are emitted in phase from two sources X and Y and travel in a vacuum. The arrangement of the sources is shown in Fig. 4.1.

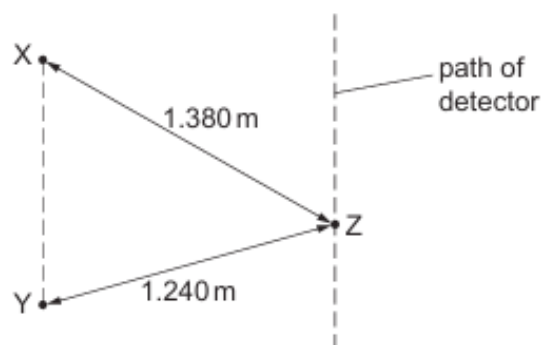


Fig. 4.1 (not to scale)

A detector moves along a path that is parallel to the line XY. A pattern of intensity maxima and minima is detected.

Distance XZ is 1.380 m and distance YZ is 1.240 m .

- (i) State the name of the region of the electromagnetic spectrum that contains the waves from X and Y.

..... [1]

- (ii) Calculate the period, in ps, of the waves.

period = ps [3]

- (iii) Show that the path difference at point Z between the waves from X and Y is 3.5λ , where λ is the wavelength of the waves.

[1]

- (iv) Calculate the phase difference between the waves at point Z.

phase difference =° [1]

- (v) The waves from X alone have the same amplitude at point Z as the waves from Y alone.

State the intensity of the waves at point Z.

..... [1]

- (vi) The frequencies of the waves from X and Y are both decreased to the same lower value. The waves stay within the same region of the electromagnetic spectrum.

Describe the effect of this change on the pattern of intensity maxima and minima along the path of the detector.

.....
..... [1]

[Total: 11]

41. 9702/23/M/J/21/Q4

- (a) State the principle of superposition.

.....
.....
..... [2]

- (b) Two waves, with intensities I and $4I$, superpose. The waves have the same frequency.

Determine, in terms of I , the maximum possible intensity of the resulting wave.

maximum intensity = I [2]

- (c) Coherent light of wavelength 550 nm is incident normally on a double slit of slit separation 0.35 mm . A series of bright and dark fringes forms on a screen placed a distance of 1.2 m from the double slit, as shown in Fig. 4.1. The screen is parallel to the double slit.



Fig. 4.1 (not to scale)

- (i) Determine the distance between the centres of adjacent bright fringes on the screen.

distance = m [3]

- (ii) The light of wavelength 550 nm is replaced with red light of a single frequency.

State and explain the change, if any, in the distance between the centres of adjacent bright fringes.

.....

.....

..... [1]

[Total: 8]

42. 9702/22/O/N/21/Q5

A tube is initially fully submerged in water. The axis of the tube is kept vertical as the tube is slowly raised out of the water, as shown in Fig. 5.1.

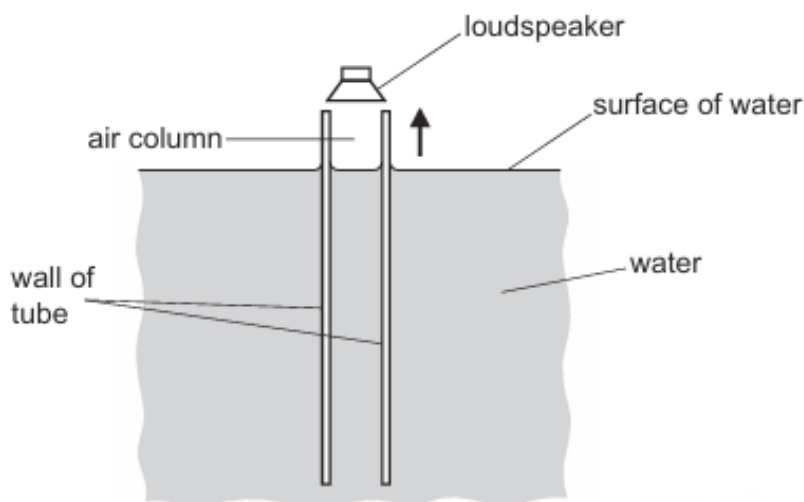


Fig. 5.1

A loudspeaker producing sound of frequency 530 Hz is positioned at the open top end of the tube as it is raised. The water surface inside the tube is always level with the water surface outside the tube. The speed of the sound in the air column in the tube is 340 m s^{-1} .

- (a) Describe a simple way that a student, without requiring any additional equipment, can detect when a stationary wave is formed in the air column as the tube is being raised.

.....
..... [1]

- (b) Determine the height of the top end of the tube above the surface of the water when a stationary wave is first produced in the tube. Assume that an antinode is formed level with the top of the tube.

height = m [3]

- (c) Determine the distance moved by the tube between the positions at which the first and second stationary waves are formed.

distance = m [1]

[Total: 5]

43. 9702/23/O/N/21/Q5

- (a) For a progressive wave on a stretched string, state what is meant by *amplitude*.

.....
..... [1]

- (b) Light from a laser has a wavelength of 690 nm in a vacuum.

Calculate the period of the light wave.

period = s [3]

- (c) A two-source interference experiment uses the arrangement shown in Fig. 5.1.

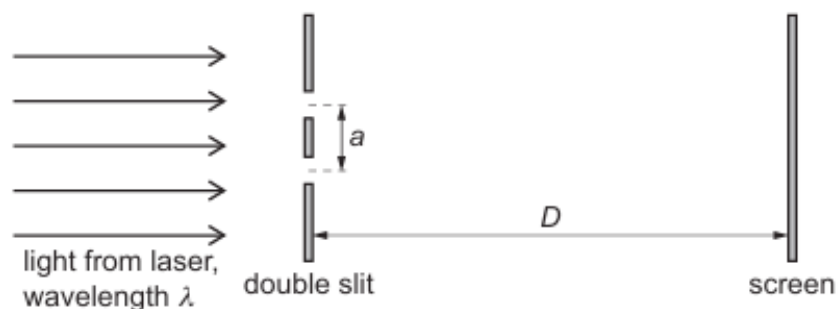


Fig. 5.1 (not to scale)

Light from a laser is incident normally on a double slit. A screen is parallel to the double slit.

Interference fringes are seen on the screen at distance D from the double slit. The separation of the centres of the slits is a . The light has wavelength λ .

The separation x of the centres of adjacent bright fringes is measured for different values of distance D .

The variation with D of x is shown in Fig. 5.2.

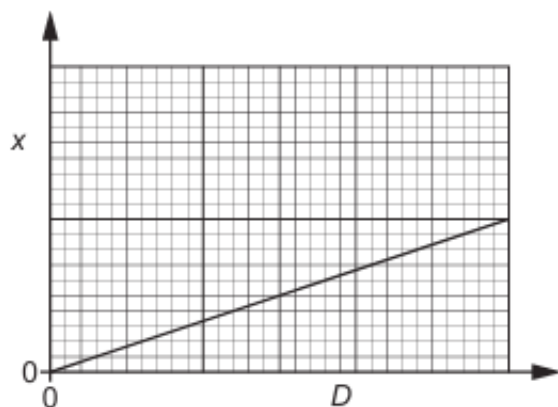


Fig. 5.2

The gradient of the graph is G .

- (i) Determine an expression, in terms of G and λ , for the separation a of the slits.

$$a = \dots\dots\dots [2]$$

- (ii) The experiment is repeated with slits of separation $2a$. The wavelength of the light is unchanged.

On Fig. 5.2, sketch a graph to show the results of this experiment. [2]

[Total: 8]

44. 9702/21/M/J/22/Q5

A horizontal string is stretched between two fixed points A and B. A vibrator is used to oscillate the string and produce an observable stationary wave.

At one instant, the moving string is straight, as shown in Fig. 5.1.



Fig. 5.1

The dots in the diagram represent the positions of the nodes on the string. Point P on the string is moving downwards.

The wave on the string has a speed of 35 ms^{-1} and a period of 0.040 s .

(a) Explain how the stationary wave is formed on the string.

.....

.....

.....

..... [2]

(b) On Fig. 5.1, sketch a line to show a possible position of the string a quarter of a cycle later than the position shown in the diagram. [1]

(c) Determine the horizontal distance from A to B.

distance = m [3]

- (d) A particle on the string has zero displacement at time $t = 0$. From time $t = 0$ to time $t = 0.060$ s, the particle moves through a total distance of 72 mm.

Calculate the amplitude of oscillation of the particle.

amplitude = mm [2]

[Total: 8]

45. 9702/22/M/J/22/Q5

Light from a laser is used to produce an interference pattern on a screen, as shown in Fig. 5.1.

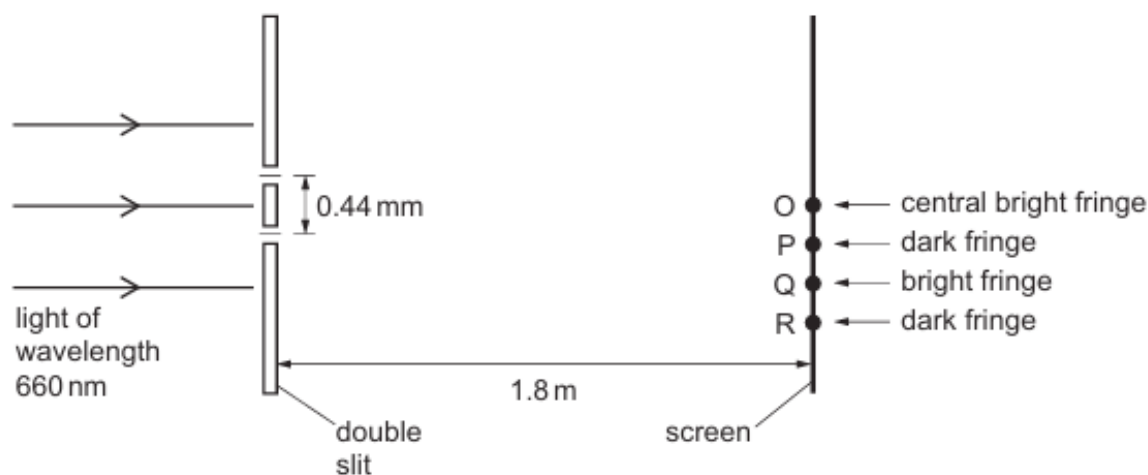


Fig. 5.1 (not to scale)

The light of wavelength 660 nm is incident normally on two slits that have a separation of 0.44 mm. The double slit is parallel to the screen. The perpendicular distance between the double slit and the screen is 1.8 m.

The central bright fringe on the screen is formed at point O. The next dark fringe below point O is formed at point P. The next bright fringe and the next dark fringe below point P are formed at points Q and R respectively.

- (a) The light waves from the two slits are coherent.

State what is meant by coherent.

.....
 [1]

- (b) For the two light waves superposing at R, calculate:

- (i) the difference in their path lengths, in nm, from the slits

path difference = nm [1]

- (ii) their phase difference.

phase difference = ° [1]

- (c) Calculate the distance OQ.

distance OQ = m [3]

- (d) The intensity of the light incident on the double slit is increased without changing the frequency.

Describe how the appearance of the fringes after this change is different from, and similar to, their appearance before the change.

.....
.....
.....
.....
..... [3]

- (e) The light of wavelength 660 nm is now replaced by blue light from a laser.

State and explain the change, if any, that must be made to the separation of the two slits so that the fringe separation on the screen is the same as it was for light of wavelength 660 nm.

.....
.....
.....
..... [2]

[Total: 11]

46. 9702/23/M/J/22/Q5

- (a) Parallel light rays from the Sun are incident normally on a magnifying glass. The magnifying glass directs the light to an area A of radius r , as shown in Fig. 5.1.

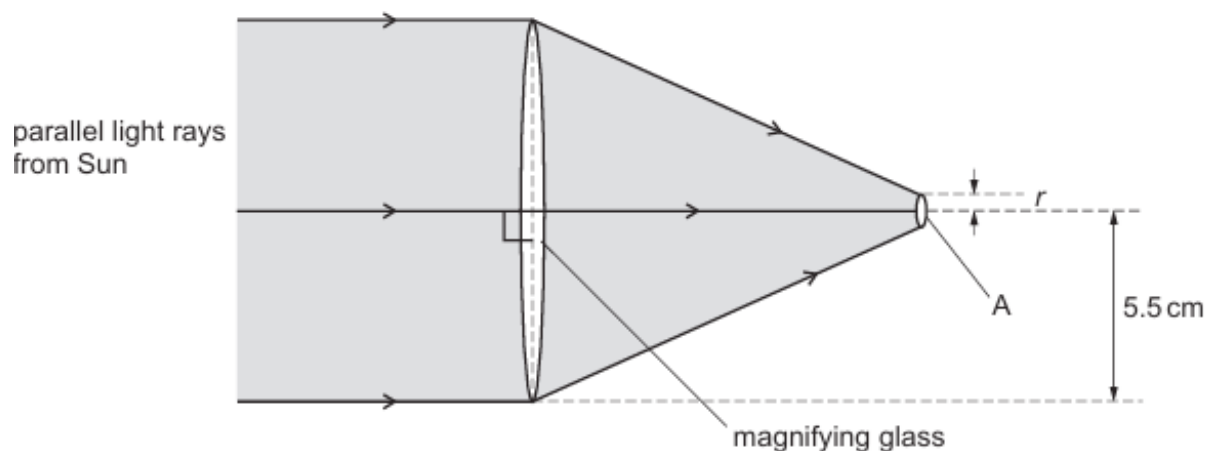


Fig. 5.1 (not to scale)

The magnifying glass is circular in cross-section with a radius of 5.5 cm. The intensity of the light from the Sun incident on the magnifying glass is 1.3 kW m^{-2} .

Assume that all of the light incident on the magnifying glass is transmitted through it.

- (i) Calculate the power of the light from the Sun incident on the magnifying glass.

power = W [2]

- (ii) The value of r is 1.5 mm.

Calculate the intensity of the light on area A.

intensity = W m^{-2} [1]

(b) A laser emits a beam of electromagnetic waves of frequency $3.7 \times 10^{15} \text{ Hz}$ in a vacuum.

(i) Show that the wavelength of the waves is $8.1 \times 10^{-8} \text{ m}$.

[2]

(ii) State the region of the electromagnetic spectrum to which these waves belong.

..... [1]

(iii) The beam from the laser now passes through a diffraction grating with 2400 lines per millimetre. A detector sensitive to the waves emitted by the laser is moved through an arc of 180° in order to detect the maxima produced by the waves passing through the grating, as shown in Fig. 5.2.

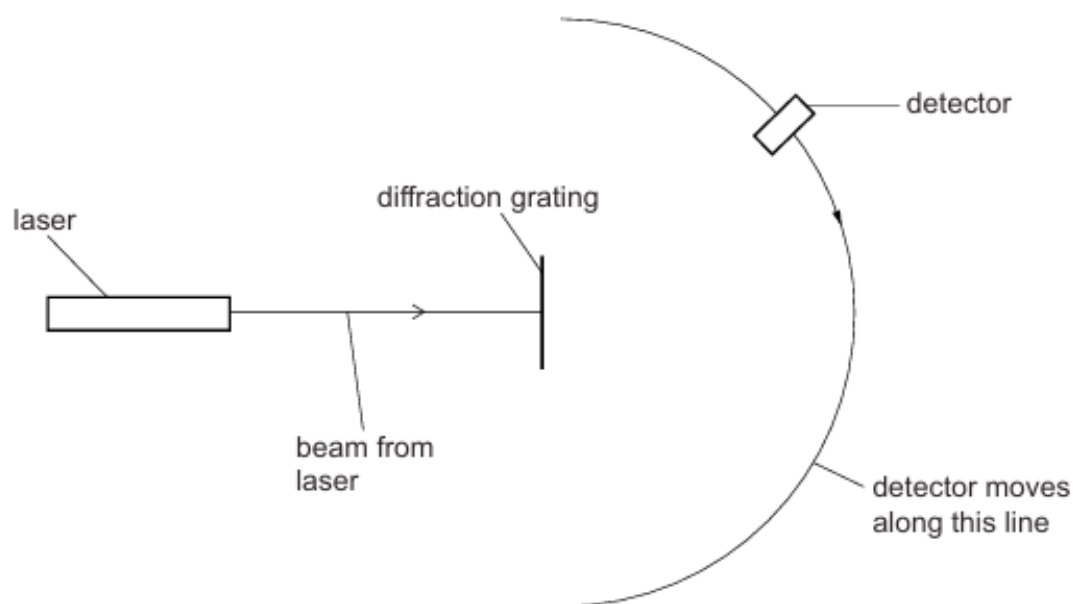


Fig. 5.2

Calculate the number of maxima detected as the detector moves through 180° along the line shown in Fig. 5.2. Show your working.

number of maxima detected = [4]

- (iv) The laser is now replaced with one that emits electromagnetic waves with a wavelength of 300 nm.

Explain, without calculation, what happens to the number of maxima now detected. Assume that the detector is also sensitive to this wavelength of electromagnetic waves.

.....
.....
..... [2]

[Total: 12]

47. 9702/22/F/M/23/Q5

- (a) A microphone and cathode-ray oscilloscope (CRO) are used to analyse a sound wave of frequency 5000 Hz. The trace that is displayed on the screen of the CRO is shown in Fig. 5.1.

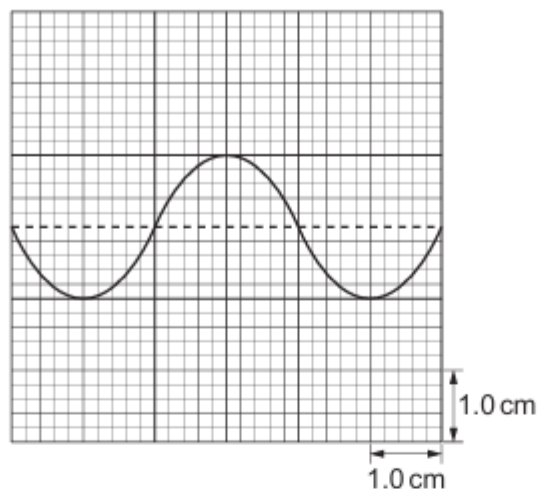


Fig. 5.1

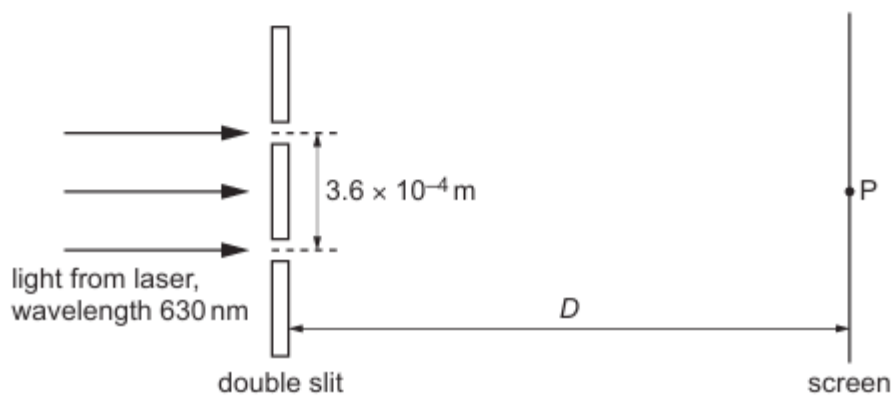
- (i) Determine the time-base setting, in s cm^{-1} , of the CRO.

time-base setting = s cm^{-1} [2]

- (ii) The intensity of the sound detected by the microphone is now increased from its initial value of I to a new value of $3I$. The frequency of the sound is unchanged. Assume that the amplitude of the trace on the CRO screen is proportional to the amplitude of the sound wave.

On Fig. 5.1, sketch the new trace shown on the screen of the CRO. [3]

- (b) An arrangement for demonstrating interference using light is shown in Fig. 5.2.



The wavelength of the light from the laser is 630 nm. The light is incident normally on the double slit. The separation of the two slits is 3.6×10^{-4} m. The perpendicular distance between the double slit and the screen is D .

Coherent light waves from the slits form an interference pattern of bright and dark fringes on the screen. The distance between the centres of two adjacent bright fringes is 4.0×10^{-3} m. The central bright fringe is formed at point P.

- (i) Explain why a bright fringe is produced by the waves meeting at point P.

.....
 [1]

- (ii) Calculate distance D .

$D =$ m [3]

- (c) The wavelength λ of the light in (b) is now varied. This causes a variation in the distance x between the centres of two adjacent bright fringes on the screen. The distance D and the separation of the two slits are unchanged.

On Fig. 5.3, sketch a graph to show the variation of x with λ from $\lambda = 400$ nm to $\lambda = 700$ nm. Numerical values of x are not required.

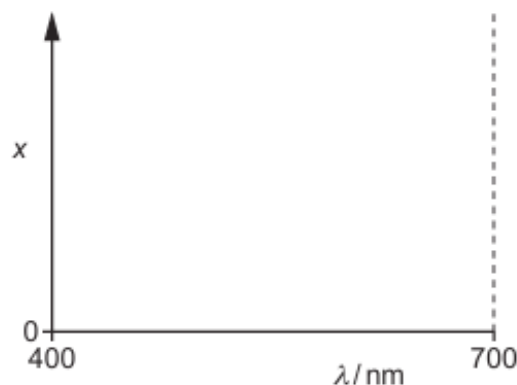


Fig. 5.3

[1]

[Total: 10]

48. 9702/21/M/J/23/Q5

(a) An electromagnetic wave in a vacuum has a wavelength of $8.4 \times 10^{-6} \text{ m}$.

(i) State the name of the principal region of the electromagnetic spectrum for the wave.

..... [1]

(ii) Calculate the frequency, in THz, of the wave.

frequency = THz [2]

(b) An arrangement that uses a double slit to demonstrate the interference of light from a laser is shown in Fig. 5.1.

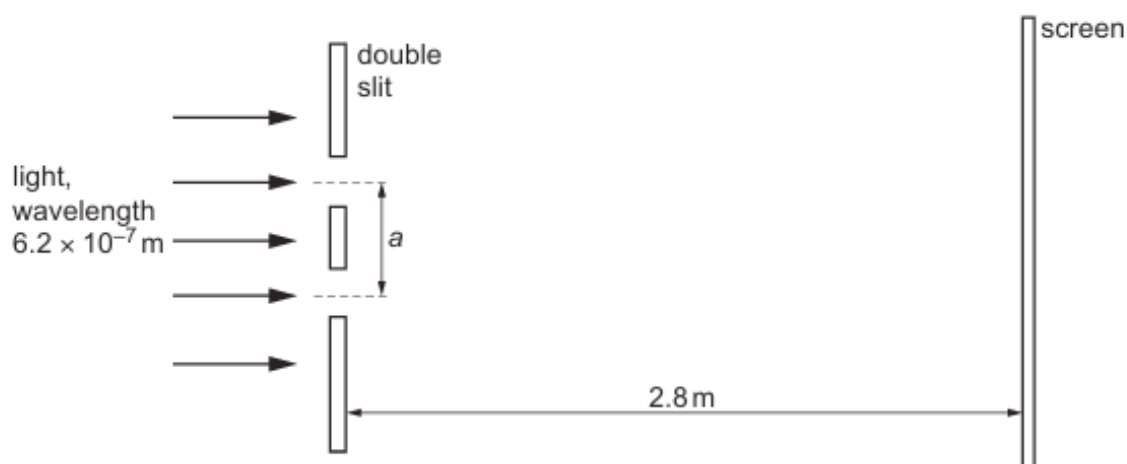
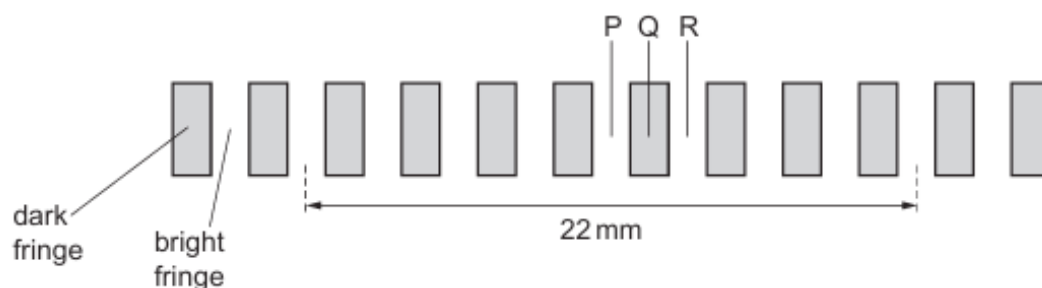


Fig. 5.1 (not to scale)

The light from the laser has a wavelength of $6.2 \times 10^{-7} \text{ m}$ and is incident normally on the slits. The separation of the two slits is a . The slits and screen are parallel and separated by a distance of 2.8 m .

An interference pattern of bright fringes and dark fringes is formed on the screen. The distance on the screen across 8 bright fringes is 22 mm , as illustrated in Fig. 5.2.



- (i) The light waves emerging from the two slits are coherent.

State what is meant by coherent.

.....
..... [1]

- (ii) Calculate the separation a of the slits.

$a =$ m [3]

- (c) Fringe P is the central bright fringe of the interference pattern in (b). Fringe Q and fringe R are the nearest dark fringe and the nearest bright fringe respectively to the right of fringe P, as shown in Fig. 5.2.

- (i) Calculate the difference in the distances (the path difference) from each slit to the centre of fringe Q.

difference in the distances = m [1]

- (ii) State the phase difference between the light waves meeting at the centre of fringe R.

phase difference =° [1]

[Total: 9]

49. 9702/23/M/J/23/Q4

- (a) For a progressive wave, state what is meant by the frequency.

.....
..... [1]

- (b) A loudspeaker, microphone and cathode-ray oscilloscope (CRO) are arranged as shown in Fig. 4.1.

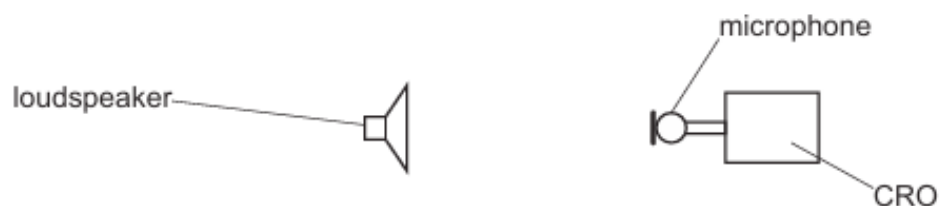


Fig. 4.1

The loudspeaker is emitting a sound wave which is detected by the microphone and displayed on the screen of the CRO as shown in Fig. 4.2.

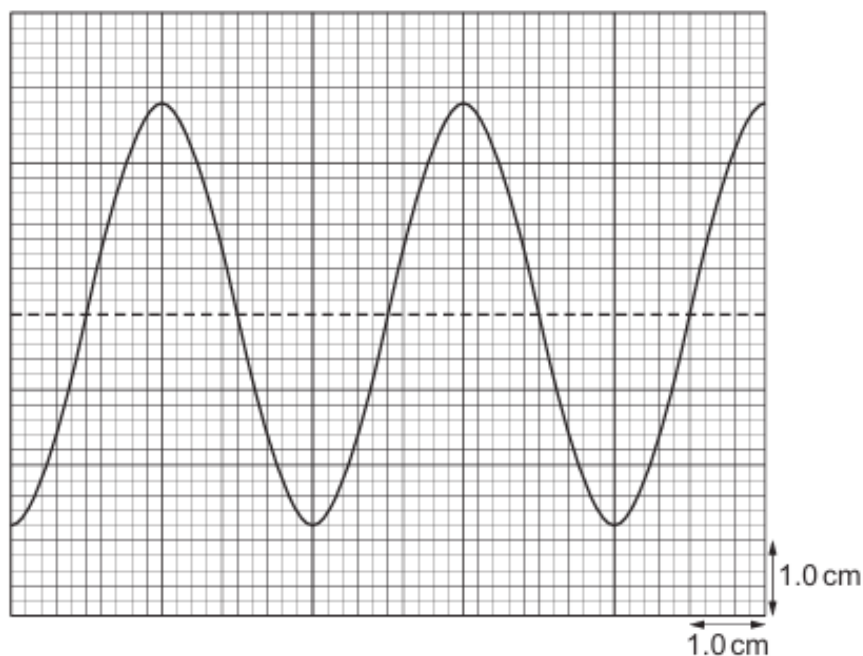


Fig. 4.2

The time-base on the CRO is set to 0.50 ms cm^{-1} and the y -gain is set to 0.20 V cm^{-1} .

Calculate:

- (i) the frequency of the sound wave

frequency = Hz [2]

- (ii) the amplitude of the signal received by the CRO.

amplitude = V [1]

- (c) The intensity of the sound wave in (b) is reduced to a quarter of its original intensity without a change in frequency. Assume that the amplitude of the signal received by the CRO is proportional to the amplitude of the sound wave.

On Fig. 4.2, sketch the trace that is now seen on the screen of the CRO. [3]

(d) A metal sheet is now placed in front of the loudspeaker in (b), as shown in Fig. 4.3.

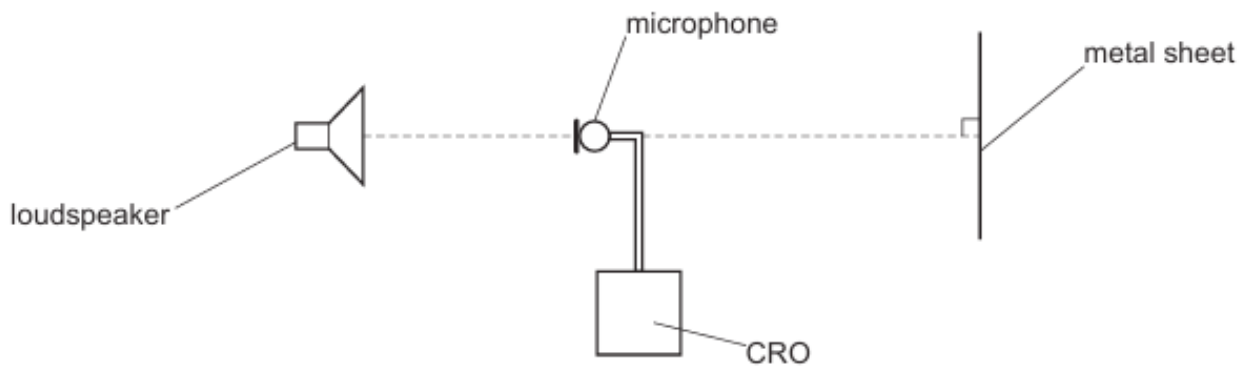


Fig. 4.3

A stationary wave is formed between the loudspeaker and the metal sheet.

(i) State the principle of superposition.

.....
.....
..... [2]

(ii) The initial position of the microphone is such that the trace on the CRO has an amplitude minimum. It is now moved a distance of 1.05 m away from the loudspeaker along the line joining the loudspeaker and metal sheet.

As the microphone moves, it passes through three positions where the trace has an amplitude maximum before ending at a position where the trace has an amplitude minimum.

Determine the wavelength of the sound wave.

wavelength = m [2]

(iii) Use your answers in (b)(i) and (d)(ii) to determine the speed of the sound in the air.

speed = ms^{-1} [2]

[Total: 13]

50. 9702/21/O/N/23/Q4

(a) State the principle of superposition.

.....

 [2]

(b) Coherent light is incident normally on two identical slits X and Y. The diffracted light emerging from the slits superposes to produce an interference pattern on a screen positioned at a distance of 1.9 m from the slits.

Fig. 4.1 shows the arrangement and the central part of the interference pattern of bright and dark fringes formed on the screen.

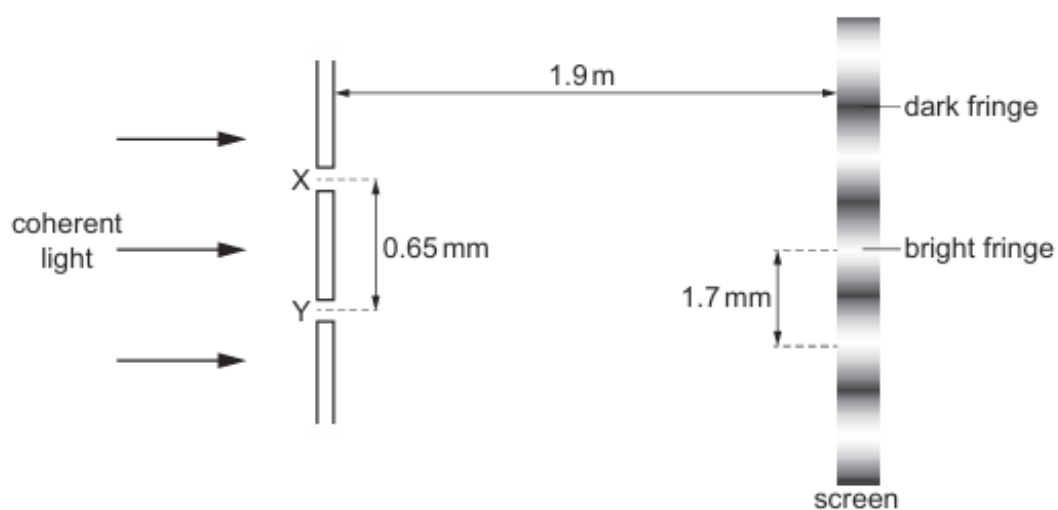


Fig. 4.1 (not to scale)

The separation of the slits is 0.65 mm. The distance between the centres of adjacent bright fringes is 1.7 mm.

Calculate the wavelength λ of the light.

$\lambda =$ m [3]

- (c) Light waves from slits X and Y in (b) arrive at a point between adjacent bright fringes on the screen. Fig. 4.2 shows the variation of displacement with time for the waves arriving at the point where they meet.

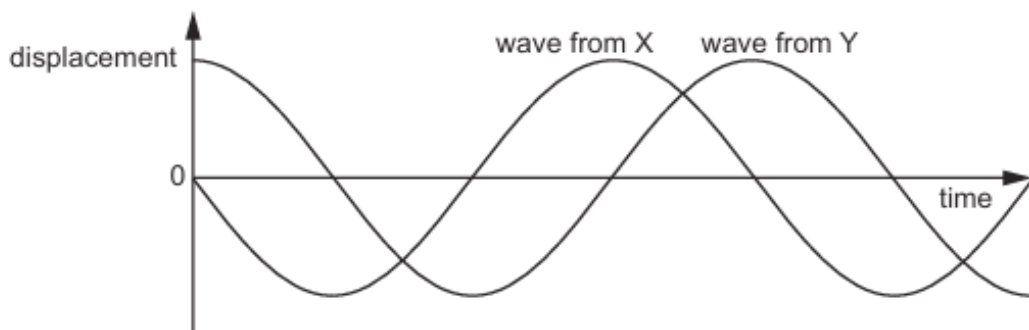


Fig. 4.2

A student makes two statements about the waves at this point:

Statement 1: 'The phase difference between the waves is 90° .'

Statement 2: 'The amplitude of the resultant wave is zero.'

- (i) Explain how statement 1 is correct.

.....

 [1]

- (ii) State and explain whether statement 2 is correct.

.....

 [1]

- (d) The width of each slit in (b) is decreased by the same amount. There is no change to the separation of the slits.

Describe and explain the effect, if any, of this change on the appearance of the interference pattern.

.....

 [2]

[Total: 9]

51. 9702/22/O/N/23/Q5

- (a) A beam of vertically polarised light is incident normally on a polarising filter, as shown in Fig. 5.1.

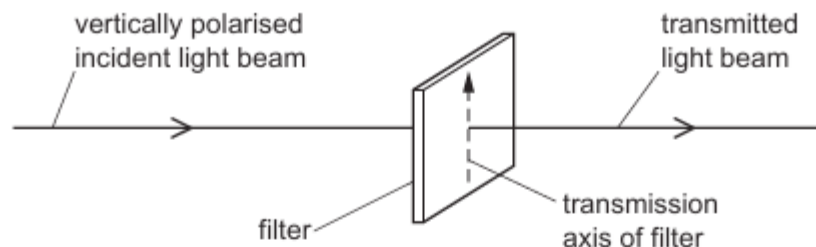


Fig. 5.1

- (i) The transmission axis of the filter is initially vertical. The filter is then rotated through an angle of 360° while the plane of the filter remains perpendicular to the beam.

On Fig. 5.2, sketch a graph to show the variation of the intensity of the light in the transmitted beam with the angle through which the transmission axis is rotated.

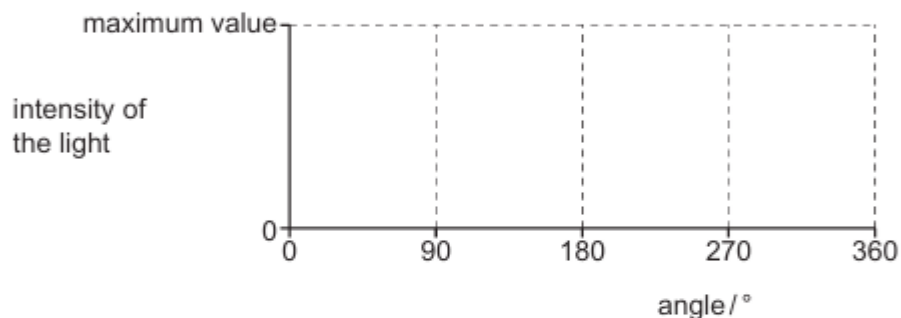


Fig. 5.2

[2]

- (ii) The intensity of the light in the incident beam is 7.6 W m^{-2} . When the transmission axis of the filter is at angle θ to the vertical, the light intensity of the transmitted beam is 4.2 W m^{-2} .

Calculate angle θ .

$$\theta = \dots\dots\dots^\circ \quad [2]$$

- (b) State what is meant by the diffraction of a wave.

.....

 [2]

- (c) A beam of light of wavelength $4.3 \times 10^{-7} \text{ m}$ is incident normally on a diffraction grating in air, as shown in Fig. 5.3.

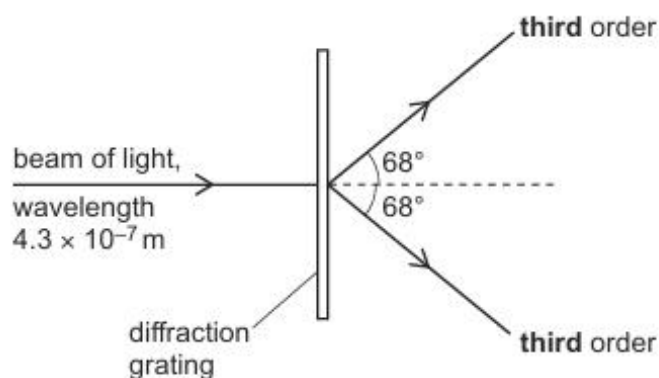


Fig. 5.3 (not to scale)

The **third**-order diffraction maximum of the light is at an angle of 68° to the direction of the incident light beam.

- (i) Calculate the line spacing d of the diffraction grating.

$d = \dots\dots\dots \text{m}$ [2]

- (ii) Determine a different wavelength of **visible** light that will also produce a diffraction maximum at an angle of 68° .

wavelength = $\dots\dots\dots \text{m}$ [2]

[Total: 10]

52. 9702/23/O/N/23/Q5

Two point sources, A and B, produce coherent electromagnetic waves. The waves from A and B are emitted in phase and have wavelength λ , as shown in Fig. 5.1.

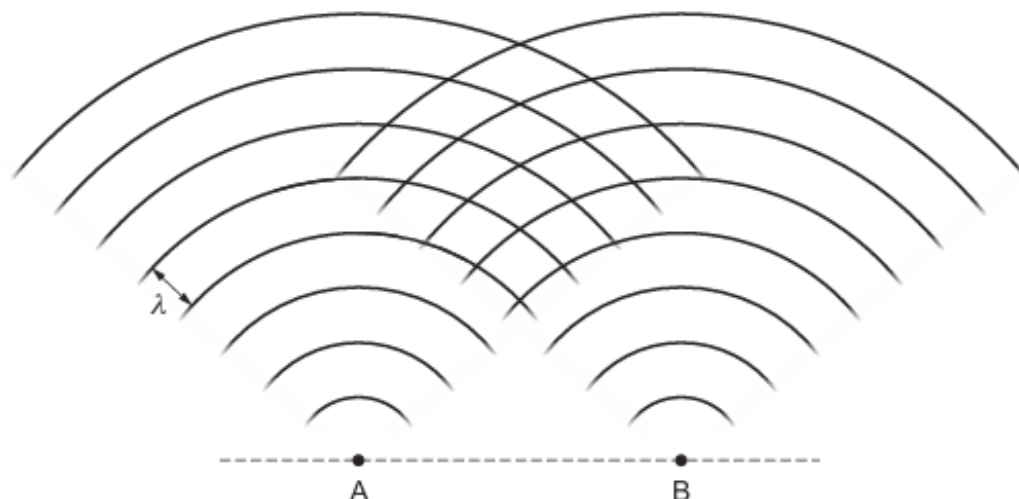


Fig. 5.1 (not to scale)

The lines on Fig. 5.1 represent wavefronts. All the points on a wavefront are in phase.

(a) On Fig. 5.1, mark with a cross (×):

- (i)** the position of an interference maximum (label this cross Y) [1]
- (ii)** the position of an interference minimum (label this cross Z). [1]

(b) The waves in air have a wavelength of 2.9×10^{-5} m.

An interference pattern is detected along a line parallel to AB and at a perpendicular distance of 140 m from AB. The spacing between adjacent interference maxima is 1.2 cm.

(i) Calculate the separation a of the sources A and B.

$$a = \dots\dots\dots \text{m} \quad [3]$$

(ii) State the principal region of the electromagnetic spectrum to which the waves belong.

..... [1]

[Total: 6]

ANSWERS

1	(a) constant <u>phase difference</u>	B1	[1]
	(b) allow wavelength estimate 750 nm → 550 nm	C1	
	separation = $\lambda D / x$	C1	
	= $(650 \times 10^{-9} \times 2.4) / (0.86 \times 10^{-3})$		
	= 1.8 mm	A1	[3]
	(allow 2 marks from inappropriate estimate if answer is in range 10 cm → 0.1 mm)		
	(c) no longer complete destructive interference / amplitudes no longer completely cancel	M1	
	so dark fringes are lighter	A1	[2]
2	(a) when a <u>wave</u> (front) passes by/incident on an edge/slit	M1	
	wave bends/spreads (into the geometrical shadow)	A1	[2]
	(b) $\tan \theta = \frac{38}{165}$		
	$\theta = 13^\circ$	C1	
	$d \sin \theta = n\lambda$	C1	
	$d = 2.82 \times 10^{-6}$	C1	
	number = $(1/d) = 3.6 \times 10^5$	A1	[4]
	(c) P remains in same position	B1	
	X and Y rotate through 90°	B1	[2]
	(d) <i>either</i> screen not parallel to grating <i>or</i> grating not normal to (incident) light	B1	[1]
3	(a) when a wave passes through a slit / by an edge the wave spreads out / changes direction	M1 A1	[2]
	(b) diagram: wavelength unchanged	M1	
	wavefront flat at centre, curving into geometrical shadow	A1	[2]
	(c) $d \sin \theta = n\lambda$	C1	
	for $\theta = 90^\circ$		
	$1 / (650 \times 10^3) = n \times 590 \times 10^{-9}$	M1	
	$n = 2.6$		
	number of orders is 2	A1	[3]
	(d) intensity / brightness decreases (as order increases)	B1	[1]
4	(a) when waves overlap / meet, (resultant) displacement is the sum of the individual displacements	B1	[1]
	(b) (i) two (ball-type) dippers	(M1)	
	connected to the same vibrating source /motor	(A1)	
	<i>or</i>		
	one wave source described	(M1)	
	with two slits	(A1)	[2]
	(ii) lamp with viewing screen on opposite side of tank	B1	
	means of freezing picture e.g. strobe	B1	[2]
	(c) (i) two correct lines labelled X	B1	[1]
	(ii) correct line labelled N	B1	[1]

5 (a)	(i) to produce coherent sources or constant phase difference	B1	[1]
(ii)	1. $360^\circ / 2\pi \text{ rad}$ allow $n \times 360^\circ$ or $n \times 2\pi$ (unit missing –1)	B1	[1]
	2. $180^\circ / \pi \text{ rad}$ allow $(n \times 360^\circ) - 180^\circ$ or $(n \times 2\pi) - \pi$	B1	[1]
(iii)	1. waves overlap / meet (resultant) displacement is sum of displacements of each wave	B1	[2]
	2. at P crest on trough (OWTTE)	B1	[1]
(b)	$\lambda = ax / D$ $= 2 \times 2.3 \times 10^{-3} \times 0.25 \times 10^{-3} / 1.8$ $= 639 \text{ nm}$	C1	
		C1	
		A1	[3]
6 (a)	waves overlap (resultant) displacement is the sum of the displacements of each of the waves	B1	[2]
(b)	waves travelling in opposite directions overlap / incident and reflected waves overlap (allow superpose or interfere for overlap here) waves have the same speed and frequency	B1	[2]
(c) (i)	time period = $4 \times 0.1 \text{ (ms)}$ $f = 1 / T = 1 / 4 \times 10^{-4} = 2500 \text{ Hz}$	C1	[2]
		A1	
(ii)	1. the microphone is at an antinode and goes to a node and then an antinode / maximum amplitude at antinode and minimum amplitude at node	B1	[1]
	2. $\lambda / 2 = 6.7 \text{ (cm)}$ $v = f\lambda$ $v = 2500 \times 13.4 \times 10^{-2} = 335 \text{ ms}^{-1}$ incorrect λ then can only score second mark	C1	
		C1	
		A1	[3]
7 (a)	(i) coherence: constant phase difference between (two) waves	M1	[2]
	(ii) path difference is <i>either</i> λ or $n\lambda$ or phase difference is 360° or $n \times 360^\circ$ or $n2\pi \text{ rad}$	B1	[1]
	(iii) path difference is <i>either</i> $\lambda/2$ or $(n + \frac{1}{2}) \lambda$ or phase difference is odd multiple of <i>either</i> 180° or $\pi \text{ rad}$	B1	[1]
(iv)	$w = \lambda D / a$ $= [630 \times 10^{-9} \times 1.5] / 0.45 \times 10^{-3}$ $= 2.1 \times 10^{-3} \text{ m}$	C1	
		C1	
		A1	[3]
(b)	no change to <u>dark</u> fringes no change to separation/fringe width <u>bright</u> fringes are brighter/lighter/more intense	B1	
		B1	
		B1	[3]

8	(a) (i)	diffraction bending/spreading of light at edge/slit this occurs at each slit	B1 B1	[2]
	(ii)	constant phase difference between each of the waves	B1	[1]
	(iii)	(when the waves meet) the resultant displacement is the sum of the displacements of each wave	B1	[1]
	(b)	$d \sin \theta = n \lambda$ $n = d / \lambda = 1 / 450 \times 10^3 \times 630 \times 10^{-9}$ $n = 3.52$ hence number of orders = 3	C1 M1 A1	[3]
	(c)	λ blue is less than λ red more orders seen each order is at a smaller angle than for the equivalent red	M1 A1 A1	[3]
	9 (a)	waves pass through the elements / gaps / slits in the grating spread into geometric shadow	M1 A1	[2]
	(b) (i) 1.	displacements add to give resultant displacement each wavelength travels the same path difference or are in phase hence produce a maximum	B1 B1 A0	[2]
	2.	to obtain a maximum the path difference must be λ or phase difference $360^\circ / 2\pi$ rad λ of red and blue are different hence maxima at different angles / positions	B1 B1 A0	[2]
	(ii)	$n \lambda = d \sin \theta$ $N = \sin 61^\circ / (2 \times 625 \times 10^{-9}) = 7.0 \times 10^5$	C1 A1	[2]
	(iii)	$n \lambda = 2 \times 625$ is a constant (1250) $n = 1 \rightarrow \lambda = 1250$ outside visible $n = 3 \rightarrow \lambda = 417$ in visible $n = 4 \rightarrow \lambda = 312.5$ outside visible $\lambda = 420$ nm	C1 A1	[2]
10	(a)	waves overlap / meet ² superpose coherence / constant phase difference (<i>not constant λ or frequency</i>) path difference = 0, λ , 2λ or phase difference = 0, 2π , 4π same direction of polarisation/unpolarised	(B1) (B1) (B1) max. 3 (B1)	[3]
	(b)	$\lambda = v / f$ $f = 12 \times 10^9$ Hz $\lambda = 3 \times 10^8 / 12 \times 10^9$ (<i>any subject</i>) = 0.025 m	C1 C1 M1 A0	[3]
	(c)	maximum at P <u>several</u> minima or maxima between O and P 5 maxima / 6 minima between O and P or 7 maxima / 6 minima including O and P	B1 B1 B1	[3]
	(d)	slits made narrower slits put closer together (<i>not just 'make slits smaller'</i>) Allow tilting the slits	B1 B1 M1	[2]
		and explanation of axes of rotation	A1	

11	(a) when waves overlap / meet the resultant displacement is the sum of the individual displacements of the waves	B1	
		B1	[2]
(b)	(i) 1. phase difference = $180^\circ / (n + \frac{1}{2}) 360^\circ$ (allow in rad)	B1	[1]
	2. phase difference = $0 / 360^\circ / (n360^\circ)$ (allow in rad)	B1	[1]
	(ii) $v = f\lambda$ $\lambda = 320 / 400 = 0.80 \text{ m}$	C1	
		A1	[2]
	(iii) path difference = $7 - 5 = 2 \text{ (m)}$ $= 2.5\lambda$		
	hence minimum or maximum if phase change at P is suggested	M1	
		A1	[2]
12	(a) <u>waves</u> from the double slit are coherent/constant phase difference	B1	
	<u>waves</u> (from each slit) overlap/superpose/meet (not interfere)	B1	
	maximum/bright fringe where path difference is $n\lambda$		
	or phase difference is $n360^\circ / 2\pi n \text{ rad}$		
	or minimum/dark fringe where path difference is $(n + \frac{1}{2})\lambda$		
	or phase difference is $(2n + 1) 180^\circ / (2n + 1)\pi \text{ rad}$	B1	[3]
(b)	$v = f\lambda$ $\lambda = (3 \times 10^8) / 670 \times 10^{12} = 448 \text{ (or 450) (nm)}$	C1	
		M1	[2]
(c)	$w = 12 / 9$ $a (= D\lambda / w) = (2.8 \times 450 \times 10^{-9}) / (12 / 9 \times 10^{-3})$ [allow nm, mm] $= 9.5 \times 10^{-4} \text{ m}$ [$9.4 \times 10^{-4} \text{ m}$ using $\lambda = 448 \text{ nm}$]	C1	
		C1	
		A1	[3]
(d)	(red light has) larger/higher/longer wavelength (must be comparison) fringes further apart/larger separation	M1	
		A1	[2]
13	(a) difference: vibration/oscillation (of particles)/displacement of particles is parallel to energy transfer/wavefronts in longitudinal and perpendicular for transverse	B1	
	or transverse can be polarised, longitudinal cannot be polarised		
	similarity: both transfer/propagate energy	B1	[2]
(b)	(i) waves from <u>slits</u> are coherent/constant phase relationship	(B1)	
	waves overlap (at screen) with a phase difference or have a path difference	(B1)	
	maxima where phase difference is integer $\times 360^\circ$ (or $\times 2\pi \text{ rad}$)		
	or path difference is integer $\times \lambda$		
	or equivalent explanation of minima e.g. $(n + \frac{1}{2}) \times 360^\circ$ max. 2	(B1)	[2]
	(ii) maxima spacing = $\lambda D / a$ $= (6.3 \times 10^{-7} \times 2.5) / 0.35 \times 10^{-3}$ $= 4.5 \times 10^{-3} \text{ m}$	C1	
		A1	[2]
(c)	(ultra-violet has) shorter wavelength, hence smaller separation/distance	A1	[1]

14	(a)	diffraction is the spreading of a wave as it passes through a slit or past an edge	B1	
		when two (or more) waves superpose/meet/overlap	M1	
		resultant displacement is the sum of the displacement of each wave	A1	[3]
(b)		$n\lambda = d \sin \theta$ and $v = f\lambda$	C1	
		max order number for $\theta = 90^\circ$		
		hence $n (= f/vN) = 7.06 \times 10^{14} / (3 \times 10^8 \times 650 \times 10^3)$	M1	
		$n = 3.6$		
		hence number of orders = 3	A1	[3]
(c)		greater wavelength so fewer orders seen	A1	[1]
15	(a)	(i)	coherent: constant phase difference	B1
			interference is the (overlapping of waves and the) sum of/addition of displacement of two waves	B1 [2]
		(ii)	wavelength = 3.2 m (allow ± 0.05 m)	M1
			$f (= v/\lambda = 240 / 3.2) = 75$ Hz	A1 [2]
		(iii)	90° (allow $\pm 2^\circ$) or $\pi/2$ rad	A1 [1]
		(iv)	sketch has amplitude 3.0 ± 0.1 cm	M1
			correct displacement values at previous peaks to produce correct shape	A1 [2]
	(b)	(i)	$\lambda = ax/D$	C1
			$x = (546 \times 10^{-9} \times 0.85) / 0.13 \times 10^{-3} (= 3.57 \times 10^{-3} \text{ m})$	C1
			$AB = 8.9 (8.93) \times 10^{-3} \text{ m}$	A1 [3]
		(ii)	shorter wavelength for blue light so separation is less	B1 [1]
16	(a)	(i)	Displacement of particles perpendicular to direction of energy propagation	B1
		(ii)	waves meet/overlap (at a point)	B1
			(resultant) displacement is sum of the individual displacements	B1
	(b)	(i)	$\lambda = vT$ or $\lambda = v/f$ and $f = 1/T$	C1
			$\lambda = 4.0 \times 1.5$	
			$\lambda = 6.0$ (cm)	A1
17	(i)		gradient = $(4.5 - 2.4) \times 10^{-3} / (3.25 - 1.75) [= 1.4 \times 10^{-3}]$	
			wavelength = $0.45 \times 10^{-3} \times 1.4 \times 10^{-3}$	C1
			= 6.30×10^{-7} (m)	C1
			= 630 nm	A1 [4]
	(ii)		(gradient is equal to λ/a therefore) gradient of line is reduced	B1
			value of x will be reduced for all values of D	
			or new line is completely below old line	
			or intercept is less	B1 [2]

18 (a) diffraction: <u>spreading/diverging</u> of <u>waves/light</u> (takes place) at (each) slit/ element/gap/aperture	B1	
interference: overlapping of waves (from coherent sources at each element)	B1	
path difference λ /phase difference of $360(^{\circ})/2\pi$ (produces the first order)	B1	[3]
(b) $d \sin \theta = n\lambda$ or $\sin \theta = Nn\lambda$	C1	
$d = (2 \times 486 \times 10^{-9}) / \sin 29.7^{\circ} (= 1.962 \times 10^{-6})$	C1	
number of lines = 510 (509.7) mm^{-1}	A1	[3]
19 (a) wave incident on/passes by or through an aperture/edge wave spreads (into geometrical shadow)	B1	
(b) $n\lambda = d \sin \theta$	C1	
substitution of $\theta = 90^{\circ}$ or $\sin \theta = 1$	C1	
$4 \times 500 \times 10^{-9} = d \times \sin 90^{\circ}$		
line spacing = $2.0 \times 10^{-6} \text{ m}$	A1	[3]
(c) wavelength of red light is <u>longer</u> (than 500 nm) (each order/fourth order is now at a greater angle so) the fifth-order maximum cannot be formed/not formed	M1	
	A1	[2]
20 (a) wave incident on/passes by or through an aperture/edge wave spreads (into geometrical shadow)	B1	
(b) (i) waves (from slits) overlap (at point X)	B1	
path difference (from slits to X) is zero/ phase difference (between the two waves) is zero (so constructive interference gives bright fringe)	B1	[2]
(ii) difference in distances = $\lambda/2 = 580/2$ = 290 nm	A1	[1]
(iii) $\lambda = ax/D$	C1	
$D = [0.41 \times 10^{-3} \times (2 \times 2.0 \times 10^{-3})] / 580 \times 10^{-9}$ = 2.8 m	C1	
	A1	[3]
(iv) same separation/fringe width/number of fringes bright fringe(s)/central bright fringe/(fringe at) X less bright dark fringe(s)/(fringe at) Y/(fringe at) Z brighter contrast between fringes decreases		
<i>Any two of the above four points, 1 mark each</i>	B2	[2]

- 21 (a) (i) waves at (each) slit/aperture spread **B1**
 (into the geometric shadow) wave(s) overlap/superpose/sum/meet/intersect **B1**
 (ii) there is not a constant phase difference/coherence (for two separate light source(s)) **B1**
 or waves/light from the double slit are coherent/have a constant phase difference
- (b) $x = \lambda D / a$ **C1**
 $\lambda = (36 \times 10^{-3} \times 0.48 \times 10^{-3}) / (16 \times 2.4)$ **C1**
 $= 4.5 \times 10^{-7} \text{ m}$ **A1**
- (c) (i) no movement of the water/water is flat/no ripples/disturbance **B1**
 the path difference is 2.5λ or the phase difference is 900° or 5π rad **B1**
- (ii) 1. surface/water/P vibrates/ripples **and** **B1**
 as (waves from the two dippers) arrive in phase
2. surface/water/P vibrates/ripples **and** **B1**
 as amplitudes/displacements are no longer equal/do not cancel
-

- 22 (a) (i) waves spread at (each) slit/gap **B1**
 (ii) constant phase difference (between (each of) the waves) **B1**
- (b) (i) $n\lambda = d \sin \theta$ **B1**
 $d \sin \theta$ is the same **and** $3\lambda_1 = 4\lambda_2$ so $\lambda_2 / \lambda_1 = 0.75$ **A1**
- (ii) $\lambda_2 / \lambda_1 = 0.75$ and $\lambda_1 - \lambda_2 = 170$ **A1**
 $\lambda_1 = 680 \text{ nm}$
-

- 23 (a) intensity $\propto$ (amplitude)² **B1**
- (b) (i) $v = f\lambda$ or $c = f\lambda$ **C1**
 $f = 3.00 \times 10^8 / 0.060$ **A1**
 $= 5.0 \times 10^9 \text{ Hz}$
- (ii) (at X path) difference = 3λ **M1**
 (at X phase) difference = 0 or 1080° **M1**
 so intensity is at a maximum/it is an intensity maximum **A1**
- (iii) 1. decrease in the distance between (adjacent intensity) maxima/minima **B1**
 2. (intensity) maxima and minima exchange places **B1**
-

- 24 (a)** when (two or more) waves meet (at a point) **B1**
 (resultant) displacement is the sum of the individual displacements **B1**
- (b)(i)** constant phase difference (between the waves) **B1**
- (ii)** 1. phase difference = 360° **or** 0 **B1**
 2. path difference = 1.5λ **A1**
 $= 1.5 \times 610$
 $= 920 \text{ nm}$
- (iii)** $\lambda = ax/D$ **C1**
 $x = 22/4 (= 5.5 \text{ mm})$ **or** $22 \times 10^{-3}/4 (= 5.5 \times 10^{-3} \text{ m})$ **C1**
 $a = (610 \times 10^{-9} \times 2.7)/(5.5 \times 10^{-3})$ **A1**
 $= 3.0 \times 10^{-4} \text{ m}$
- (iv)** shorter wavelength **and** (so) separation decreases **B1**
- (v)**
 - no change to fringe separation/fringe width/number of fringes
 - bright fringes are brighter
 - dark fringes are unchanged**B2**
- Any two of the above three points, 1 mark each.*

- 25 (a)** $n\lambda = d \sin \theta$ **C1**
 $\lambda = 640 \times 10^{-9} \text{ (m)}$ **C1**
 $2 \times 640 \times 10^{-9} = 1.7 \times 10^{-6} \times \sin \theta$ so $\theta = 49^\circ$ **A1**
- (b)** $2 \times 640 \times 10^{-9} = 3 \times \lambda$ **C1**
or
 $1.7 \times 10^{-6} \times \sin 49^\circ = 3 \times \lambda$
 $\lambda = 4.3 \times 10^{-7} \text{ m}$ **A1**

26 (a)	graph with x-axis labelled 'distance' and wavelength/ λ correctly shown	B1
	graph with x-axis labelled 'time' and period/ T correctly shown	B1
	graph with y-axis labelled 'displacement' and amplitude/ A correctly shown	B1
(b)(i)	wave (moves along string and) reflects at fixed point/Y/X/end/wall/boundary	B1
	the incident and reflected waves interfere/superpose	B1
(ii)	100/40 or 2.5 (cycles/periods/ T)	C1
	1. displacement = 0	B1
	2. distance = 130 mm	A1
(iii)	1. $f = 1/40 \times 10^{-3}$	A1
	= 25 Hz	
	2. $v = f\lambda$ or $\lambda = vT$	C1
	$\lambda = 30/25$ or $30 \times 40 \times 10^{-3}$ (= 1.2 m)	C1
	distance = 1.2×1.5	A1
	= 1.8 m	
27 (a) (i)	1. $N\lambda$	B1
	2. N/f	B1
(ii)	v (= distance / time) = $N\lambda / (N/f)$ so $v = f\lambda$	B1
(b)	$T = 4.0 \times 0.20 = 0.80$ (ms) or 8.0×10^{-4} (s)	C1
	$f = 1/8.0 \times 10^{-4}$	A1
	= 1300 Hz	
(c) (i)	constant phase difference (between the waves)	B1
	(ii) 180°	A1
(iii)	path difference = 2λ or $S_1Y - S_2Y = 2\lambda$	C1
	distance = $7.40 + (0.85 \times 2)$	A1
	= 9.1 m	

28 (a) (resultant) force proportional/equal to/is rate of change of momentum **B1**

(b)(i) distance = area under graph **or** $s = \frac{1}{2} (u + v) t$ **C1**
 $= \frac{1}{2} \times (9 + 13) \times 10$

or

$$s = ut + \frac{1}{2}at^2$$
$$= (9 \times 10) + (\frac{1}{2} \times 0.40 \times 10^2)$$

or

$$s = vt - \frac{1}{2}at^2$$
$$= (13 \times 10) - (\frac{1}{2} \times 0.40 \times 10^2)$$

or

$$v^2 = u^2 + 2as$$
$$13^2 = 9^2 + (2 \times 0.40 \times s)$$
$$\text{distance} = 110 \text{ m}$$

A1

(ii) 1. $a = \text{gradient}$ **or** $a = (v - u) / t$ **or** $a = \Delta v / (\Delta)t$ **C1**
e.g. $a = (14 - 9) / 12.5$ **or** $(13 - 9) / 10$
 $a = 0.40 \text{ m s}^{-2}$ **A1**

2. resultant force = 850×0.40 **A1**

$$= 340 \text{ N}$$

3. $(F =) 510 + 440 + 340 = 1300 \text{ (N)}$ **A1**

4. $P = Fv$ **C1**

$$= 1300 \times 13$$
 A1

$$= 1.7 \times 10^4 \text{ W}$$

(c) $E = \sigma / \epsilon$ **C1**

$$E = (F / A) / (\Delta L / L) \text{ or } E = FL / A\Delta L$$
 C1

$$\Delta L = (480 \times 0.48) / (3.0 \times 10^{-4} \times 2.2 \times 10^{11})$$
 A1

$$= 3.5 \times 10^{-6} \text{ m}$$

(d) $f_o = f_s v / (v - v_s)$ **C1**

$$480 = f_s \times 340 / (340 - 14)$$

$$f_s = 460 \text{ Hz}$$
 A1

29.

4(a)(i)	distance (in a specified direction of particle/point on wave) from the equilibrium position	B1
4(a)(ii)	the maximum distance (of particle/point on wave) from the equilibrium position or the maximum displacement (of particle/point on wave)	B1
4(b)	$I \propto A^2$	C1
	$I_R / I = (3.6 - 1.2)^2 / (1.2)^2$ resultant intensity = $4.0I$	A1
4(c)(i)	as wave(s) pass through the slit(s)	B1
	wave(s) spread (into geometric shadow)	B1
4(c)(ii)	$n\lambda = d \sin \theta$	C1
	$3\lambda = d \sin 90^\circ$ or $3\lambda = d$	C1
	$d = 3 \times 630 \times 10^{-9}$ $= 1.9 \times 10^{-6} \text{ m}$	A1
4(c)(iii)	wavelength of blue light is shorter (than 540 nm/630 nm/wavelengths of original light)	M1
	(so) third order diffraction maximum is produced	A1

30.

5(a)	(incident) wave reflects at end/top of tube	B1
	(incident) wave and reflected wave interfere/superpose	B1
5(b)	line has maximum value of amplitude at $h = 0$ and $h = 0.60 \text{ m}$ only	B1
	line has minimum/zero value of amplitude at $h = 0.30 \text{ m}$ only	B1
5(c)(i)	vertical/along length of tube/along axis of tube	B1
5(c)(ii)	phase difference = 0	A1
5(d)	$v = f\lambda$	C1
	$v = 340 / (2 \times 0.60)$ $= 280 \text{ Hz}$	A1
5(e)	$f = 340 / 0.60$ $= 570 \text{ Hz}$	A1

31.

5(a)(i)	the dippers are connected to the same vibrator/motor	B1
5(a)(ii)	(the overlapping waves have) similar/same amplitude	B1
5(b)	any means of 'freezing' the pattern e.g. use a stroboscope/strobe	B1
5(c)	$vT = \lambda$ or $v = f\lambda$ and $f = 1 / T$	C1
	$T = 0.060 / 0.40$ $= 0.15 \text{ s}$	A1
5(d)(i)	path difference = 3.0 cm	A1
5(d)(ii)	phase difference = 180°	A1
5(e)	line drawn joining points where only maxima are observed (i.e. through points where wavefronts intersect) of length at least 4 cm	B1

32.

5(a)(i)	(coherence means) constant phase difference (between waves)	B1
5(a)(ii)	(interference is) the sum/addition/combination of the displacements of overlapping/meeting waves	B1
5(b)(i)	$n\lambda = d \sin \theta$	C1
	$\lambda = \sin 51^\circ / (2 \times 6.7 \times 10^5)$	A1
	$= 5.8 \times 10^{-7} \text{ m}$	
5(b)(ii)	smaller angle (corresponding to second order maxima and so) shorter distance (between second order maxima spots)	B1

33.

4(a)(i)	vibrations (of particles) are parallel to direction of energy propagation	B1
4(a)(ii)	waves meet/overlap (at a point)	B1
	(resultant) displacement is sum of individual displacements	B1
4(b)(i)	$\lambda = ax / D$	C1
	$= (3.7 \times 10^{-4} \times 4.3 \times 10^{-3}) / 2.3$	C1
	$= 6.9 \times 10^{-7} \text{ (m)}$	A1
	$= 690 \text{ nm}$	
4(b)(ii)	- no change to fringe separation/fringe width/number of fringes - bright fringes are darker - dark fringes are brighter Any two marking points, 1 mark each	B2

34.

4(a)	progressive waves transfer energy or stationary waves do not transfer energy	B1
4(b)(i)	0.32 m	A1
4(b)(ii)	$v = \lambda / T$ or $v = f\lambda$ and $f = 1 / T$	C1
	$v = 0.32 / 0.020$ or 50×0.32	A1
	$= 16 \text{ m s}^{-1}$	
4(b)(iii)	450° or 90°	A1
4(b)(iv)	(P has) maximum downward displacement at 0.005 s	B1
	returns to original position/point (at 0.010 s)	B1
4(c)(i)	(position where) zero amplitude	B1
4(c)(ii)	2	A1
4(c)(iii)	180°	A1
4(c)(iv)	string drawn between X and Y with one antinode midway along the string	B1

35.

6(a)	the waves (of the same type) move in opposite directions and overlap	B1
	the waves have the same (speed and) frequency/wavelength	B1
6(b)(i)	zero amplitude	B1
6(b)(ii)	distance = 6.0×4 = 24 cm	A1
6(b)(iii)	180°	A1

36.

5(a)	$v = f\lambda$ or $c = f\lambda$	C1
	$f = 3.0 \times 10^8 / 0.040$	C1
	= 7.5×10^9 (Hz)	A1
	= 7.5 GHz	
5(b)(i)	path difference = 0.020 m	A1
5(b)(ii)	phase difference = 180°	A1
5(c)	(intensity) increases	C1
	(intensity) increases by a factor of 4	A1
5(d)(i)	minimum moves closer to the maximum or decrease in separation of maximum and minimum	B1
5(d)(ii)	the maximum and minimum exchange places or the maximum becomes a minimum and the minimum becomes a maximum	B1

37.

5(a)(i)	$T = 2.0 \times 10^{-5} \times 6.0$ (= 1.2×10^{-4} s)	C1
	$f = 1 / (2.0 \times 10^{-5} \times 6.0)$	A1
	= 8300 Hz	
5(a)(ii)	new trace shows the same period	B1
	new trace shows amplitude of 10 small squares	B1
5(a)(iii)	(trace is a) vertical line	B1
5(b)(i)	$n\lambda = d \sin \theta$	C1
	$\lambda = (3.4 \times 10^{-6} \times \sin 16^\circ) / 2$	A1
	= 4.7×10^{-7} m	
5(b)(ii)	$n = 3.4 \times 10^{-6} (\times \sin 90^\circ) / 4.7 \times 10^{-7}$ or $2 (\times \sin 90^\circ) / \sin 16^\circ$ (= 7.2 or 7.3)	C1
	highest order = 7	A1

38.

4(a)	(two or more) waves <u>meet</u> (at a point)	B1
	(resultant) displacement is the sum of the individual displacements	B1
4(b)(i)	it is a (wave) reflector / it reflects (the wave)	B1
4(b)(ii)	$v = f\lambda$ or $c = f\lambda$	C1
	$f = 3.0 \times 10^8 / 0.040$ $= 7.5 \times 10^9$ (Hz) $= 7.5 \times 10^9 / 10^9$ (GHz)	C1
	$= 7.5$ GHz	A1
4(b)(iii)	1 distance = 0.020 m	A1
	2 number = 5	A1

39.

4(a)	distance moved by wavefront/energy during one cycle/oscillation/period (of source) or <u>minimum</u> distance between two wavefronts or distance between two <u>adjacent</u> wavefronts	B1
4(b)	$v = \lambda / T$ or $v = f\lambda$ and $f = 1 / T$	C1
	$T = 460 \times 10^{-9} / 3.00 \times 10^8$	C1
	$= 1.5 \times 10^{-15}$ s	A1
4(c)	waves pass through/enter the slit(s)	B1
	waves spread (into geometric shadow)	B1
4(d)(i)	$n\lambda = d \sin \theta$	C1
	$G = \sin \theta / \lambda$	A1
	$d = 4 / G$	
4(d)(ii)	straight line from 400 nm to 700 nm that is always below printed line	M1
	straight line has smaller gradient than printed line and is 5 small squares high at wavelength of 700 nm	A1

40.

4(a)	time for one oscillation/vibration/cycle or time between <u>adjacent</u> wavefronts (passing the same point) or <u>shortest</u> time between two wavefronts (passing the same point)	B1
4(b)	(when two or more) waves meet/overlap (at a point)	B1
	(resultant) displacement is sum of the individual displacements	B1
4(c)(i)	microwave(s)	B1
4(c)(ii)	$v = \lambda / T$ or $v = f\lambda$ and $f = 1/T$	C1
	$T = 0.040 / 3.00 \times 10^8$	C1
	$= 1.33 \times 10^{-10}$ (s)	A1
	$= 1.33 \times 10^{-10} / 10^{-12}$ (ps) $= 130$ ps	

4(c)(iii)	$(1.380 - 1.240) / 0.040 = 3.5$ or $1.380 / 0.040 - 1.240 / 0.040 = 3.5$	A1
4(c)(iv)	phase difference = 1260° or 180°	A1
4(c)(v)	(always) zero	A1
4(c)(vi)	increase in distance between (adjacent intensity) maxima/minima	A1

41.

4(a)	(when two or more) waves meet/overlap (at a point)	B1
	(resultant) displacement is sum of the individual displacements	B1
4(b)	intensity $\propto$ amplitude ²	C1
	maximum intensity = $9I$	A1
4(c)(i)	$x = \lambda D / a$	C1
	$= (550 \times 10^{-9} \times 1.2) / (0.35 \times 10^{-3})$	C1
	$= 1.9 \times 10^{-3} \text{ m}$	A1
4(c)(ii)	red light has longer wavelength (than 550 nm) so distance (between fringes) increases	B1

42.

5(a)	a (much) louder sound can be heard	B1
5(b)	$v = f\lambda$	C1
	$\lambda = 340 / 530$ (= 0.64 m)	C1
	height = $0.64 / 4$ = 0.16 m	A1
5(c)	distance = $0.64 / 2$ or 0.16×2 or $(\frac{3}{4} \times 0.64 - \frac{1}{4} \times 0.64)$ = 0.32 m	A1

43.

5(a)	maximum displacement (of a point/particle on string/wave)	B1
5(b)	$v = \lambda / T$ or $v = f\lambda$ and $f = 1 / T$	C1
	$T = 690 \times 10^{-9} / 3.00 \times 10^8$	C1
	$= 2.3 \times 10^{-15} \text{ s}$	A1
5(c)(i)	$\lambda = ax / D$	C1
	$G = x / D$	A1
	(so) $a = \lambda / G$	
5(c)(ii)	straight line from origin always below printed line	M1
	line is half the height of printed line at maximum D	A1

44.

5(a)	wave(s) (travel along string and) reflect at fixed point / A / B / end	B1
	incident and reflected waves superpose / interfere or two waves travelling / with speed in opposite directions superpose / interfere	B1
5(b)	line has an approximate sinusoidal shape with maximum downward displacement at P and zero displacement at each node	B1
5(c)	$v = \lambda / T$ or $v = f\lambda$ and $f = 1 / T$	C1
	$\lambda = 35 \times 0.040$ or $35 / 25$ (= 1.4 m)	C1
	distance = 1.4×2.5 = 3.5 m	A1
5(d)	(number of periods / cycles) ($= t / T$) = $0.060 / 0.040$ (= 1.5)	C1
	amplitude = $72 / 6$ = 12 mm	A1

45.

5(a)	constant phase difference (between the waves)	B1
5(b)(i)	path difference = 1.5×660 = 990 nm	A1
	phase difference = $360^\circ \times 1.5$ = 540°	A1
5(c)	$\lambda = ax / D$	C1
	$x = (660 \times 10^{-9} \times 1.8) / 0.44 \times 10^{-3}$	C1
	= 2.7×10^{-3} m	A1
5(d)	bright fringes are brighter	B1
	no change to dark fringes	B1
	no change to (fringe) separation / (fringe) spacing	B1
5(e)	(blue light has) shorter wavelength	M1
	(so) decrease (slit) separation	A1

46.

5(a)(i)	power = intensity $\times$ area	C1
	= $1.3 \times 10^3 \times (\pi \times 0.055^2)$ = 12 W	A1
5(a)(ii)	intensity = power / area	A1
	= $12 / (\pi \times 0.0015^2)$ = 1.7×10^6 W m ⁻²	
5(b)(i)	($\lambda =$) v / f or c / f	C1
	($\lambda =$) $3.0 \times 10^8 / 3.7 \times 10^{15} = 8.1 \times 10^{-8}$ (m)	A1
5(b)(ii)	ultraviolet	A1

5(b)(iii)	$d \sin \theta = n\lambda$ or $(1/N) \times \sin \theta = n\lambda$	C1
	$d = 1 / 2400 \times 10^3 \text{ (m)}$ $= 4.2 \times 10^{-7} \text{ (m)}$ or $N = 2400 \times 10^3 \text{ (m}^{-1}\text{)}$	C1
	$n = 4.2 \times 10^{-7} \times \sin 90^\circ / 8.1 \times 10^{-8}$ or $\sin 90^\circ / 2400 \times 10^3 \times 8.1 \times 10^{-8}$ $n = 5.2$ or 5.1 or when $n = 5$, $\theta = 76.4^\circ$ and when $n = 6$, $\sin \theta > 1$ (so) $n = 5$	B1
	number of maxima = $(2 \times 5) + 1$ $= 11$	A1
5(b)(iv)	the wavelength has increased	M1
	(so) number of maxima decreases	A1

47.

5(a)(i)	period or $T = 1 / 5000 (= 2 \times 10^{-4} \text{ s})$	C1
	time-base setting = $1.5 \times 2 \times 10^{-4} / 6.0$ or $2 \times 10^{-4} / 4.0$ $= 5 \times 10^{-5} \text{ s cm}^{-1}$	A1
5(a)(ii)	new trace drawn with same period as original trace	B1
	new trace drawn with amplitude greater than 1.0 cm	M1
	new trace drawn with amplitude of 1.7 cm	A1
5(b)(i)	path difference (from slits to P) is zero or phase difference (between waves at P) is zero (so constructive interference)	B1
5(b)(ii)	$\lambda = ax / D$	C1
	$D = (3.6 \times 10^{-4} \times 4.0 \times 10^{-3}) / 630 \times 10^{-9}$	C1
	$= 2.3 \text{ m}$	A1
5(c)	upward sloping straight line starting from a non-zero value of x at $\lambda = 400 \text{ nm}$	B1

48.

5(a)(i)	infrared	B1
5(a)(ii)	$v = f\lambda$ or $c = f\lambda$	C1
	$f = 3.0 \times 10^8 / 8.4 \times 10^{-6}$ $= 3.6 \times 10^{13} \text{ (Hz)}$ $= 36 \text{ THz}$	A1
5(b)(i)	constant phase difference (between the waves) (with time)	B1
5(b)(ii)	$\lambda = ax / D$	C1
	$x = 22 / 8$ or 2.75 (mm) or $22 \times 10^{-3} / 8$ or $2.75 \times 10^{-3} \text{ (m)}$	C1
	$a = (6.2 \times 10^{-7} \times 2.8) / (22 \times 10^{-3} / 8)$ $= 6.3 \times 10^{-4} \text{ m}$	A1

5(c)(i)	difference in distances = $6.2 \times 10^{-7} / 2$ = 3.1×10^{-7} m	A1
5(c)(ii)	phase difference = 360°	A1

49.

4(a)	the number of wavefronts/crests/troughs passing a fixed point per unit time or the number of oscillations per unit time (of source / point on wave / particle of medium)	B1
4(b)(i)	$T = 4 \times 0.50 \times 10^{-3}$ (= 2.0×10^{-3} s)	C1
	$f = 1 / 2.0 \times 10^{-3}$ = 500 Hz	A1
4(b)(ii)	amplitude = 2.8×0.20 = 0.56 V	A1
4(c)	period same as original trace	B1
	sinusoidal wave of constant amplitude less than 2.8 cm throughout	M1
	amplitude 1.4 cm	A1
4(d)(i)	when (two or more) waves <u>meet</u> (at a point)	B1
	(resultant) displacement is the sum of the individual displacements	B1
4(d)(ii)	node-to-node separation is $\lambda / 2$ or microphone moves through 3 node-to-node separations or $d = 1.5\lambda$	C1
	$\lambda = 1.05 / 1.5$ = 0.70 m	A1
4(d)(iii)	$v = f \times \lambda$	C1
	= 500×0.70	A1
	= 350 m s^{-1}	

50.

4(a)	(when two or more) waves meet / overlap (at a point)	B1
	(resultant) displacement is the sum of the individual displacements	B1
4(b)	$\lambda = ax / D$	C1
	$\lambda = [(0.65 \times 10^{-3}) \times (1.7 \times 10^{-3})] / 1.9$	C1
	$\lambda = 5.8 \times 10^{-7}$ m	A1
4(c)(i)	waves are out of phase by a quarter of a cycle / period or when one wave has maximum/minimum displacement, the other wave has zero displacement	B1
4(c)(ii)	(statement 2 is) not correct (because) waves do not have phase difference of 180° / not in antiphase or one wave has some displacement when other has no displacement or the displacements of the waves are not always equal and opposite	B1

4(d)	more diffraction (of light/waves by slits) or light/waves are more spread (by slits) or light/waves (from slits) have less intensity	B1
	more (bright/dark) fringes or bright fringes are less bright / are dimmer / have lower intensity or no change to fringe spacing/separation/width	B1

51.

5(a)(i)	light intensity has maximum value at 0° , 180° , 360° and zero intensity at 90° , 270°	M1
	'sinusoidally-shaped' curve	A1
5(a)(ii)	$4.2 = 7.6 \cos^2 \theta$	C1
	$\theta = 42^\circ$	A1
5(b)	wave passes (through) an aperture or wave passes (by / through / around) an edge	B1
	wave spreads (into geometrical shadow)	B1
5(c)(i)	$n\lambda = d \sin \theta$	C1
	$d = (3 \times 4.3 \times 10^{-7}) / \sin 68^\circ$ $= 1.4 \times 10^{-6} \text{ m}$	A1
5(c)(ii)	$1.4 \times 10^{-6} \times \sin 68^\circ = 2 \times \lambda$ or $3 \times 4.3 \times 10^{-7} = 2 \times \lambda$	C1
	$\lambda = 6.5 \times 10^{-7} \text{ m}$	A1

52.

5(a)(i)	cross labelled Y drawn: at any position where wavefronts cross or centrally in a 'diamond' shape formed between any adjacent wavefronts from A and B	B1
5(a)(ii)	cross labelled Z drawn on a wavefront from one source at a point midway between adjacent wavefronts from the other source	B1
5(b)(i)	$\lambda = ax / D$	C1
	$a = (2.9 \times 10^{-5} \times 140) / (1.2 \times 10^{-2})$	C1
	$= 0.34 \text{ m}$	A1
5(b)(ii)	infrared	A1